S.O.L.A.R.I.S.

Solar-**O**ptimized **L**and **A**utonomous **R**echarge & **I**ntelligence **S**ystem

Authors:
(Group # 12)

Rafael Puig
Electrical Engineering

Ryan Markowitz
Computer Engineering

Nathan Hammond
Electrical Engineering

Garrett Fortier
Computer Engineering

Reviewer Committee:

Dr. Maddox	ECE	Adjunct Professor
Dr. Rakshan	ECE	Assistant Professor
Dr. Sahawneh	ECE	Lecturer

Mentor:
Dr. Suboh Suboh

Chapter 2 - Project Description

2.1 Project Background and Motivation

In senior design, constraints define a project's limitations in terms of functionality and capabilities. Our project aims its sights on industry-relevant processes and standards, which serve as a learning opportunity. Solar-Optimized Land Autonomous Recharge & Intelligence System (*SOLARIS*), our project, can be split into two different categories. The main objectives of interest are solar optimization and autonomous intelligence systems. With these ideas in mind, we plan to tackle power-conscious design of a system intended to be self-sufficient. A major constraint yet to be stated for the team is cost. By splitting the cost between all team members, it remains important to find the balance between expenditure, low-power, and net positive charging capability. The key educational value remains an important portion as it relates heavily to our academic studies and our own personal interest as Electrical Engineer (EE) and Computer Engineering (CpE) majors. As we dive further into motivations for the project, we see software and hardware equally important to achieving this final goal.

Software design, firmware included, is one such interest that drives a large portion of the project. Embedded architecture has long since dominated consumer electronics markets and continues to prove itself valuable time and time again. An embedded system would be nothing without firmware or a software interface to connect it to the internet. That is why it is a motivation of ours to not only design a robust control system but also interact with it through a wireless connection. This not only provides an interesting learning experience but also adds valuable and relevant experience to the large toolkit expected from engineers. The combination provides an interaction between hardware constraints and software design in real-world applications. Integration of wireless communications reflects industry-relevant practices where modularity, scalability, and maintainability are essential. To emphasize, mobile and desktop applications for control and statistics serve as a valuable experience in developing a bridged embedded system. These embedded applications showcase engineering skills that are indicative of common workflows, frequent pairing with companion applications, and automated control systems.

Solar energy adoption continues to accelerate as improvements in battery storage and solar panel efficiency make renewable systems increasingly practical. This project offers hands-on experience designing and implementing a solar-powered system in a real-world context. At the same time, the growing fields of robotics and automation demand engineers who can effectively integrate electromechanical hardware with reliable control strategies. Through this project, the team will gain experience in motor and actuator selection, control system design and tuning, and system-level power management. By addressing efficiency, variable loading, and practical implementation constraints, the project will develop skills applicable to both energy systems and robotics and automation engineering roles.

2.2 Current Commercial Technologies and Existing Projects

2.2.1 Jackery Solar Mars Bot

Figure 1. *Jackery Solar Mars Bot*

Currently, there is no commercially realized product directly comparable to *SOLARIS*. This is primarily due to the high cost of solar panel technology, the reduced effectiveness of mobile robots burdened with large-area solar panels, and the greater efficiency of traditional stationary solar panel installations for battery charging. However, one company that specializes in clean, portable power solutions, Jackery, has proposed a concept for a future autonomous solar-farming mobile robot known as the “Solar Mars Bot” [1].

While the Solar Mars Bot has not yet been released as a consumer product, its existence as a concept is significant. Jackery describes the system as an autonomous robot capable of navigating toward regions of stronger sunlight and repositioning itself when light conditions deteriorate. The design integrates foldable photovoltaic panels with onboard energy storage, allowing the robot to both generate and manage its own power. When deployed, the panels are rated to deliver up to 600 W of charging power to the onboard 5 kWh battery.

Although technical details regarding its sensing, control algorithms, and navigation methods are not publicly available, the concept itself demonstrates a growing commercial interest in autonomous, mobile solar collection. This proposal reinforces the idea that *SOLARIS* occupies an emerging design space. The Solar Mars Bot suggests that industry is beginning to explore robotic platforms that can actively seek out optimal light conditions rather than relying on fixed installations. Even in its conceptual form, it validates the direction of this project and highlights the relevance of developing efficient, autonomous systems that can adapt their position in response to changing environmental conditions.

2.2.2 SolarX: DIY STEM Solar Powered Robot

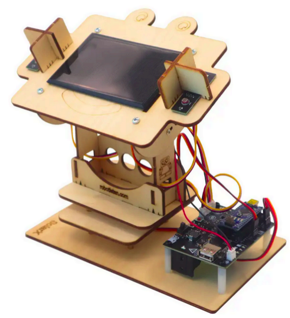


Figure 2. *DIY STEM Solar Powered Robot*

The SolarX: DIY STEM Solar Powered Robot is one of many consumer products designed as learning tools for directing a solar panel toward a light source, typically the sun, as effectively as possible. It uses photoresistive sensors to measure incoming light and determine how the panel should be oriented. Two servo motors, mounted on perpendicular axes, control the horizontal and vertical position of the solar panel, allowing it to continuously adjust until it is facing the brightest direction. [3]

Although this product represents only a small subset of what *SOLARIS* aims to accomplish, it felt important to include it as a design reference for that specific subpart of the project. A core goal of *SOLARIS* is to improve power generation by responding to changing light conditions, and this kit provides a concrete example of how sensor-driven solar alignment can be implemented in practice.

One major difference is in how motion will be achieved. The SolarX relies on servos, which require continuous power to hold their position. For *SOLARIS*, the intent is to explore a brushless motor actuator or similar mechanism that can reposition the panel without needing constant power to maintain its orientation. Since power efficiency is a primary constraint of the system, reducing energy draw is critical. While the actuation method will differ, the underlying design principle remains the same: use light feedback to determine how the panel should be oriented and adjust it accordingly to maximize available power.

2.2.3 Summary

At this time, there is no consumer product that fully matches the scope of *SOLARIS*. However, existing concepts and educational devices show that the core ideas behind the project are both feasible and relevant. The Jackery Solar Mars Bot highlights a growing commercial interest in autonomous, mobile solar platforms that can reposition themselves in response to changing light conditions. Even as a concept, it supports the idea that mobile robots can be used to actively seek out better solar exposure rather than relying on fixed installations.

On a smaller scale, products such as the SolarX DIY STEM Solar Powered Robot demonstrate how light sensors and actuators can be used to orient a solar panel toward the brightest direction. While these systems are limited and primarily educational, they provide a clear and proven example of sensor-driven solar alignment, which directly aligns with one of *SOLARIS*' core goals.

SOLARIS takes these existing ideas and pushes them further by turning solar tracking into a fully autonomous, mobile behavior. Instead of only adjusting the angle of a panel, the system is designed to decide when orientation alone is no longer sufficient and physically move to a better location. Power efficiency is treated as a core constraint, so every action is driven by the goal of gaining more energy than it consumes. By combining light sensing, autonomous decision making, and low-power actuation, *SOLARIS* explores how a robot can actively respond to its environment to improve its own energy generation.

2.3 Goals and Objectives

2.3.1.1 Basic Goals Summarized

- Create a robot capable of autonomously moving out of shade on flat ground.
- Rotate the solar panel about two axes to orient itself in peak sunlight conditions.
- Energy gained from moving the robot must be greater than an equivalent, fixed solar panel over a full daylight cycle.
- Measure and report information to the end user such as distance traveled, energy used, energy gained, state of charge, etc.
- Create a phone app to serve as the primary means of communication between the user and the robot.

2.3.2.1 Advanced Goals Summarized

- Allow users to manually select between two different modes through the phone app, stationary, autonomous, and manual.
    - In stationary, the robot will not move, however it will still move the solar panel.
    - In autonomous mode, full control is given to the robot.
- The phone app should display all information we collect from the robot to the user.
- The robot should be capable of avoiding collisions. In the event of a collision, it should be able to tell, back up and turn.
- The robot will be able to reliably deliver regulated electrical power from its onboard energy storage system to a user's personal electronic device.

2.3.3.1 Stretch Goals Summarized

- Create a website that mimics the phone app's interface. It will not be capable of connecting to the robot, but if users want to see their data collected from the robot, they can sign in and see it.
- Use IR sensors to detect going onto a curb / road.
- Add GPS functionality to track position throughout the day, and use that information to have a more robust model of where good spots are at given times.

2.3.1.2 Basic Goals

At the project's core, our main goal is to create an autonomously moving robot capable of solar farming. The solar farming robot should be able to rotate and angle a solar panel for peak sunlight positioning. *SOLARIS* needs to be capable of operating in outdoor environments like flat yards or open fields, as opposed to traditional solar harvesting methods. As it moves, there will be a low-level obstacle avoidance protocol causing the robot to traverse non-ideal environments. With both basic goals in mind, we want to expand capabilities that support our broader goal, maximizing solar energy collection without requiring external intervention. The overarching goal is what supports the basic idea to design a new kind of solution that is more practical.

Another central goal that is explored in the senior design project is providing a net positive energy benefit when compared to stationary solar harvesting approaches. This robot will evaluate the tradeoffs seen between energy expended for movement and the additional energy gained by repositioning itself. It is without question that aiming the solar panel at the source of sunlight will increase electrical intake; however, it is yet unknown whether movement can also play a role in this collection process. Therefore, success is defined by achieving greater net energy generation over a full daylight cycle compared to an equivalently sized fixed solar panel.

Though autonomous in nature, our goals do not completely neglect human interaction with the project. For these reasons, we need observable data that gives us insight into efficiency and mobility. The robot should also serve as a platform for observing some of its most basic operating behaviors. Key metrics like battery life, charging rate, and distance traveled should be available to the user through a wireless connection. This system should also support safe operation, as it is meant to be autonomous in nature but observable by a user. The robot should be able to identify an event, such as a collision, and perform a maneuver to avoid the continuation of the event. These goals ensure autonomous operation, which can remain practical and robust for a real-world setting while also keeping the project innovative and fresh.

2.3.2.2 Advanced Goals

An advanced goal of the project is to ultimately enhance user interaction through the usage of a mobile application. The mobile application will serve as the primary interface between the user and the robot. With this platform, we can provide performance metrics, system behavior, and influence robot operation without touching the robot physically. In the mobile application, users should be able to select between one of two modes. These two modes exist transparently but functionally work the same. One mode allows the bot to become stationary, and the bot will function as a traditional solar harvesting setup with a rotating solar panel. The other mode would be default, as the robot's full autonomous nature would allow it to navigate on its own and relocate if necessary.

This fully autonomous behavior highlights an advanced goal for this project as the robot is given complete control over its own decisions. The robot will be responsible for its energy management, navigation, and overall decision-making. In this mode, the robot will systematically evaluate conditions to determine if movement is necessary and then avoid obstacles on its way to a perceived optimal solution. All while a user can watch over system metrics via a wireless connection to the robot. This presents the last advanced goal of our project, which is to present relevant data to the user's device. On the device, metrics like distance traveled, temperature, and charging rate can all be displayed. Displaying this data allows a user to understand the robot's behavior while answering an important question about the robot's functionality.

2.3.3.2 Stretch Goals

To expand upon our advanced goals, we would like to introduce additional functionality that can enhance user control, system intelligence, and long-term data analysis. With these features, we could apply a more sophisticated system to explore decision-making capabilities that expand upon the original core system requirements for functionality. One such stretch goal is to expand upon a manual control system from the user interface of the application. This could include finer control over movements, adjustments for speed, and the ability to directly interact with solar panel orientation. Ultimately, these adjustments would provide users with greater flexibility in their own use of the platform. Similarly, this would require a much more robust design, allowing for a much more fine-tuned system entirely.

In addition to mobile application enhancements, there could be an integration of higher-level control logic or the mounting of an artificial intelligence to improve autonomous behavior. This next level of autonomy could enhance decision-making behavior capable of strategizing navigation. The robot would also be capable of a higher level battery management with adaptive battery level thresholds and movement statistics. These upgrades would allow the system to better respond to changing environmental conditions, which could produce increased performance over time. To expand on connectivity, the robot could alternatively connect to a web-based interface where the structure of the app would be mirrored, but would apply a more practical interaction. The website would not provide direct communication with the robot, but would alternatively allow for more detailed logging of events and statistics for later consumption. All stretch goals mentioned support a long-term analysis and provide a more accessible platform for interaction with the robot.

2.3.4 Basic Objectives

- Design and fabricate a mobile robotic platform driven by DC motors and a motor driver capable of traversing flat outdoor terrain such as yards or open fields.
- Implement a mechanically actuated solar panel mounting system using a motorized joint to enable controlled rotation and tilt for sunlight alignment.
- Integrate obstacle detection sensors (ultrasonic sensors) and low-level motor control logic to support autonomous avoidance of environmental obstacles.
- Incorporate onboard energy storage, charge regulation, and efficient power distribution circuitry to support simultaneous solar charging and system operation.

- Develop embedded control firmware on a microcontroller to manage motion control, sensor input, and autonomous decision-making.
- Measure and record battery voltage, charging current, and power consumption using onboard sensing circuitry.
- Implement wireless communication hardware to transmit system metrics and operational status to an external user device.
- Detect collision or near-collision events using physical or proximity sensors and trigger corrective motor commands to prevent sustained contact.
- Ensure the system can operate autonomously without external power or physical user input after deployment.

2.3.5 Advanced Objectives

- Support multiple operational modes selectable via wireless communication, including:
    - A stationary mode where only the solar panel actuation hardware is active.
    - A fully autonomous mode where the drive motors and sensors are continuously engaged.
- Integrate wheel encoders or equivalent motion sensing to quantify distance traveled during autonomous operation.
- Design the electronics and firmware architecture to allow modular expansion of sensors or actuators without redesigning core subsystems.
- Integrate a regulated power delivery system capable of safely converting stored onboard energy into standardized output to charge consumer electronic devices with reliable performance and protection mechanisms.

2.3.6 Stretch Objectives

- Implement manual motor and panel control through the application by directly commanding motor drivers and actuators.
- Develop a web-based interface to visualize logged system data independent of direct robot communication.
- Use hall effect sensors to detect collisions as a bumper instead of checking if an axle has difficulty moving through the encoder.

2.4 Required Specifications

2.4.1 Overall System Specifications

Table 1. *Energy Specifications*

Energy Specifications		
Specification	**Description**	**Target Value and Unit(s)**
Energy for User	After a sunny, optimal day, the system shall provide the end user with at least 15 watt-hours (Wh) of usable energy from the battery.	≥15 Wh/day
Energy Generated	After a sunny, optimal day, the system shall generate at least 75 watt-hours (Wh) of electrical energy through solar harvesting.	75 Wh/day
Power Output Connection	A USB-C Power Delivery (PD) connection shall be provided for user energy extraction from the system.	USB-C Power Delivery (PD)
Power Input Connection	A USB-C Power Delivery (PD) connection shall be used to charge the system battery.	USB-C Power Delivery (PD)

Table 2. *Mobility Specifications*

Mobility Specifications		
Specification	**Description**	**Target Value and Unit(s)**
Max Forward Robot Speed	Maximum designed robot speed (forward).	0.3 m/s
Average Robot Speed	Normal forward robot speed.	0.1 m/s
Max Reverse Robot Speed	Maximum designed robot speed (reverse).	0.3 m/s
Weight		<30 lbs

Mobility Specifications		
Solar Panel Rotation	Pan axis range of motion.	360 Degrees
Solar Panel Tilt	Tilt axis range of motion.	45 Degrees

2.4.2 Component System Specifications

Table 3. *Component Specifications*

Component	Parameter	Subsystem and Description	Target Value and Unit(s)
Geared BDC Motor w/ Planetary Gearbox	Torque	4 Motors that provide the means of rotating the wheels for movement.	24 V, 48 W, 23 RPM
Solar Panel	Power	Primary source of energy for battery charging.	20 Watts
Solar Panel	Voltage	Voltage level that solar panel operates at.	12 V
Battery	Capacity	Powers all devices on the robot system.	128 Wh
Pneumatic Knobby Tires	Size	Wheels that each motor will be linked to via a shaft.	4”
High Torque Worm Gear BDC Motor w/ Self Locking	Torque	Motor for pan-axis rotation of solar panel.	24 V or 12 V 5 RPM
Bluetooth Low Energy (BLE)	Wireless Communication Range	Communication subsystem. Used to send data from the robot to a mobile app.	1-6 meters
Linear Actuator	Torque	Tilt-axis motion for solar panel	200 N-M, 4” Stroke
PWM H-Bridge Motor Drive	Amps	3 boards for driving the motors and linear actuator	10 A

5 V Regulator	Volts	Regulator for sensors (ultra-sonic sensor)	5 V
3.3 V Regulator	Volts	Regulator for photoresistors	3.3 V

2.4.3 Hardware and Software Diagrams + Prototype illustration

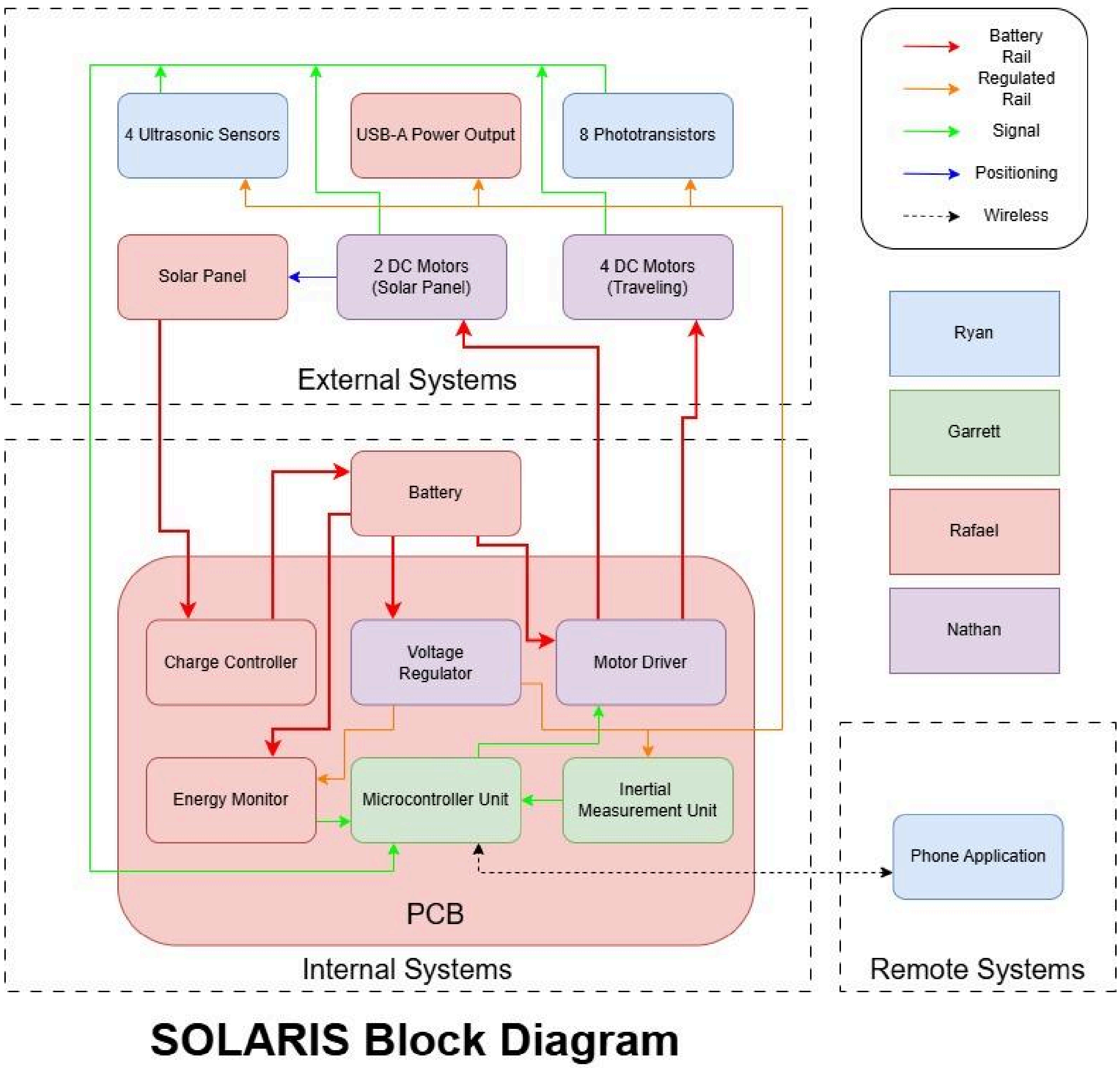


Figure 3. *Hardware Block Diagram*

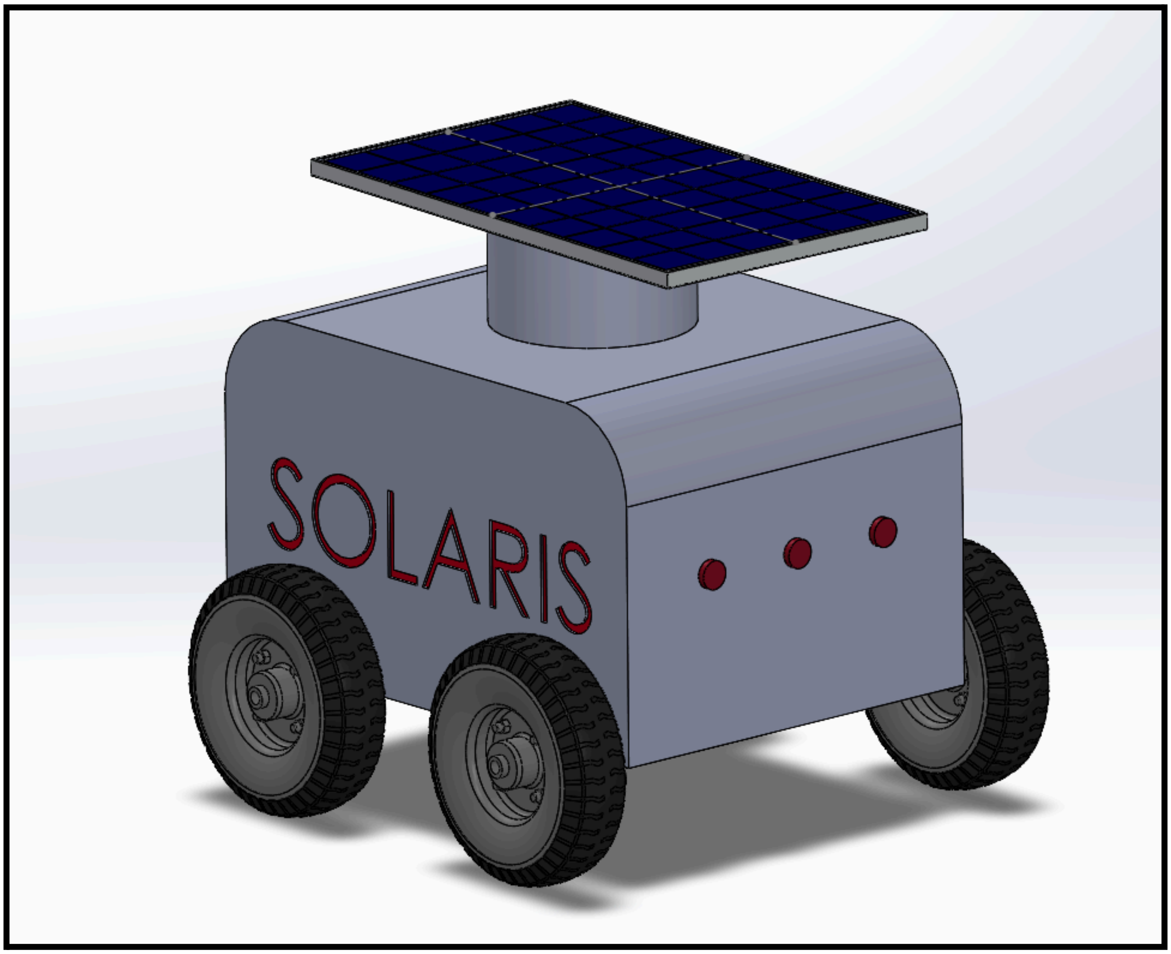


Figure 4. *SOLARIS Prototype Illustration*

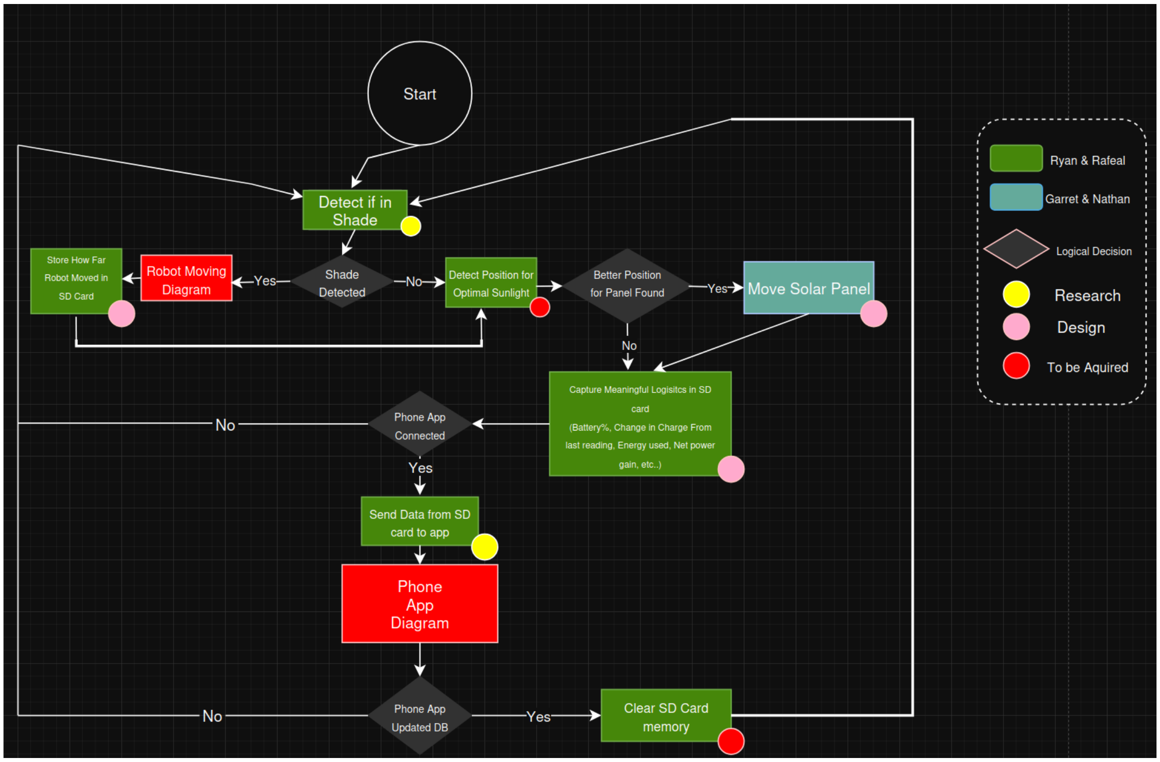


Figure 5. *SOLARIS Main Function Software Diagram*

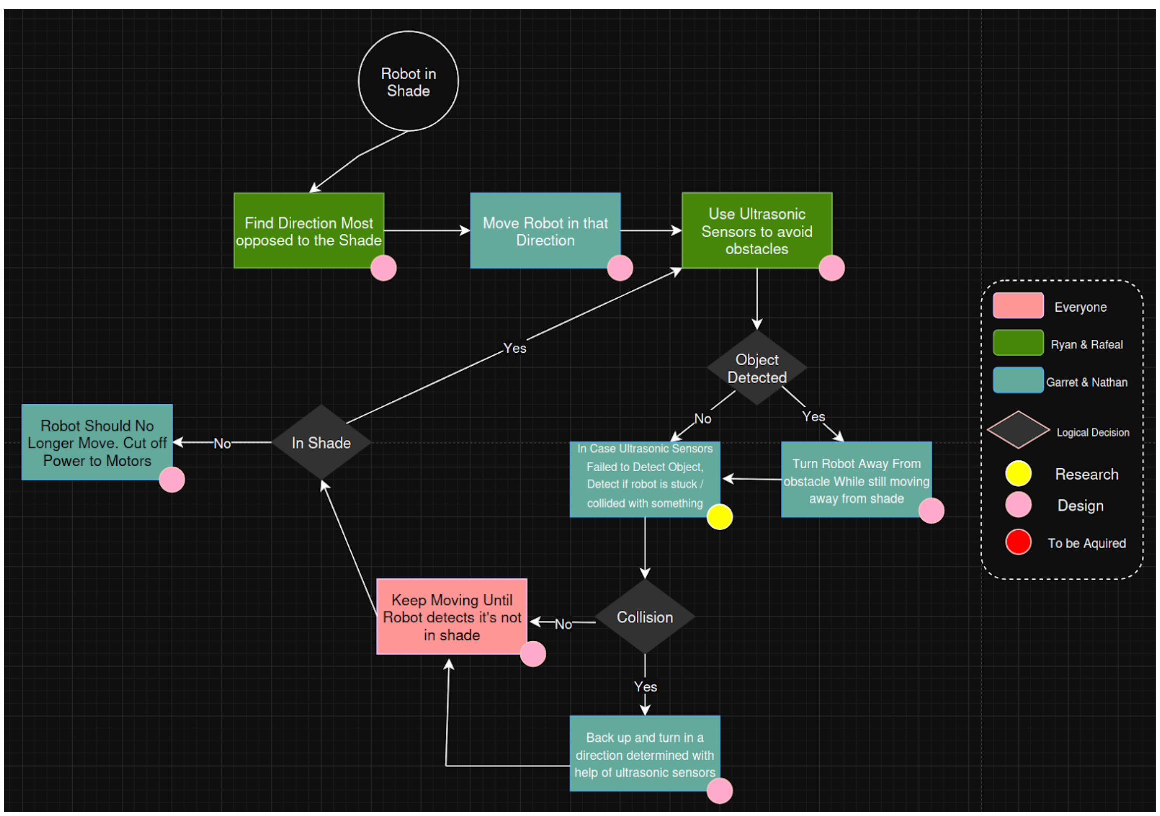


Figure 6. *SOLARIS Robot Driving Software Diagram*

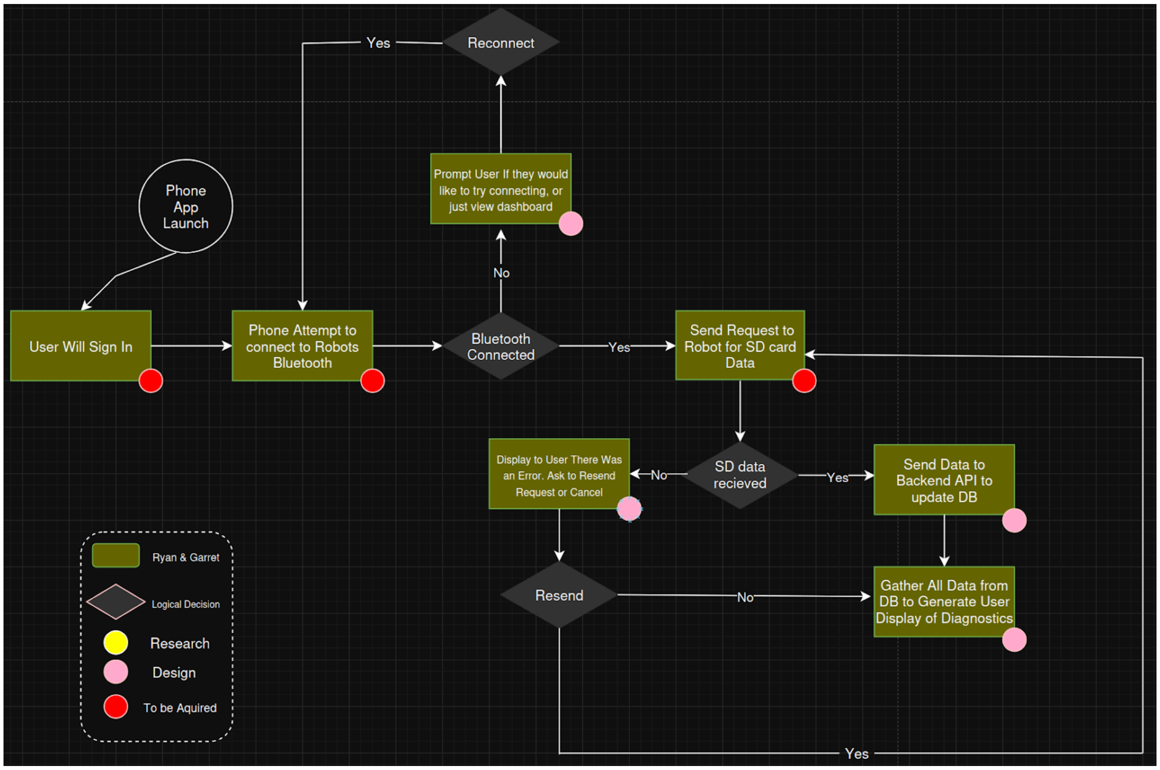


Figure 7. *SOLARIS phone app Software Diagram*

2.5 Design Requirements

2.5.1 Marketing & Engineering Requirements

Table 4. *Justification Chart for Requirements*

Marketing Requirements	Engineering Requirements	Justification
1, 2	1. Rover should reposition itself autonomously toward higher light intensity using sensors and the control system.	The ability to autonomously reposition corrects inefficiencies in fixed solar harvesting systems while reducing the need for manual intervention.

1, 2, 4	2.	The system should operate in low-power sleep mode while charging, consuming <300mW in the idle state.	Low-power operation lets the system maximize energy gain while extending operational lifetime; the bot can outlast suboptimal weather conditions entirely.
1, 2, 3	3.	The rover should detect obstacles within a minimum range of 20-50 cm accurately using ultrasonic sensors to avoid collision.	Automated obstacle avoidance ensures reliability in dynamic real-world environments, whether outdoors or indoors.
2, 3	4.	The system should provide real-time battery voltage, charging current, and charge state data.	Being able to monitor energy metrics allows users to evaluate system performance like distance traveled, battery level, and charging statistics.
2, 3, 4	5.	The rover should communicate wirelessly with a smartphone via Bluetooth Low Energy (BLE).	Bluetooth Low Energy (BLE) enables low-power communication with user devices, making it suitable for field deployment and demonstrations.
1, 3	6.	Rover should have four wheels with encoders to estimate distance traveled and improve distance traveled accuracy.	Feedback from encoders provides improved navigational efficiency while allowing precise movements to be made for travel.
3	7.	The system should require no external infrastructure for normal operation.	An independent infrastructure makes deployability a priority, allowing for remote or agricultural environments.
4	8.	Total system hardware cost shall not exceed $1000	Cost constraints enable feasibility for deployment while also aligning with the competitive market for low-cost energy monitoring.
3, 4	9.	The system should be implemented on a single custom PCB with modular sensor connections.	Custom PCB demonstrates design project integrity, while the modularity of sensors allows for an easier debugging process.
4, 5	10.	The charging system for solar-battery should exceed an efficiency of >50% under nominal operating conditions.	An efficient charging system is achievable using commercial MPPT-based Li-ion charging ICs and demonstrates the balance between complexity, cost, and performance.

5	11.	The System should operate reliably in outdoor lighting and temperature without user intervention.	The reliability of an unattended rover in a working environment increases the number of practical applications.
1, 2, 4	12.	The rover should operate for a minimum of 72 hours without solar input under normal conditions.	Ensuring continued operation even during sub-optimal conditions demonstrates robust power design and applicability in differing environments.
Marketing Requirements **1. The system should maximize solar energy harvesting through autonomous behavior** **2. The system should operate reliably with minimal human intervention** **3. The system should be easy to deploy and monitor** **4. The system should be low-cost and energy-efficient** **5. The system should be reliable in a real-world operating condition**			

2.5.2 House of Quality

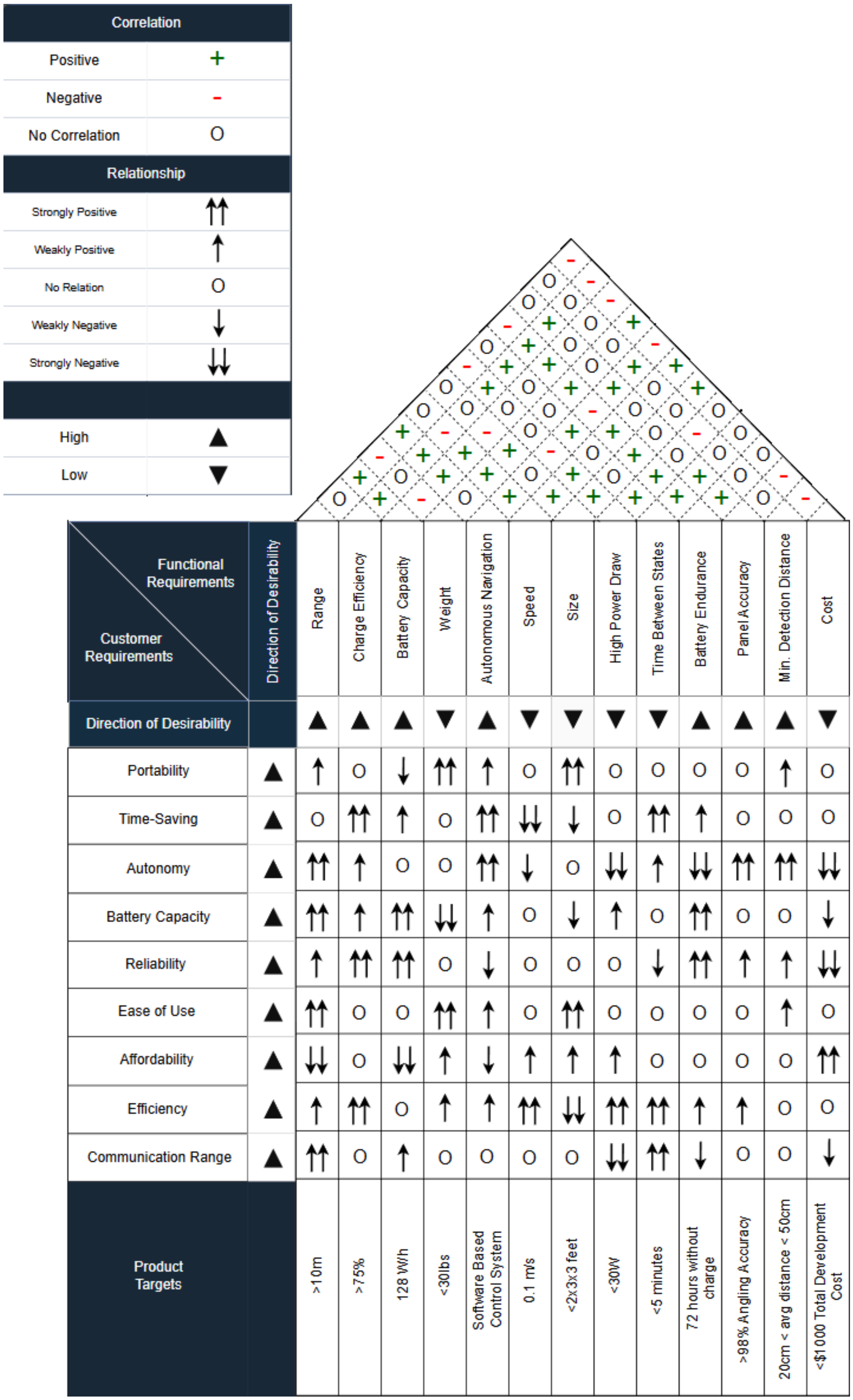


Figure 8. *House of Qualities*

Chapter 3 - Hardware & Software Research

3.1 Hardware Technologies

3.1.1 Brief

Up until this point we have shared our goals and requirements for this project. Mentions of hardware usage have been stated but the reasoning for their choice have not been stated in detail. In this section we look over the valuable research used in making those decisions. It is not only valuable to look into the technologies we want to use but to also be open-minded as to why we did not choose others given sufficient investigation. This presents itself as a strong opportunity to deepen our understanding of the hardware technologies available to us for best fit within our project *SOLARIS*.

3.1.2 Control Unit

3.1.2.1 What is a MCU?

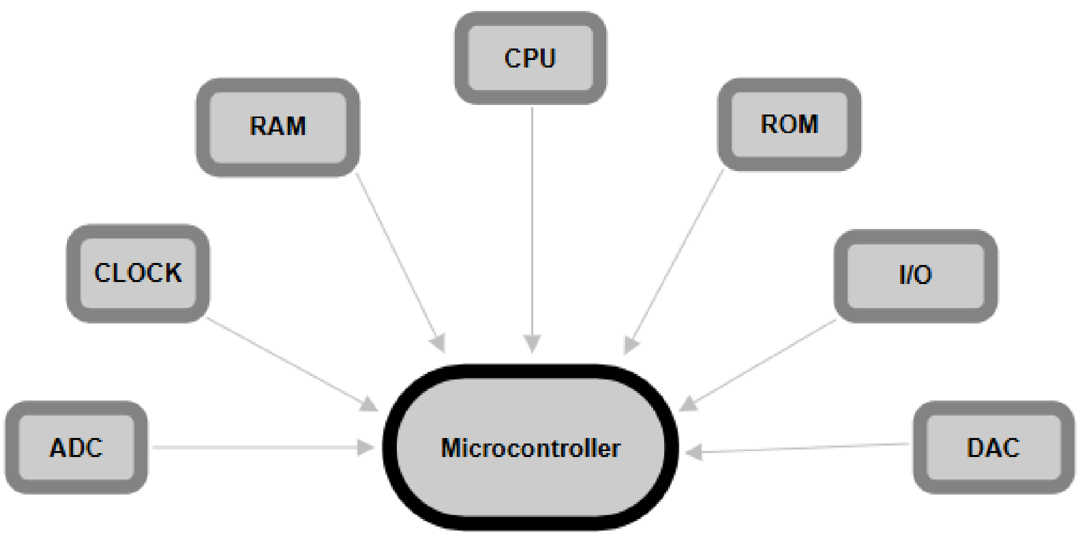


Figure 9. *Microcontroller Hardware Layout*

Our control unit consideration begins with understanding microcontrollers (MCU). Designed as a small computer on a single chip, MCUs are made for embedded systems without requiring an operating system. This is important because this combines all effective traditional hardware into one compact integrated circuit (IC). As compared to a microprocessor (MPU), MCUs integrate processing, memory, and input/output (I/O) peripherals like timers, counters and analog-to-digital converters (ADCs) into one standalone unit [35]. The general makeup of a microcontroller goes as follows: it receives one or more inputs, processes the data, and generates system outputs. These inputs and outputs are generally voltages which the microcontroller can recognize and output. While the I/O interacts with the external environment, code generally written in C/C++ is stored in local memory alongside stored data from a running program [36]. This allows for a bare metal connection of user control for device programming. In a nutshell, MCUs are high speed devices which can operate at low power or on battery powered systems for self-contained systems like telephones, appliances, vehicles and other computer systems.

Table 5. *MCU Comparisons*

Specification	ESP32-S3	ESP32-P4	MSP430FR6989	STM32U0
Architecture	Xtensa LX7 Dual-core	RISC-V Dual HP + LP	16-bit RISC	Cortex-M0+
Max Clock Speed	240 MHz	400 MHz	16 MHz	56 MHz
Flash Memory	up to 16 MB	up to 32 MB	128 KB	64KB ≤ 256KB
Bluetooth/Wi-Fi	BLE/Wi-Fi 4 (802.11 b/g/n)	None	None	None
ADC	2 x 12-bit, 20 ch	2x 12-bit, 8 ch	12-bit, 16 ch	12-bit, 10 ch
UART / SPI / I2C	3/4/2	3/8/3	2/2/1	4/3/2
Low Power Mode	Modern Sleep, Light/Deep Sleep	Multiple LP modes	LPM0-LPM4.5	Stop / Standby / Shutdown
Active Current	~340 mA peak	~123 mA peak	~275 µA	~25uA/MHz
Low Power Current	~7 µA	~25 µA	~0.5 µA	Varies < ~0.4 µA
Operating Temp	-40 to 85ºC	-40 to 85ºC	-40 to 85ºC	-40 to 85ºC
Operating Voltage	3.0-3.6V	3.0-3.6V	1.8-3.6V	1.71-3.6V
Development Ecosystem	ESP-IDF, Arduino	ESP-IDF	TI Code Composer	STM32CubeIDE
GPIO Count	~45	~54	~83	~29

Unit Cost	~$6.10	~$10.50	~$11.20	~$3.50

Source: *[37]-[42]*

3.1.2.1.1 ESP32-S3

The ESP32-S3 can be considered the low-power family of processors from Espressif Systems. This processor utilizes a dual-core Xtensa LX7 processor which runs at 240 MHz. The integration of a 2.4 GHz Wi-Fi and Bluetooth 5 LE radio makes this chip highly capable for Internet of Things (IoT) applications. Designed for workloads in artificial intelligence (AI) and machine learning (ML), the processor uses a five-stage pipeline which supports varieties of code density. The ultra low power (ULP) coprocessors are complemented by a ULP-FSM used with deep sleep operations making this chip incredibly power efficient and responsive. The ESP32-S3 comes with an internal USB controller meaning it can act as a native USB device without using any external silicon. Opening a range of possibilities for flashing and debugging the device. The microcontroller is rich in I/O with almost 45 GPIO ready for configuration in a multitude of designs.

Major considerations that make the ESP32-S3 stand out compared to the other options in research are the integrated RF handling and low power settings. There are four operating modes defined by a deep-sleep drawing around 7 µA which makes it competitive for a battery powered device. Espressifs support for open source design goes as far as their framework ESP-IDF and integration for use with Arduino. The ESP32-S3 receives a large portion of community support with online access to projects and libraries to make programming a simpler process. Configurability is not an issue since it has 3 hardwired UART lines and 4 hardwired SPI lines with 2 multiple configurable I2C lines to be used. It is also important to have on board storage and this microcontroller comes in around the middle of the pack with different variations of ROM and SRAM which reduce complexity and allow for storage of data components for export wirelessly or wired.

3.1.2.1.2 ESP32-P4

Another one of Espressif Systems offerings, this option comes in similarly to the S3 model. Some key differences here are the lack of Bluetooth and Wi-Fi support but a plethora of GPIO and horsepower. The absence of integrated wireless means that the P4 is wired only, relying on Ethernet MAC for network connectivity or pairing with an external wireless module. Sitting above traditional MCUs the P4 uses integrated image processing pipelines to reduce component count and bill of materials (BOM) cost for camera and display applications. Making its strongest cases in multimedia applications and edge vision, the combinations of image processing hardware and generous integrated memory simplify this process to one chip. One big issue with design is the lack of community support due to this microprocessor's unique use case.

Where the ESP32-P4 makes sense for SOLARIS is its strong dual-core RISC-V architecture which constitutes real time motor and sensor control. The processing power also would support high level autonomous logic, communication, and decision making without blocking each other. The P4 also comes with two full Motor Control PWM (MCPWM) blocks. Each of these blocks has three timers, fault detection, and capture submodules. For a vehicle with drive motors and panel positioning this would be purpose built rather than making a more customized motor control system. In the case of computer-vision based shadow detection or for more complicated processing the P4 is a perfect option. Similar hardware like the ADC with 14 analog input channels which can be routed across two 12-bit SAR gives headroom for current/voltage sensing, light detection, and battery monitoring at the same time.

3.1.2.1.3 MSP430FR6989

The MSP430FR6989 is purpose built for minimizing energy consumption. When running in active mode it only draws around 275 µ. Beyond this it offers seven distinct low-power modes (LPM0 - LPM4.5) which allows for an application to sleep between sensing intervals and then wake instantly on interrupts. In the application of a solar vehicle like SOLARIS, these key power saving features are critical. Like many other MCUs it supports a 12-bit ADC for applications like an array of phototransistors or reading battery state-of-charge. Onboard timer peripherals provide hardware level pulse width modulation (PWM) generation to drive motor controls. This avoids CPU overhead with hardware specific modules for performance scenarios. Bluetooth communication is not native but highly supported with use of a Bluetooth-to-serial module. Via a UART interface, the bluetooth module appears as a simple serial port as the app sends commands in ASCII or as binary packets while the MSP430 parses and executes them.

Something to appreciate about Texas Instrument as the developers of this MCU is the toolset for use with the MSP430FR6989. TI Code Composer Studio is industry developed software with support for a multitude of libraries making it easy to set up and flash a microcontroller with ease. Features like per line of code energy usage statistics make it a massive win for designing low power systems. Another major win for this MCU is its non-volatile FRAM storage. This local storage allows for unlimited write cycles for logs of solar stats, distance traveled, and battery charge level even after power loss. The MSP430FR6989 could be a well matched choice for this project due to its low power drawn with flexible sleep modes. The main limitation is its lack of native Bluetooth or Wi-Fi and 16MHz processor speed capacity.

3.1.2.1.4 STM32U0

The STM32U0 is a family of entry level low power microcontrollers from the company STMicroelectronics. Built on the ARM Cortex-M0+, this family boast static power consumptions of 160 nA in standby with as low as 16nA in full shutdown. Between the different models they go from 20 to 81 GPIO pins and offer between 64 KB and 256 KB of flash memory. In active mode the chip draws 25 µA/MHz at 3.0 V. With a tiered sleep architecture, the MCU can be put into sleep mode for low power draw and then woken quickly for a superior wake-up performance. From the analog perspective this MCU uses one 12-bit ADC with up to 16 channels, a digital to analog convertor (DAC), an analog comparator, and an operational amplifier all on one chip. For example the comparator can be use to watch an array of phototransistors for shadow detection while the CPU sleeps.

Similar to the MSP430FR6989, the STM32U0 family includes a 16-bit PWM timer which is dedicated specifically for motor control. For wireless communication like Bluetooth the STM32U0 uses UART connections to remain active in receiving Bluetooth commands. This means a separate chip for serial from Bluetooth is not necessary but a receiver is still required for use. ST has also developed an ecosystem for programming their devices which stands tall around many. Key functionality like a graphical pin configuration and code generation tools make programming the device visually simple. Overall this MCU contains a large timer functionality, dedicated motor control hardware, a mature software ecosystem for development, and hierarchical low power design.

3.1.2.2 What is a MPU?

Table 6. *MPU Comparison*

Feature	*Nvidia Jetson Nano*	*BeagleBone Black*
Core Architecture	128-core Nvidia Maxwell GPU	TI Sitara AM335x
Primary Strength	AI and neural networks	Deterministic execution and PWM control
Specialized Hardware	Integrated Maxwell GPU for vision tasks	Programmable Real-Time Units (PRUs) & 12-bit ADC
Best Use Case	Real-time object detection adn self-driving stacks	Motor control and ultrasonic sensors
Power	5W and 10W modes	~2.5W at idle

Sources: [47], [49], [50]

In the context of the technology, a microprocessor is a foundational design for use in modern computing. This hardware serves as an custom designed, all in one Central Processing Unit (CPU). Unlike a Microcontroller (MCU) a CPU integrates typical core components like an Arithmetic Logic Unit (ALU) and Control Unit (CU), to work in tandem with high-speed cache systems with minimal latency [47]. This architectural shift can be dated back to 1971 with the introduction of the Intel 4004 which significantly improved upon system reliability. These days, a microprocessor acts as the brain for a computer for examples like autonomous vehicles or even remote controls. This is made possible by a level of reliability standards which exponentially increased processing power over other similar designs. The high level processor uses a pipeline of fetching data from memory, decoding it into specific commands, executing the required operations, and storing the results to memory. While on the software level an operating system virtualizes the CPU for more modern task and avoids low level programming.

The use of a microprocessor in our project *SOLARIS* would benefit from the real-time navigation and energy efficiency. Core components of a microprocessor like the ALU and CU can work together in tandem to process sensor data with minimal latency. This makes the architecture of a MPU critical for a responsive system like that of the potential solar robot. Modern multi-core microprocessors allow for parallel processing similar to that of an FPGA. This can be understood as one core handling pathfinding algorithms while the other monitors the solar to battery charging levels. Another added benefit of a microprocessor is the ease of integration with specialized types of MCUs including Digital Signal Processors (DSPs). For vision tasks these can be optimized into a self driving stack where split second safety decisions can be made while sticking to strict power constraints.

3.1.2.2.1 Jetson Nano

The Nvidia Jetson Nano is a specialized microprocessor (MPU) which is designed for use with artificial intelligence. For a project involving a solar-powered autonomous system this MPU acts as a sophisticated brain to handle perception and high-level logic, which would be complementary to that of an FPGA. The biggest differentiator with the Jetson Nano compared to other offerings is its 128-core Nvidia Maxwell GPU, which provides 472 GFLOPS of accelerated computing power [49]. Unlike a standard MPU which would rely solely on the CPU for tasks the Nano can run neural networks on its GPU. For autonomous navigation tasks like real-time object detection this becomes a strong competitor. The biggest concern with a system like this is its power management system. Though it is capable of 5W and 10W modes respectively, this is too power hungry for use in a solar vehicle where every bit of power matters. It is capable of digesting video to detect obstacles with ease but this level of performance conflicts with the low power design needed to make a lightweight system work.

3.1.2.2.2 BeagleBone Black

The BeagleBone Black is another high-performance microprocessor (MPU) based on the Texas Instruments Sitara AM335x [50]. It serves as a capable Linux brain for a project. The unique architecture makes it significantly more effective for a semi-autonomous vehicle as apposed to a standard MPU. Dedicated hardware on chip called Programmable Real-Time Units (PRUs) remove the unpredictable “jitter” when timing signals in Linux. This provides deterministic execution which is critical when defining PWM signals for motor control or ultrasonic senors where microsecond accuracy is required. The PRUs also allow for an FPGA like flexibility being that they provide a bridge between high-level code and physical hardware. Beyond the real-time capabilities of the MPU, the BeagleBone Black comes equipped with a integrated 8-channel, 12-bit Analog-to-Digital Converter (ADC). Similar to the Jetson Nano, this MPU unfortunately finds itself being too powerful for use in a solar powered system. Generally drawing around 2.5 W in idle it does not have the benefits of a sleep mode like microcontrollers (MCUs).

3.1.2.3 What is a FPGA?

Table 7. *FPGA Technology Comparison*

FPGA Technology	***Configuration Style***	***Typical Speed***	***Power Use***	***Cost***	***Common Use Cases***
SRAM-based FPGA	SRAM configuration cells	Very High	Higher	Moderate (best cost/performance)	Prototyping, DSP, AI/ML acceleration, and mainstream development
Flash-based FPGA	On-chip flash memory	High	Low	Moderate	Low-power embedded systems and industrial systems
Antifuse-based FPGA	Permanent programmable connections	Very High	Very Low	High (specialized markets)	Aerospace, military, radiation-hardened sytems for ultra reliable hardware
Heterogeneous FPGA	Typically SRAM or Flash	Very High	Moderate	Moderate (depends on device class)	Signal processing, machine learning, and high bandwidth computing systems

Source: *[8]*

A Field Programmable Gate Array (FPGA) and how it compares to Microcontrollers (MCU) or Microprocessors (MPU) is a careful distinction that needs to be made early on. Unlike MCUs and MPUs an FPGA is capable of parallel operation where processing resources are not split between single cores [6]. Typically with microcontrollers and microprocessors, reconfigurability is reduced to software programming using a High-Level Language (HLL) like that of C/C++ or Java. This distinction makes FPGAs a close competitor to Application-Specific Integrated Circuits (ASICs) while still maintaining feasibility for functionality in many projects. In the design process, users write code in Hardware Description Languages (HDLs), commonly VHDL or Verilog, then synthesize and implement it using specialized tools [5]. This reconfiguration of hardware allows for adaptability and in turn makes them indispensable for addressing computational challenges in diverse environments [5]. As the name suggest, FPGAs are designed for their ability to be reprogrammed in the field for many use cases.

Though FPGAs first emerged during the appearance of Programmable Array Logic (PAL) and Complex Programmable Logic Devices (CPLDs) these were merely precursors to true FPGA technology [5]. Many companies sought to make their own takes on the architecture and sparked the beginning of the modern era. Two main companies emerged as key players, Xilinx (now part of AMD) and Altera (now part of Intel) which introduced successive generations of FPGAs [5]. The FPGA consist of two main parts which are similar in all architectures. The first component known as a Look-Up Table (LUT) and the second component known as a Flip-Flop (FF). Using these devices as key building blocks for integrated memory these devices could store applications more efficiently and improve performance with reduced latency. These key advancements by the end of the 90s allowed FPGAs to take common place in automotive, consumer, and industrial applications [6].

Different types or flavors of FPGAs also exist to fill the various types of power consumption, configurability, and use specific needs of many applications. Some key types include something known as Antifuse technologies which are configurable but not reconfigurable [7]. This means that the device can be used as a one-time programmable device for security reasons or for a higher performance design due to hard wired connections. The most common and flexible technologies are always SRAM-based FPGAs [7]. These use Static Random-Access Memory (SRAM) to store configuration instructions but they need external memory to store this code. In the present and moving forward into the future we see a more common player known as the System-on-chip (SOC) variant. These devices integrate the programmable logic with processor cores, therefore combining the functionality onto one silicon chip for multiple systems in one unit [7].

Table 8. *Xilinx Artix-7 Vs. Altera Cyclone V*

Specification	***Artix-7***	***Cyclone V***
Process Technology	28nm HPL (High Performance, Low Power)	28nm LP (Low Power)
Logic Architecture	6-input LUT	8-input ALM
Max Logic Density	~215k Logic Cells	~301k Logic Elements
DSP Resources	>740 DSP Slices	>342 DSP Blocks
Transceiver Speed	>6.6 Gbps	>6.144 Gbps
Memory Bandwidth	1,066 Mbps DDR3	800 Mbps DDR3
Analog Integration	Dual 12-bit, XADC	None
Processor Support	MicroBlaze (Soft Core)	Nios II (Soft Core)
Design Software	Vivado Design Suite	Quartus Prime

Sources: *[45-46]*

3.1.2.3.2 Artix-7

One option in the FPGA space comes from Xilinx (AMD) and boast many technical advantages and features compared to others. The Artix-7 represents a generational leap compared to its predecessors with a well priced performance-per-watt. This is due to its High-Performance, Low-Power (HPLP) process that the silicon is designed on. Loaded with high speed transceivers up to 6.6Gb/s with an enabled 211 Gb/s in full duplex mode [45]. It is considered a more low end FPGA but also presents a good entry point for high-performance digital signal processing (DSP). Useful I/O like PCI Express lanes alongside two Analog-to-Digital Convertors (XADC) make it a great choice for computer vision. The level of high performance processing offered by a capable FPGA allows for a real time computing system for use in SOLARIS. One major positive of the Xilinx family of offerings is their own software for programming and debugging FPGAs. With support of two languages being VHDL and Verilog/SystemVerilog. It is clear to see from an architectural standpoint how the Artix-7 silicon offers significant hardware advantages for semi-autonomous systems.

3.1.2.3.3 Cyclone-V

Alteras (Intel) family being considered for *SOLARIS* has key advantages for use with autonomous systems, AI, and computer vision. The Cyclone-V uses a sophisticated Hard Processor Subsystem (HPS), featuring a dual-core ARM Cortex-A9. This system allows you to run a high level operating system while also benefiting from the direct control made possible by FPGA fabric. In an application like this variable-precision DSP blocks are designed for Edge AI. These integrations would benefit a performance real-time image recognition algorithm or pathfinding. The Cyclone-V is a low cost, high-integration solution for a power sensitive application like a semi-autonomous vehicle. One major concern is the lack of ADC on chip making the configuration and I/O take a hit. To make up for this lack of peripherals the silicon includes integrated transceivers to support data rates up to 6.144 Gbps.

3.1.2.3.3 When To Use FPGA

The research above points our project in the direction of a more complete solution like the MCU. In this stage of the project our system would not benefit enough to constitute a stronger solution like an FPGA or even an MPU. Due to the simplistic nature, *SOLARIS* requires a lightweight, low power, and cheap answer. The benefits of parallel processing only come into account if *SOLARIS* were to implement some level of computer vision or more advanced driving system. That being said, it is still incredibly important to know where an FPGA does fit into a project of this scale. Though FPGAs are commonly seen in autonomous driving it finds itself to be overkill for a simple control system. FPGAs are a great solution for digital signal processing (DSP), device controllers, and computer hardware simulation. If the project where to ever become more complicated in scope an FPGA would be more than sufficient for the computer vision or complex control system needs of an improved *SOLARIS*. With timing also being a large constraint we do not possess the time or the resources to work through the development process for an FPGA implementation.

3.1.2.4 MCU Vs. MPU

Table 9. *MCU Vs. MPU*

	Microprocessor(MPU)	***Microcontroller(MCU)***

Description	CPU integrated into a single chip, designed for general purpose computing, requiring external memory and peripherals.	Small computer on a single chip with CPU, RAM, and program memory.
Memory	External memory RAM and storage (scalable).	On-chip RAM and program memory (limited).
Speed	High (for complex tasks)	Moderate (for dedicated tasks)
Power	High (may need cooling)	Low (energy efficient)
Cost	Higher (external components)	Low (inexpensive)
Programming	Runs bare-metal firmware.	Supports full operating system (OS)
Applications	Personal Computers (PCs), laptops, and high-performance computing (HPC).	Embedded systems, IoT devices, and robotics.
Strengths	High performance, flexible, and scalable.	Integrated features, low power, and low cost.
Weaknesses	High power consumption and more expensive.	Limited processing power and memory.

Source: *[11]*

When comparing these technologies it is important to note that MCUs are more similar to MPUs than FPGAs are to either. While similar they all differ in architecture as well as performance capability. Between MCUs and MPUs, the key difference lies in MCUs containing all additional peripherals with integrated memory, input/output (I/O) components and other varied peripherals [11]. Typically more cost effective, small in size, and low in power, microcontrollers are optimized for an all in one functionality. On the other side MPUs are designed to support higher performance in demanding applications like personal computing and graphics processing [11]. Therefore a MCU can succeed in an area of real-time and high-speed computing tasks while a MPU sees itself as reliable and scalable. This leaves microcontrollers to be typically used for handling fast signal processing while microprocessors are favored for more demanding high-performance computing (HPC) related tasks [11].

3.1.2.5 MCU Vs. FPGA

Table 10. *MCU Vs. FPGA*

	Microcontroller(MCU)	*Field-Programmable Gate Array (FPGA)*
Description	Small computer on a single chip with CPU, RAM, and program memory.	A configurable integrated circuit composed of programmable logic blocks designed for reprogramable hardware
Key Components	Single chip computer with CPU, memory, and built in peripherals	Configurable Logic Blocks (CLBs), programmable interconnects, and external memory interface
Speed	Sequential Processing (executes instructions one at a time)	Parallel processing (can perform multiple operations simultaneously)
Power	Low (energy efficient)	Higher (configuration dependant)
Cost	Low (inexpensive)	Higher (hardware and development)
Programming	Software-based (C/C++, Python, Assembly)	Hardware Description Languages (Verilog VHDL)
Applications	Embedded systems, IoT devices, and robotics.	Digital signal processing (DSP), rapid prototyping, and ASIC development.
Strengths	Integrated features, low power, and low cost.	Highly flexible, parallel-processing, hardware optimization.
Weaknesses	Limited processing power and memory.	Requires special knowledge with a longer development cycle.

Source: *[10]*

Though all can be thought of as small computers it is still important to compare FPGAs as well. Unlike a microcontroller, FPGAs offer a level of reprogrammability at a hardware level. MCUs operate in a sequential pattern by reading lines of code whereas FPGAs are unique with capabilities of handling parallel inputs simultaneously. An FPGA can be programmed to perform the function of a microcontroller; however, a microcontroller cannot be reprogrammed to perform as an FPGA [10]. While FPGAs appear to be a clear winner it would certainly be overlooking many constraints like cost, ease of programming, energy efficiency, and best fit for application. MCUs are cheaper and require basic conceptual knowledge of software languages. MCUs also consume much less power than FPGAs allowing them to be suitable for battery powered applications and embedded systems [10]. This means that small, affordable, and non-volatile microcontrollers are ubiquitous in modern electronics [10]. For our application it may be safe to assume that an FPGA would be overkill as the major processing will certainly see itself reduced to a simple control system with calculations.

3.1.2.6 Control Unit Part Selection

ESP32 S3: After the consideration between microcontroller, microprocessor, and field-programmable gate arrays it is a smart decision to implement a microcontroller as the brains of our system. In particular the ESP32 S3 is one of the best choices for three main reasons. Built in Bluetooth and Wi-Fi capabilities make this a strong choice for pairing with a potential companion app and control of the system wirelessly. This eliminates the need for an external module which could potentially complicate design constraints. When it comes to self-autonomy the balance between computational power and amount of GPIO also make this microcontroller shine as the best option. The ESP32 S3s dual core processor is ample for semi-autonomous logic, sensor fusion, and PWM motor control. Lastly the ability to program this microcontroller easily is a huge advantage over other options seeing as the Espressif line is strictly open source. There are many libraries online with large communities of people making the support for programming this device overwhelmingly helpful. The ESP32 S3 also utilizes low power modes similar to other microcontrollers. For our project *SOLARIS* this option excels in all areas relevant to our project and does not lack in basic feature sets common with other MCUs.

3.1.3 Drive Motors

3.1.3.1 Brushed DC Motor

Brushed DC motors are commonly used in robotic systems due to their simple control requirements, low cost, and high starting torque, making them practical for drive mechanisms and auxiliary actuators. From a system design perspective, these motors are advantageous because their speed can be controlled directly using voltage or pulse-width modulation without complex commutation electronics, reducing controller overhead and hardware complexity. The torque produced is proportional to armature current, allowing straightforward current-based torque regulation for motion control. However, mechanical commutation via brushes introduces friction losses, electrical noise, and gradual component wear, which reduce efficiency and limit operational lifetime. These factors must be considered when designing battery-powered robots, as energy losses translate directly into reduced runtime. Consequently, brushed motors are most suitable for applications where simplicity, cost, and ease of integration outweigh long-term efficiency or durability considerations.

3.1.3.2 Brushless DC Motor

Brushless DC motors provide significant advantages in robotic systems where efficiency, reliability, and dynamic performance are critical design priorities. Unlike brushed motors, commutation is performed electronically using a motor driver that energizes stator windings in sequence based on rotor position feedback, creating a rotating magnetic field that drives the rotor. This eliminates brush friction and electrical arcing, resulting in higher efficiency, reduced thermal losses, and longer service life. In robotic platforms with limited power resources, these efficiency improvements can substantially extend operating time which is a primary goal with *SOLARIS*. Brushless motors also offer higher torque density and smoother operation, which improves motion precision and reduces mechanical vibration transmitted to the chassis. The primary tradeoff is increased system complexity, as they require dedicated electronic speed controllers and more sophisticated firmware for commutation and feedback processing. Despite this added complexity, brushless motors are a good choice due to their increased overall efficiency and longer lifespan.

3.1.3.3 Stepper Motor

Stepper motors are small motors that produce movement to a desired angular position that is precisely controlled by the given control system. The main difference is that these changes are given in small step increments. These motors are typically used in small applications such as robot arms due to their small size, weight and highly precise nature. They were not selected due to their small nature and inability to handle *SOLARIS*' load demands.

3.1.3.4 AC Motor

AC motors are widely used in industrial automation and large scale robotic systems because of their robustness, high efficiency, and ability to operate continuously under heavy loads. They generate torque through a rotating magnetic field produced by alternating current, which induces rotor motion either through electromagnetic induction or synchronous magnetic coupling. From a design standpoint, AC motors are quite advantageous in applications that require sustained high power output, such as industrial robotic arms or automated manufacturing equipment. However, they typically require variable frequency drives for speed control, which significantly increases complexity and cost for just the control system. In small or mobile robotic platforms minimizing power consumption requirements, an AC motor is less desirable due to its larger size, weight and overall power requirements when compared to DC motors.

3.1.3.5 AC Motors Vs. DC Motors

Table 11. *AC Motors vs. DC motors*

	AC Motors	***DC Motors***
Description	Robust, high efficiency motors that operate well under heavy loads, typically used in large scale robotics and industrial applications.	Robust and easier to control using PWM signals from MCU. Also provides a high starting torque which is ideal for robot drive applications.
Key Components	Motor, Variable Frequency Drive, inverter, controls, AC power source.	Motor, DC power source, motor driver, MCU interface.
Speed	Depends on supply frequency and poles. Requires a variable frequency drive for speed control.	Proportional to applied voltage, allowing for easier control via a pulse width modulator signal from the MCU.
Power	Continuous high power and efficient operation under heavier loads.	Lower power systems that provide a high torque at low speeds.
Cost	Higher cost due to variable frequency drive.	Lower cost due to simpler control technology.
Applications	Large industrial robots, electric vehicles, manufacturing equipment.	Small actuators, robotic arms, small robotic drives.
Strengths	High efficiency at high loads, long lifespan.	Simple control, cheaper, compact size, high torque at low speeds.
Weaknesses	More complex controls, large size, higher weight.	Lower efficiency, brush wear, noisier, lower lifespan.

Sources: [127-129]

3.1.3.6 Drive Motor Type Selection

DC Motor: While AC motors have a higher lifespan and efficient continuous operation, the advantages that these properties afford come with significant drawbacks for our application. AC motors have inherently larger sizes and would require an inverter to convert the DC from the battery to AC and then back to DC in the variable frequency drive to produce the required signal to rotate the motor. This chain of equipment raises cost and, in particular the variable frequency drive, will require lots of space inside of the robot and add to the weight. DC motors are directly compatible with our already low voltage battery without requiring additional power conversion. The high torque at low speeds is desirable for efficiency and for outdoor terrain. Additionally, their compact size and straightforward integration with embedded microcontrollers simplify electrical and mechanical implementation. For a battery powered robotic system where efficiency, controllability, and minimal hardware overhead are the focus, DC motors offer a more practical and scalable solution than AC alternatives.

3.1.3.7 Brushed Vs. Brushless DC Motors

Brushed and brushless DC motors differ in how they rotate and control rotation. Brushed motors work by switching current to the windings using a physical brush on a commutator meaning they will spin naturally when voltage is applied which is significantly easier to implement and control using a device such as an H-bridge controller. The main drawback of brushed DC motors stems from the inherent mechanical contact of the brush riding on the commutator which introduces electrical noise, brush wear and efficiency. Brushless motors use solid-state switching through the windings which requires the implementation of an external controller that creates the desired signal for rotation and control of the motor. While more complex in its implementation and control, this allows brushless motors to have smoother torque, more precise speed regulation and minimal wear. While brushed motors are simpler and able to respond very predictably for a given input, they demonstrate lower efficiency over time and performance under heavier loads and prolonged use. Brushless motors can better adapt to varying load conditions because the external controller actively regulates the current and timing. Overall, brushed motors offer simplicity and direct operation while brushless motors offer controller performance, adaptability and consistency and higher efficiency.

3.1.3.7 DC Motor Type Selection

Brushed Motor: Brushed DC motors were selected for driving the robot because the efficiency and precision gains offered by brushless motors were not in the project scope for *SOLARIS*. While brushless motors allow for greater electrical efficiency, their implementation and operation requires dedicated controllers, more complex control, and more integration concerns. All of these factors raise the risk of increasing development time and system operation risk. Brushed DC motors are typically driven with simpler pulse width modulation (PWM) control methods which allows for rapid prototyping, reliable operation with minimal setup required. Since *SOLARIS* does not operate continuously and only in short bursts, the efficiency difference is not likely to significantly impact the battery life or efficiency of the robot. The simplified wiring and reduced control requirements with ease of predictability is desired to allow for ease of troubleshooting and allow focus on higher level system design and implementation. The benefits of brushed DC motors will allow us to achieve our desired robustness, and system reliability while meeting all of our performance requirements.

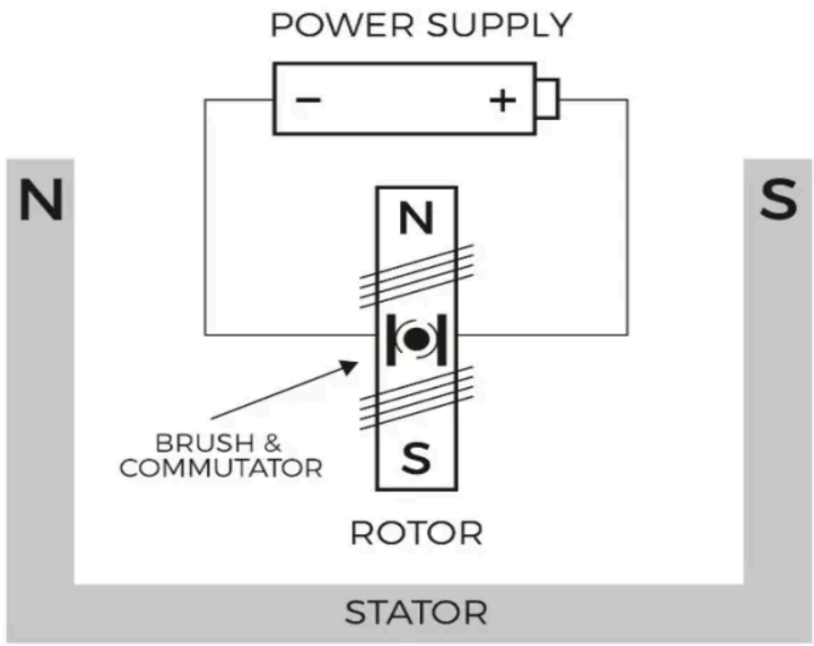


Figure 10. *Two-pole DC Motor Model [83]*

3.1.3.8 Drive DC Motor Selection

When selecting specific motors for *SOLARIS* the first major factor is ensuring that the motors can handle the load of the entire robot. To meet this end, there are several factors that must be taken into consideration.The first factor was finding a gearbox output speed that could ensure a substantial torque margin. The wheel size that has been selected were 4-inch diameter wheels and a desired speed of 0.3 m/s the required wheel speed is approximately 53 RPM after gearbox reduction. A motor RPM of 60-80 RPM was determined to be the target range that would meet the desired performance and margins.

The second major design consideration is the torque under load that the motors must be able to handle during routine, expected operation. *SOLARIS* must be able to roll on various grass types and heights, startup inertia, and friction losses from a differential steering scheme; all of these obstacles will affect the desired torque requirements. The next consideration is stall torque. Stall torque is the maximum capability torque that a motor can sustain. Running torque should be at a margin approximately 20-30% below this maximum torque to ensure efficient operation and avoid high current spikes. The motors were considered based on their rated running torque with an error margin to what is the minimum required for *SOLARIS*.

The third major design consideration was the gear ratio and gearbox construction for the robot for the required performance. Higher gearbox ratios will increase the output torque while reducing speed, making them better for heavier, lower speed robots. Too high of a reduction scheme can yield undesirable mechanical losses and backlash which would affect responsiveness and efficiency. Motors with planetary gearboxes were considered most heavily because they distribute the load evenly on the gears which offers improved torque density and robustness when compared to similar gearbox configurations such as spur gears.

The final major consideration were the electrical attributes of the motor. A 12 V motor was selected because it is the voltage level that the battery for *SOLARIS output* and thus will not require any additional power conditioning to allow the motors to operate. THe current rating must also be able to handle current surges from motor stalls under unexpected conditions to work with the motor drivers, wiring and battery. Since 4 motors are desired with a differential steering configuration, startup inrush current and turning currents can place high stress on the power delivery systems so the motor driver must be able to accommodate these worst-case events. Additionally, most of the considered motors come with encoders already built in which is very desirable for ease of implementation with simplified control design.

Table 12. *DC Motor Comparison*

Brushed DC Motor Model	***Voltage***	***Rated Torque***	***Rated Speed***	***Gear Ratio***	***Gearbox***	***Encoder***	***Cost***
StepperOnline F5D60-12GN-30S-5GN50K	**12V**	**6.27 Nm**	**60 RPM**	**50:1**	**Spur**	**No**	**$52**
Duma Dynamics 13PPR Encoder 60RPM DC Gearmotor	**12V**	**5 Nm**	**60 RPM**	**104:1**	**Planetary**	**Yes**	**$59**

goBILDA 5204 Series Yellow Jacket Planetary Gear Motor	**12V**	**18.14 Nm**	**60 RPM**	**139:1**	**Planetary**	**Yes**	**$56.99**

Sources: [93], [123-124]

3.1.3.9 StepperOnline F5D60-12GN-30S-5GN50K brushed gearmotor

This motor is a relatively inexpensive, simple and conservative option that is an industrial adjacent drive train motor for *SOLARIS*. The motors 12V rating with a torque of 6.27 NM at 60 RPM post-reduction is a very adequate fit for the robot and will more than carry the robot's estimated weight of 15-20 lb across grass at the desired speed of ~0.3 m/s. This motor is the strong and robust option making it a good choice where a lot of resistance is expected for the environment or terrain. With a priority being simplicity, the spur gearbox makes the drivetrain integration easier due to the conventional gearing inside of the box at the expense of efficiency. The main drawback of this motor is that it does not come with an encoder which is a massive consideration when selecting motors. Including an encoder is strongly desired to minimize extra wiring and reduce complexity.

3.1.3.10 E-S Motor 36D Planetary Gearmotor

This motor offers a strong balance between torque capability and control functionality. With an output torque around 5 Nm and a gearbox ratio of 104:1 and a speed near the desired operating range, it provides more than enough mechanical capability for grass traversal while remaining within budget constraints. Its integrated encoder simplifies implementation of feedback control, allowing for more precise speed regulation and improved navigation accuracy without additional hardware. Functionally, this makes it attractive for autonomous systems that rely on wheel feedback for distance estimation or correction. Compared to the StepperOnline motor, it provides slightly less torque but still maintains ample safety margin for real-world operation.

3.1.3.11 goBILDA 5204 Series Yellow Jacket Planetary Gear Motor

This motor stands on its own as a highly capable choice for a small outdoor robotic drivetrain. Its planetary gearbox architecture provides a strong torque density, meaning it can deliver a large output torque relative to its size and weight along with providing more efficiency, and the 60 RPM gearbox output aligns well with a robot's desired cruising speed when paired with a wheel size of both 4 and 6 inches. This motor also includes an integrated quadrature encoder that runs on 3.3–5 V logic, making it easy and straightforward to interface with the ESP32's PWM pins for closed-loop speed control and odometry without needing external sensors which would drive up the integration complexity. The documented stall torque and stall current give ample headroom for traction on grass and small disturbances such as rocks or uneven terrain, and goBILDA's product pages include downloadable specifications that support formal engineering justification. Additionally, the Yellow Jacket series is widely used in competitive robotics (FTC, FIRST, VEX communities), so its mechanical behavior and integration patterns are well understood, making it a reliable and well-supported choice.

3.1.3.12 Drive DC Motor Selection

goBILDA 5204 Series Yellow Jacket Planetary Gear Motor: The goBILDA strikes a strong balance between performance, documentation, and system integration. The StepperOnline F5D60-12GN-30S-5GN50K motor has a respectable rated torque point but is a spur gearbox design without an integrated encoder, which adds complexity for encoder feedback and makes closed-loop control harder to implement directly. On the other hand, the Duma Dynamics 13PPR Encoder 60 RPM DC Gearmotor does offer encoder feedback and decent torque, but it tends to be more expensive and less commonly used in the robotics community, which can make finding mounting options and support more difficult. The Yellow Jacket's planetary gearbox gives it an inherent mechanical advantage in torque density and smoothness over simpler spur gear alternatives, and its integrated encoder reduces wiring and sensor overhead relative to motors without feedback. For *SOLARIS*' needs, budget, and ESP32 architecture, the Yellow Jacket provides the best blend of torque margin, encoder support, documentation quality, and community familiarity.

3.1.4 Solar Panel Pan/Tilt Motor Selection

Table 13. *Brushed DC Motor Comparison*

Brushed DC Motor Model	*Voltage*	*Rated Torque*	*Rated Speed*	*Gearbox*	*Encoder*	*Cost*
5203 Series Yellow Jacket Planetary Gear Motor (188:1 Ratio, 24mm Length 8mm REX® Shaft, 30 RPM, 3.3 - 5 V Encoder)	**12 V**	**24.5 Nm**	**30 RPM**	**Planetary**	**Yes**	**$54.99**
E-S Motor 12 V	**12 V**	**3.92 Nm**	**54 RPM**	**Worm**	**Yes**	**$27.50**

105 RPM Worm Gearmotor with Encoder				**(Self Locking)**		
FeeTech FT5330M Digital Servo (180°)	**7.4V**	**3.48 Nm**	**0.3 sec/60 degrees**	**Servo**	**Yes**	**$20.74**

Sources: [92] [125-126]

3.1.4.1 Solar Panel Pan/Tilt Motor Selection

When selecting a motor for the solar panel pan-and-tilt system, the main goals were to ensure the motor could provide enough torque to move and hold the panel, operate from the project's electrical system, and still remain practical for a senior design build. Since the panel is mounted near its center, the torque demand is lower than a worst-case edge-mounted estimate, but the motor still needs enough margin to account for imbalance, friction, and wind loading. The three options considered here were chosen because they each provide some combination of geared torque multiplication, position feedback, and manageable size or cost. Other important considerations included whether the motor could be controlled in a servo-like manner, whether it could be integrated easily into the mechanical design, and whether reliable documentation was available for design justification. Because the same motor is desired for both pan and tilt, torque margin, controllability, and documentation quality became more important than simply choosing the cheapest possible option.

3.1.4.2 goBILDA 5203 Series Yellow Jacket Planetary Motor

This is a strong candidate because it offers a high torque rating of 24.5 N·m at 12 V while still maintaining a moderate speed of 30 RPM. Its planetary gearbox gives it much better torque density than a small un-geared motor and helps reduce the effort required from the motor itself during holding and positioning. The built-in encoder also makes it well suited for closed-loop control, allowing it to be used in a servo-style system with a motor driver and controller. This motor greatly exceeds the estimated torque requirement, which gives a design margin for imbalance, wind disturbance, and startup torque. At the same time, it is not so large that it becomes unrealistic for the project's scale, making it a very balanced choice at an affordable price.

3.1.4.3 E-S Motor 12V worm gearmotor with encoder

This motor is appealing because its worm gearbox offers a self-locking characteristic, which is desirable for holding position with reduced backdriving. This is desired because of efficiency. With a self-locking mechanism, there will be no need to send a hold signal to the motors and thus wasting power. Its listed torque of 3.92 N·m is close to the original design target, so on paper it appears capable of handling the required load with a modest margin. The built-in **e**ncoder is also useful for feedback and position control, which makes it possible to implement controlled motion on both the pan and tilt axes. However, its higher 54 RPM speed is somewhat faster than desired, and more importantly, motors in this class are often limited by weaker documentation and fewer supporting ecosystem parts.

3.1.4.3 FeeTech FT5330M Digital Servo Motor

This is the simplest option from a control perspective because it is already a packaged servo, meaning that it includes internal feedback and control electronics. Its torque rating of 3.48 N·m is within the general range of the project's needs, and its rated speed of 0.3 sec/60 degrees indicates that it is fast enough for slow solar tracking adjustments. Because it is a servo, it is easy to command directly without requiring a separate feedback loop implementation, which can reduce software and integration effort. However, it runs at 7.4 V instead of 12 V which increases integration complications, which would require an additional regulator, and servos typically hold position by continuously applying torque rather than by mechanically resisting motion.

3.1.4.4 Solar Panel Pan/Tilt Motor Part Selection

goBILDA 5203 Series Yellow Jacket Planetary Gear Motor: Overall, this is the best choice because it offers the best combination of torque margin, control flexibility, documentation quality, and mechanical practicality. It is not a worm gearmotor, but truly well-documented 12 V worm gear motors with encoder support within a reasonable price range are very difficult to find.. The 188:1 gear ratio is valuable because it means the motor only needs to generate a small fraction of the output torque at the motor shaft, which reduces the current needed to hold the panel in place, so the remaining power draw is mostly electrical loss, which is minimized by the large gear ratio. A mechanical brake can be easily and cheaply added to assist the holding of the motor if desired. On top of that, goBILDA provides strong documentation, and a large mounting and shaft ecosystem making integration much easier and troubleshooting than other options.

3.1.5 Motor Drivers

3.1.5.1 Unidirectional Drivers

Unidirectional motor drivers control speed but do not allow reversal of rotation. THis method usually utilizes a ground side N-channel MOSFET switch that uses the PWM voltage signal to regulate the motor speed. High-side switch drivers operate in a similar manner but control the supply ride of the motor instead which adds safety to the design. Linear drivers control voltage by operating the MOSFET in the linear region which provides smoother control but at the cost of power losses as heat. While these approaches are electrically simple and cheap they do not offer the bidirectional functionality required by the robots drive system.

3.1.5.2 Half-Bridge Drivers

A half-bridge driver consists of two switching devices that control a single side of the motor and is a part of the H-bridge circuit topology. Half-bridge drivers do not offer bidirectional motion on their own but can be used in more specialized drive configurations due to their modularity. Half bridges provide good speed control with PWM voltage but are insufficient alone for the desired bidirectional movement.

3.1.5.3 Relay-Based Drivers:

Relay motors drivers are drivers that use physical switching relays to reverse polarity and control the motor direction. While relays are excellent for discrete control in many industrial applications due to their robustness and simplicity, Relays switch slowly compared to solid state options and take up much more space within the robot to implement the same task. The slower switching is not generally well suited for speed control and mechanical wear is also a factor with continuous rapid switching. They can be combined with FET transistors for speed control improvements but the added cost and complexity makes the simplicity of the option hard to consider.

3.1.5.4 H-Bridge Driver

H-Bridge Drivers are common for brushed DC motor control due to their efficient and fast switching speeds, bidirectional motion and good speed control. The driver works by having 4 N-channel MOSFET transistors arranged in an H-shape that has the motor commutator in the center of the bridge. The MOSFETS switch so that the polarity of the voltage applied to the motor can be reversed which allows for bidirectional motion of the motor. H-Bridge drivers also use pulse-width modulation to regulate the voltage supply to the motor which enables precise motor speed control. H-Bridge drivers also provide electrical breaking which allows faster stopping of the motor when its usage is no longer desired.

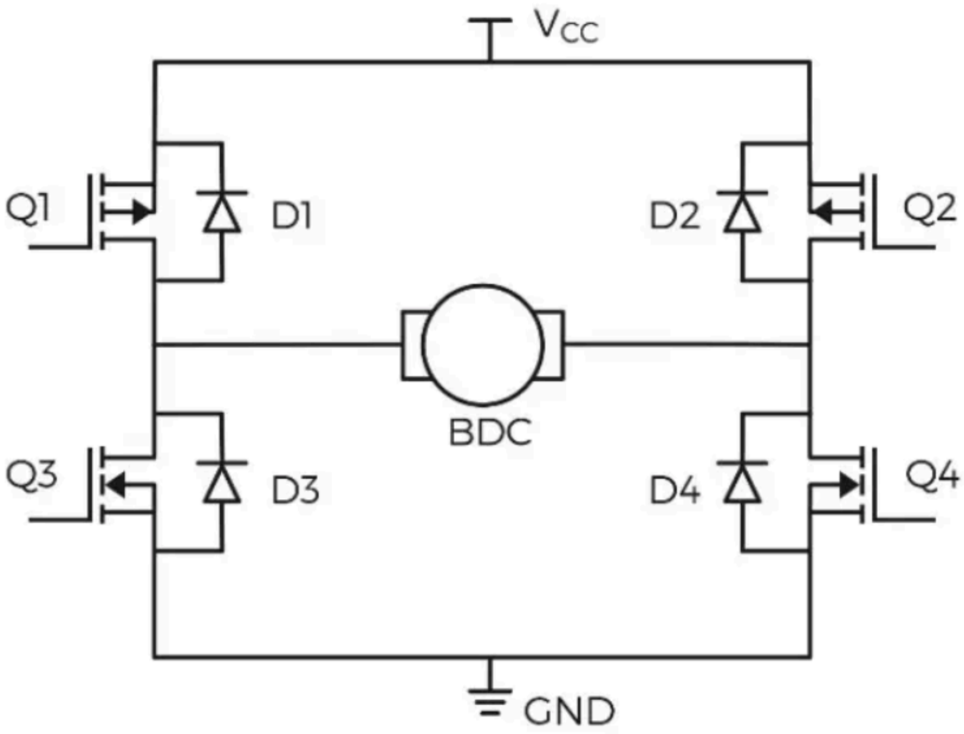


Figure 11. *H-Bridge Topology [83]*

3.1.5.5 Motor Driver Selection

H-Bridge Driver: The H-Bridge driver was selected for *SOLARIS* due to the balance of cost, efficiency and functional requirements. While unidirectional drivers are efficient and allow for precise speed control, they do not allow for the reversal of the voltage polarity thus lack the ability to reverse the direction of the motor. The H-Bridge's switch configuration allows for bidirectional motion which enables the robot to reposition itself as required for optimal sunlight exposure. Relay based drivers did not provide the switching speed required to precisely control the speed of the motor as desired and also introduced the unreliability of mechanical wear into the driver system. The PWM based speed control allows for precision and electrical braking which will enable more precise movement which will meet the requirements for robot navigation over the variable terrain. The H-Bridge driver is the most balanced solution that allows for all of the control, simplicity and ease of integration requirements that the project demands.

3.1.5.6 Motor Driver Components

Table 14. DC Motor Driver Models

Driver Model	*Channels*	*Voltage Range*	*Current*	*Logic Voltage*	*PWM/Control*	*Cost*

Pololu TB6612FNG Dual Motor Driver Carrier	**2**	**4.5-13.5 V**	**1 A/Channel continuous, 3 A peak**	**2.7-5.5 V**	**Up to 100 kHz PWM**	**$4.95**
STMicroelectronics VNH5019ATR-E	**1**	**5.5 - 24 V**	**30 A**	**3.3/5 V**	**In/Out control 20 kHz**	**$9.22**
Pololu VNH5019 Motor Driver	**1**	**5.5 - 24 V**	**12 A Continuous, 30 A peak**	**2.5 - 5 V**	**Up to 20 kHz PWM, current-sense output**	**$29.95**

Source: [94]

When selecting an H-bridge motor driver for our DC motor, the most critical consideration is current handling capacity. The driver must safely support both the motor's continuous operating current and its higher startup surge or stall current without overheating or triggering protection faults. In multi-motor robots like *SOLARIS*, cumulative current draw during acceleration or turning maneuvers can momentarily spike, so driver selection must account for worst-case conditions rather than nominal cruising current. Undersizing the driver risks voltage drop, thermal shutdown, or permanent damage, while oversizing may increase cost and board footprint unnecessarily. Therefore, a balanced selection ensures sufficient headroom while remaining efficient and cost effective.

Another important factor is logic compatibility and control interface to ensure compatibility with the ESP32. The driver must accept 3.3 V logic signals or explicitly support mixed-voltage systems to avoid the need for level shifting hardware. Control schemes such as PWM + direction or locked antiphase control affect how easily the system can implement speed and directional control in firmware. Additionally, maximum PWM frequency support influences smoothness of operation. Selecting a driver that integrates cleanly with the ESP32 simplifies firmware programming and improves overall system robustness.

Protection features and electrical robustness also play a significant role in driver selection. Features such as over-current protection, thermal shutdown, under-voltage lockout, and reverse polarity protection can prevent catastrophic failure during testing or unexpected operating conditions. In outdoor mobile robotics, transient loads and battery voltage fluctuations are common, making integrated safeguards particularly valuable. Some drivers also include current-sense outputs, which can be leveraged for monitoring motor load or detecting stall conditions. These protections improve safety, increase system durability, and reduce the risk of damage during iterative development.

Finally, practical integration considerations must be evaluated alongside electrical performance. The number of motor channels per board affects packaging, wiring complexity, and overall system layout, especially in a four-motor drivetrain. Board size, heat dissipation requirements, and mounting options influence mechanical integration within the chassis. Cost per channel must also be considered to maintain project budget constraints. Ultimately, driver selection requires balancing electrical capacity, control compatibility, protection features, packaging practicality, and cost to ensure reliable and efficient operation of the robot's drivetrain.

3.1.5.7 Motor Driver Component Selection

VNH5019ATR-E: The STMicroelectronics VNH5019ATR-E was selected over the other drivers due to its better power performance and system flexibility. The TB6612FNG is limited to low-current applications, making it an unideal for *SOLARIS*, whereas the VNH5019 supports much higher current and a much wider voltage range. Compared to the Pololu VNH5019 board, the standalone IC provides the same electrical performance but allows for a custom PCB design with improved thermal management and optimized layout for higher current functionality. Overall, the VNH5019ATR-E offers the robustness, scalability, and integration flexibility required for reliably driving motors in a 20–30 lb mobile robot without concerns for current and overheating or efficiency.

3.1.6 Solar Panel Technologies

Table 15. *Solar Panel Technology Comparison*

Solar Panel Technology	*Efficiency*	*Weight*	*Cost*	*Temperature Performance*	*Degradation Rate*
Monocrystalline Silicon	**Excellent**	**Good**	**High**	**Good**	**Low**
Polycrystalline Silicon	**Good**	**Good**	**Moderate**	**Moderate**	**Moderate**
Thin-Film Silicon	**Low**	**Excellent**	**Low**	**Best**	**High**

Sources: *[4],[43]*

3.1.6.1 What is a Solar Panel?

Solar photovoltaic (PV) modules operate by orienting an array of semiconductor cells toward incident solar irradiance. Upon absorption of photons within the p-n junction of the semiconductor material (typically silicon-based), electron-hole pairs are generated, producing a photovoltage via the photovoltaic effect. This voltage is scaled up by interconnecting multiple cells in series to achieve the desired output potential. Commercially available PV technologies primarily differ in cell composition, efficiency, and encapsulation methods, including the protective front glass and backsheet materials. Key selection criteria for PV modules include module efficiency, specific power (W/kg), mechanical and environmental durability (e.g., hail impact resistance and degradation rate), form factor (e.g., dimensions, flexibility, and mounting compatibility), and cost.

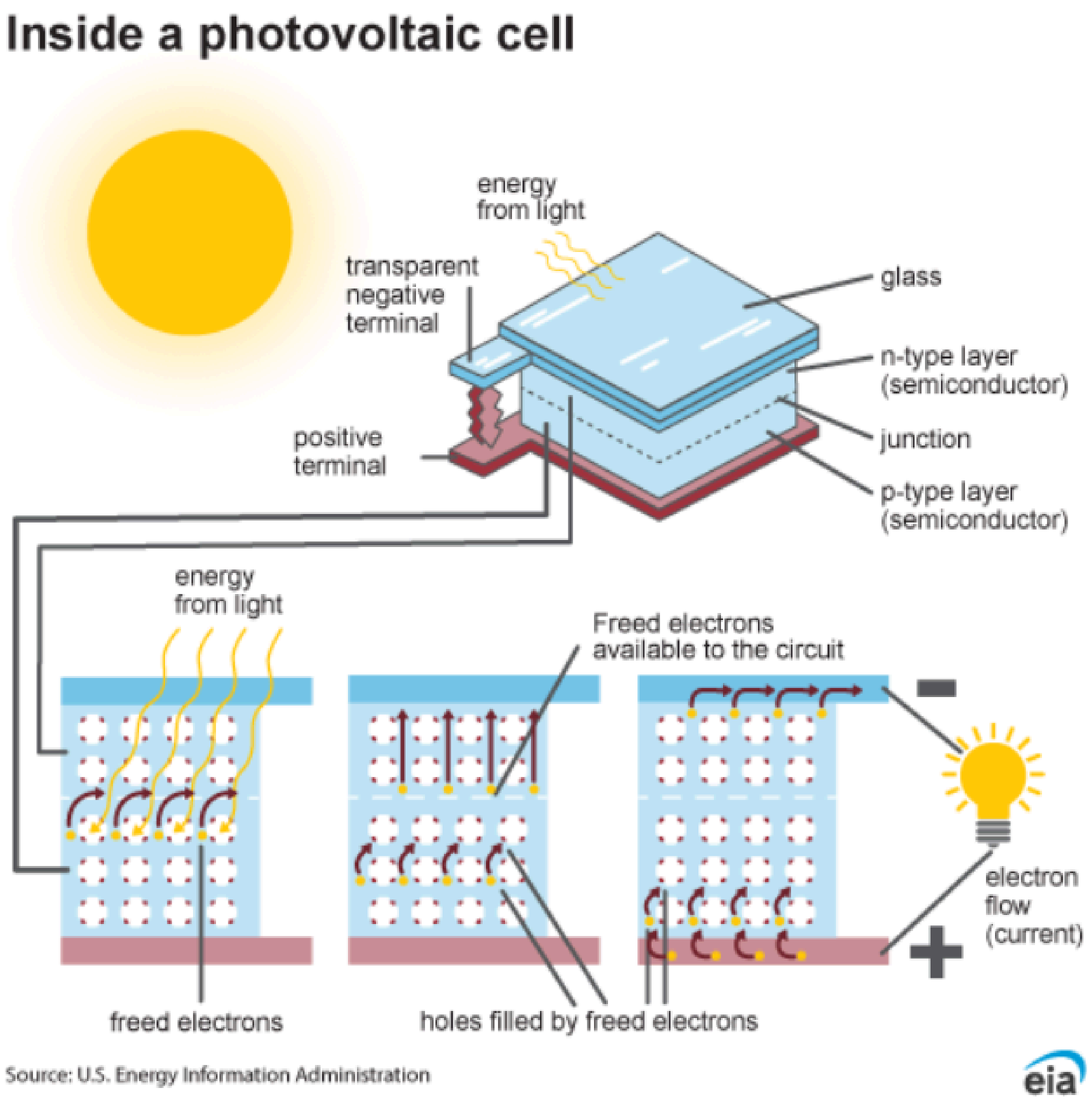


Figure 12. *Photovoltaic Cell Diagram [43]*

3.1.6.2 Monocrystalline Silicon

Monocrystalline silicon solar panels are widely regarded as the highest-performance mainstream photovoltaic technology. Because each cell is cut from a single, continuous crystal structure, electron flow is more uniform, resulting in higher efficiencies typically in the 20-23% range for commercial modules. This means more power output per square meter compared to polycrystalline or thin-film alternatives, making them ideal when roof space is limited. They also tend to have slightly lower weight per watt because fewer panels are needed to achieve the same system size, though individual modules weigh similarly to other silicon panels. The tradeoff is cost: monocrystalline panels are generally more expensive due to the energy-intensive crystal growth process and material purity requirements.

In terms of performance in real-world conditions, monocrystalline panels offer strong temperature characteristics, with lower temperature coefficients (often around -0.3 to -0.4%/°C). This means they lose less efficiency as temperatures rise, which is critical in hot climates where rooftop surfaces can become extremely warm. Additionally, they exhibit low annual degradation rates allowing them to retain over 85-90% of their original output after 25 years.

3.1.6.3 Polycrystalline Silicon

Polycrystalline silicon solar panels are a widely used and cost-effective photovoltaic technology, characterized by cells made from multiple silicon crystal fragments melted together. This manufacturing process is simpler and less energy-intensive than that of monocrystalline silicon, which translates to lower overall cost per panel. In terms of efficiency, polycrystalline panels typically operate in the 15-18% range, meaning they produce less power per unit area and require more surface space to achieve the same output as monocrystalline panels. Their weight per panel is generally comparable to other silicon-based modules, but because more panels are needed for a given system size, the total system weight can be slightly higher for equivalent power output.

Regarding performance and longevity, polycrystalline panels have slightly worse temperature performance, with temperature coefficients often around -0.38 to -0.45%/°C. As a result, their output drops more noticeably in high-temperature environments, making them somewhat less ideal for very hot climates. However, their degradation rates are still respectable, typically around 0.5-0.8% per year, allowing systems to maintain reliable output over decades.

3.1.6.4 Thin-Film Silicon

Thin-film silicon solar panels use very thin layers of photovoltaic material, deposited onto substrates like glass, metal, or flexible polymers. Because the semiconductor layer is much thinner than in crystalline panels, manufacturing can be less material-intensive and potentially lower cost at scale. However, efficiency is significantly lower, typically in the 10-13% range for silicon-based thin-film modules. This means more surface area is required to generate the same power as crystalline panels. On the positive side, thin-film panels are generally lighter and can even be flexible.

In real-world operating conditions, thin-film silicon panels often perform well at high temperatures, with relatively favorable temperature coefficients compared to polycrystalline modules. Their output tends to degrade less sharply as temperatures rise, which can partially offset their lower nominal efficiency in hot climates. However, degradation rates can be somewhat higher in the early years due to light-induced degradation effects, though performance typically stabilizes afterward. Over the long term, annual degradation may range around 0.5-1% depending on manufacturing quality.

3.1.6.5 Solar Panel Technology Selection

Monocrystalline silicon: With industry-leading conversion efficiencies typically exceeding 20%, monocrystalline silicon panels generate more electrical power per unit area than polycrystalline or thin-film alternatives. For a mobile robotic platform like *SOLARIS*, surface area is limited and every square inch of panel space must be maximized. Higher efficiency directly translates to greater energy harvest under the same sunlight conditions, enabling longer operation time, improved battery charging capability, and more reliable performance in partially shaded or lower-irradiance environments. Simply put, monocrystalline technology allows *SOLARIS* to extract the maximum possible energy from the space available.

Beyond efficiency, monocrystalline panels also offer strong temperature performance and low long-term degradation, which reinforces their suitability for an outdoor robotic system. Their lower temperature coefficient ensures that power output remains more stable in hot conditions, preventing significant efficiency losses during peak sunlight hours when energy generation matters most. While thin-film technology often exhibits more favorable temperature coefficients than monocrystalline, the increased area of solar panel that would be required for equivalent power would not be worthwhile. Additionally, their low annual degradation rate ensures sustained performance over years of operation, protecting the long-term reliability of *SOLARIS*. While the upfront cost is higher, the improved energy density, durability, and system efficiency justify the investment by delivering consistent, high-quality power generation critical to the robot's functionality and mission success.

3.1.7 Solar Panel Parts

Table 16. *Solar Panel Part Comparison*

20W 12V Solar Panel Model	*Temperature Coefficient of Pmax*	*Size*	*Weight*	*Power Density (mW/in^2)*	*Cost*
ACOPOWER HY020-12M	**-0.45%/°C**	**13.6 × 18.5 × 0.8 in**	**6.2lbs**	**79.5**	**$39.50**
Newpowa NPA20S-12J	**-0.36%/°C**	**15.6 × 12 × 0.9 in**	**3.16lbs**	**106.8**	**$39.99**

SLD TECH ST-20P-12	**-0.5%/°C**	**15.4 × 14.2 × 1.2 in**	**4lbs**	**91.5**	**$63.80**
Ameresco C1D2-420J	**-0.45%/°C**	**16.7 × 19.8 × 2 in**	**6lbs**	**60.5**	**$167.70**

Sources: *[67],[68]*

The key voltage and current parameters when selecting a solar panel are the values at the maximum power point (Vmp and Imp). A solar panel exhibits a nonlinear current–voltage (I–V) characteristic, meaning power output ($P = V \times I$) varies across the curve. There exists a specific operating voltage at which the product of voltage and current is maximized, this is known as the maximum power point (MPP). Operating at higher voltage is advantageous because, for a given power level, higher voltage corresponds to lower current. Since resistive cable losses are proportional to I^2R, reducing current significantly decreases conduction losses in wiring. Therefore, panels that allow higher operating voltage can improve system efficiency.

The temperature coefficient of Pmax quantifies the rate at which a solar module's maximum power output decreases as cell temperature rises above Standard Test Conditions (25°C cell temperature). It is typically expressed in %/°C and is negative for silicon modules, indicating power loss with increasing temperature. A coefficient closer to zero (i.e., less negative) means the panel experiences a smaller reduction in power as temperature increases, allowing it to maintain higher output under real-world operating conditions.

The Nominal Operating Cell Temperature (NOCT) specifies the expected cell temperature under defined conditions: 800 W/m² irradiance, 20 °C ambient temperature, and moderate wind speed. It provides a realistic operating temperature reference. The difference between NOCT and the 20 °C ambient used in its definition represents the typical temperature rise of the cells above ambient under those conditions. This value can then be used to estimate operating cell temperature at other ambient conditions.

For example, assume a solar panel module has a temperature coefficient of -0.45%/°C and a Nominal Operating Cell Temperature (NOCT) of 45°C at 20°C ambient. This corresponds to a 25°C temperature rise above ambient under NOCT conditions. Assuming a 35°C ambient temperature on a sunny July day and similar irradiance, the expected cell temperature would be approximately 60°C. Since STC is defined at 25°C cell temperature, this represents a 35°C increase. Applying the temperature coefficient yields a total power reduction of 15.75%, resulting in an expected maximum power output of approximately 16.85 W.

The physical size, weight, and resulting power density of the solar panel directly impact the compactness, mobility, and mechanical integration of the robot. A larger or heavier panel increases structural load, raises the center of mass, and requires greater torque from any panning or tilting mechanisms used for solar tracking. This can increase actuator power consumption, reduce stability, and complicate mechanical design. Therefore, minimizing panel size and mass is critical for maintaining efficient mobility and responsive directional control.

3.1.7.1 ACOPOWER HY020-12M

The ACOPOWER HY020-12M offers solid mechanical resilience for portable and small-scale applications, with an aluminum alloy frame, low-iron tempered glass, and EVA/TPT encapsulation. Its compact dimensions (13.6×18.5×0.8 in) and weight (6.2lbs) provide decent portability, while electrical performance includes Vmp of 17.5 V and Imp of 1.14A at STC, with a module efficiency of 10.89% and ±3% power tolerance. Temperature coefficients are balanced (Pmax -0.45%/°C, Voc -0.35%/°C, Isc +0.05%/°C), and NOCT of 45±5°C supports operation from -40°C to +85°C. The 0-100% relative humidity rating listed for this panel is particularly advantageous for deployment in central Florida, where summer conditions routinely drive relative humidity levels to 90-100%. On the downside, its lower efficiency translates to modest power density (about 79.5 mW/in²), necessitating larger arrays for higher energy needs, and the specific power (3.23 W/lbs) is unremarkable for weight-sensitive robotics.

3.1.7.2 Newpowa NPA20S-12J

The Newpowa NPA20S-12J delivers exceptional efficiency and compactness, boasting 22% module efficiency with 33 Class A high-efficiency monocrystalline cells, yielding the highest power density (106.8 mW/in²) and specific power (6.33 W/lbs) among the options. It features solid encapsulation for potential-induced degradation protection, and offers a 0-+5W positive tolerance with Vmp of 19.08V and Imp of 1.05A at STC. The panel's high reliability in harsh environments make it suitable for outdoor, mobile use in Florida's variable weather.

Weaknesses are minimal but include a slightly higher NOCT (45±2°C) that could affect peak-hour performance, though mitigated by the best temperature coefficient for power (-0.36%/°C), reducing heat-related losses. Its dimensions (15.6×12×0.9 in) and light weight (3.16lbs) are advantages. Its high performance-to-cost ratio makes this solar panel a strong candidate for *SOLARIS*.

3.1.7.3 SLD TECH ST-20P-12

The SLD TECH ST-20P-12 achieves a module efficiency of 20% with 32 monocrystalline cells, enabling strong power density (about 91.5 mW/in²) and a positive tolerance of 0-+5W for reliable output. It features heavy-duty anodized frames, high-transparent low-iron tempered glass, and high potential-induced degradation resistance. This panel is resistant to salt mist, per IEC 61701, making it ideal for coastal or harsh weather. Electrical specs include Vmp of 18.53V and Imp of 1.08A at STC.

Weaknesses include a relatively poor temperature coefficient for power (-0.5±0.05%/°C), the most negative among comparators, causing greater output loss in high temperatures. Having a NOCT of 47±2°C also indicates this solar panel operates hotter than the other ones, which further reduces power efficiency. Its dimensions (15.4×14.2×1.2 in) and weight (4lbs) yield moderate specific power (5 W/lbs).

3.1.7.4 Ameresco C1D2-420J

The Ameresco 420J excels in durability and reliability for off-grid applications, featuring a robust design proven in extreme environments like Antarctica and the Australian outback. Its mechanical construction includes a thick, scratch-resistant white polyester backsheet for enhanced insulation and protection against rough handling, along with a silver anodized aluminum frame and high-transmission 3.2mm glass front cover. Electrically, it offers a nominal 12V output with Vmp of 16.8V and Imp of 1.19A at STC, supported by certifications including IEC 61215/61730, UL1703, and suitability for Class 1 Division 2 hazardous locations.

However, the Ameresco 420J has limitations in efficiency and compactness, with a module efficiency of only 9.4% due to its 36 monocrystalline silicon cut cells, resulting in lower power density (60.5 mW/in²) and requiring more surface area for equivalent output. Its dimensions (16.7×19.8×2 in) and weight (6lbs) make it bulkier and heavier than competitors, reducing specific power (3.33 W/lbs) and suitability for space-constrained mobile applications. Temperature performance is moderate, with a Pmax coefficient of -0.45%/°C and NOCT of 47±2°C, leading to noticeable derating in hot climates like Florida, while the ±10% power tolerance introduces variability in performance.

3.1.7.5 Solar Panel Part Selection

Newpowa NPA20S-12J: This panel was selected primarily for its superior power density and optimal temperature coefficient of Pmax, making it the most suitable for the *SOLARIS* mobile robotic platform operating in Central Florida's hot and humid climate. With a power density of approximately 106.8 mW/in², it maximizes energy harvest from limited mounting space, ensuring extended autonomous operation without expanding the robot's footprint. Its Pmax temperature coefficient of -0.36%/°C, the least negative among the panels, minimizes derating during peak solar hours when ambient temperatures often exceed 35°C (cell temperatures reaching 50-65°C), preserving 5-10% more output compared to panels with -0.45% to -0.5%/°C coefficients. This combination, alongside high specific power (6.33 W/lbs) for reduced system weight, justifies the choice. Delivering reliable, high-density power in variable thermal conditions is critical for mission success, and the Newpowa NPA20S-12J panel accomplishes that goal effectively.

3.1.8 Solar Panel Charge Controller Technologies

Table 17. *Solar Panel Charge Controller Technology Comparison*

Solar Panel Charge Controller Technology	*Power Efficiency*	*Daily Energy Capture*	*Safety*	*Complexity*	*Cost*
Multi Power Point Tracker	**95-99%**	**High**	**High**	**High**	**High**
Pulse Width Modulation	**70-80%**	**Moderate**	**Moderate**	**Moderate**	**Moderate**

Sources: *[27]-[30]*

3.1.8.1 What is a Solar Panel Charge Controller?

A solar panel charge controller regulates the power flowing from a photovoltaic panel to a battery, ensuring the battery charges safely, efficiently, and without overvoltage or overcurrent damage. It prevents overcharging, limits reverse current at night, and manages charging stages to protect battery lifespan. The two primary types are PWM (Pulse Width Modulation) and MPPT (Maximum Power Point Tracking). PWM controllers are simpler and less expensive, but they force the panel to operate near battery voltage, which reduces efficiency and lowers daily energy capture. MPPT controllers are more complex and costly because they use DC-DC conversion to continuously track the panel's maximum power point, allowing higher efficiency and greater daily energy harvest. In short, PWM offers lower cost and simplicity, while MPPT offers higher efficiency and better overall energy performance.

Constant current (CC) and constant voltage (CV) charging are critical for $LiFePO_4$ batteries because they ensure the battery is charged efficiently without exceeding safe electrical limits. During the CC stage, the battery receives a controlled, steady current that allows it to charge quickly while its voltage rises in a predictable manner. Once the battery reaches its specified upper voltage limit, the charger must transition to the CV stage, holding the voltage constant while the current tapers off as the battery approaches full capacity. This prevents overvoltage stress, overheating, and long-term degradation of the cells. A solar panel charge controller should ideally manage this CC–CV profile automatically, regulating both current and voltage to match the battery's specifications while also coordinating with the battery management system (BMS). Without proper CC and CV control, the battery's lifespan, safety, and performance can be significantly compromised.

3.1.8.2 Maximum Power Point Tracker

An MPPT controller uses a DC-DC power converter to actively regulate the solar panel operating voltage at its maximum power point. Because a solar panel has a nonlinear I–V curve, there is a specific voltage (Vmp) at which the product of voltage and current is highest. The converter then steps the panel voltage down to the required battery charging voltage while conserving power (minus conversion losses). By allowing the panel to operate at its optimal voltage and current combination, the MPPT controller increases the total energy delivered to the battery compared to other solar panel charge controllers, particularly when environmental conditions vary throughout the day.

3.1.8.3 Pulse Width Modulation

A PWM solar charge controller regulates battery charging by rapidly switching the connection between the solar panel and the battery on and off at a high frequency, effectively controlling the average current flowing into the battery. Instead of converting voltage, a PWM controller essentially forces the solar panel to operate near the battery's voltage by directly coupling them during each pulse. When the battery is low, the controller allows longer "on" pulses to deliver more current. As the battery voltage approaches its target charge voltage, the controller shortens the pulse width to taper the charging current and prevent overcharging. This simple switching method approximates CC and CV charging behavior through duty cycle modulation in practice, but because the panel voltage is pulled down to the battery voltage rather than held at its maximum power point, overall energy harvest and efficiency are lower compared to MPPT systems.

3.1.8.4 Solar Panel Charge Controller Technology Selection

Multi Power Point Tracker: For the *SOLARIS* platform, an MPPT (Maximum Power Point Tracking) charge controller was selected as the interface between the photovoltaic module and the 12 V $LiFePO_4$ battery system. The primary design driver for this project is maximizing usable energy generation within strict size and power constraints. Operating the solar panel at battery voltage (as in PWM control) does not generally correspond to its maximum power point. An MPPT controller continuously adjusts the panel operating voltage to maintain operation at Vmp while performing DC–DC conversion to deliver the appropriate CC/CV charging profile to the battery. This enables higher instantaneous conversion efficiency and increased total daily energy harvest, particularly under elevated cell temperatures and varying irradiance conditions typical of outdoor operation.

Although PWM controllers offer lower cost, reduced circuit complexity, and simpler implementation, their inability to track the maximum power point results in reduced energy extraction. Even a 10–15% improvement in daily energy capture can have a significant impact on the mission duration, recharge time, and system energy balance. Given that *SOLARIS* operates under a constrained energy budget, prioritizing energy optimization over minimal cost or simplicity is justified. Therefore, the performance gains and improved system efficiency provided by MPPT technology outweigh the incremental increase in cost and design complexity.

3.1.9 Solar Panel Charge Controller Parts

Table 18. *Solar Panel Charge Controller Part Comparison*

Solar Panel Charge Controller Part	***Electrical Efficiency***	***Operating Power Consumption***	***Key Features***	***Weight***	***Cost***
GV-5-LI-14.2V	**94-99.85%**	**0.19 mA**	**Fixed charge algorithm, 10-year warranty, IP40, 6-position terminal block connection**	**0.176lbs**	**$99**
GV-5-MOD-LFP	**94-99.85%**	**0.15 mA**	**Fixed charge algorithm, 10-year warranty, 24-pin connection**	**0.062lbs**	**$80**
Victron SmartSolar MPPT 75/10	**≤98%**	**20 mA**	**Built-in Bluetooth, programmable charge algorithm, 5-year warranty, IP22, screw terminal connection**	**1.102lbs**	**$73**
RNG-CTRL-RVR 20-US	**≥95%**	**≤100 mA**	**Optional Bluetooth add-on, programmable charge algorithm, 3-year warranty, IP32, screw terminal connection**	**3.1lbs**	**$116**

Sources: *[79]-[82]*

The solar panel charge controller in *SOLARIS* is essential for regulating the power from the solar panel to safely charge the 12V LiFePO4 battery, preventing overcharging, undercharging, or damage. Selecting an optimal solar panel charge controller requires a balanced evaluation of electrical efficiency, power consumption, and integrated features to ensure system longevity. As shown in the comparison of various models, electrical efficiency typically ranges from 95% to as high as 99.85%. While efficiency is critical for maximizing energy harvest, the operating power consumption is equally vital for maintaining battery state of charge during periods of low solar input.

The final selection often hinges on specific application needs, such as connectivity and physical durability. Programmable controllers like the Victron SmartSolar and the RNG-CTRL-RVR 20-US offer built-in or optional Bluetooth capabilities, allowing for real-time monitoring and custom charge algorithms. Conversely, the Genasun models prioritize simplicity and reliability with fixed charge algorithms and an impressive 10-year warranty, compared to the 3-to-5-year warranties found on other units. Physical constraints must also be considered; weights vary significantly from the ultra-lightweight GV-5-MOD-LFP at 0.062 pounds to the more robust RNG-CTRL-RVR 20-US at 3.1 pounds.

3.1.9.1 GV-5-LI-14.2V

The Genasun GV-5-Li-14.2V is a compact MPPT solar charge controller designed for 12V lithium batteries, boasting an industry-leading 99.85% peak efficiency that extracts up to 30% more power than PWM controllers in varying conditions, such as low light or partial shading. Key features include electrolytic-free ceramic capacitors for exceptional durability, ultra-fast true MPPT tracking at 15 Hz, excellent low-light performance, and a fixed CC/CV charging profile at 14.2V. What sets it apart is its military-grade reliability suited for harsh environments like marine, RV, or off-grid applications, backed by a 10-year warranty and a lightweight 0.176 pounds form factor that ensures long-term performance without wear-out components, making it ideal for small, high-efficiency setups where longevity and maximal energy harvest are priorities.

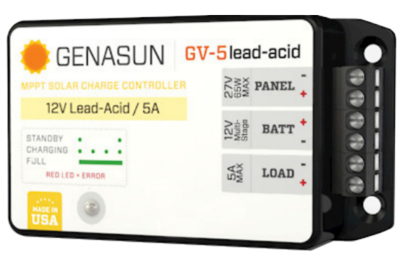


Figure 13. *Genasun GV-5-Li 5A MPPT Solar Buck Controller [79].*

3.1.9.2 GV-5-MOD-LFP

The Genasun GV-5-MOD-LFP is the more modular version of the GV-5 series, optimized for 12V (4S) Lithium Iron Phosphate batteries with a fixed CC/CV charging profile at

14.2V, offering the same 99.85% peak efficiency and 5A rated output for up to 65W panels. Notable features include a 24-pin header for seamless PCB integration, low self-consumption of 0.15mA, and protections against overvoltage, reverse polarity, and short circuits, while supporting custom voltage variations for diverse battery types. On the contrary, this component is limited by its lack of ingress protection making it require a proper enclosure to operate safely, long-term. It stands out for its OEM-focused design that reduces assembly labor and system costs in solar-powered products, providing plug-and-play reliability in embedded applications like industrial or street lighting, with a 10-year warranty and compact dimensions (3.28" x 1.88" x 0.48") that differentiate it from standard controllers by prioritizing integration ease over user-facing interfaces.

Figure 14. *Genasun GV-5-MOD-LFP 5A MPPT Solar Charge Controller [80].*

3.1.9.3 Victron SmartSolar MPPT 75/10

The Victron SmartSolar MPPT 75/10 is a high-quality MPPT charge controller for 12 V/24 V systems, delivering up to 10 A output with 98% peak efficiency, supporting up to 145 W at 12 V and 75 V max PV open circuit voltage, and featuring advanced adaptive 3-stage charging with flexible algorithms, automatic equalization, and temperature compensation via internal or external sensors. Notable features include built-in Bluetooth for wireless configuration via the VictronConnect app, VE.Direct port for data communication, VE.Smart Networking for synchronized operation with other devices, a 15 A load output with BatteryLife algorithm, and extensive protections like over-temperature and PV short circuit. It distinguishes itself through superior smart connectivity and ecosystem integration within the Victron lineup, enabling remote monitoring, firmware updates, and scalability for professional installations in boats, vans, or off-grid homes, backed by a 5-year warranty and robust build quality that emphasizes long-term reliability and user customization over basic plug-and-play simplicity.

Figure 15. *Victron SmartSolar MPPT 75/10 Charge Controller [81].*

3.1.9.4 RNG-CTRL-RVR20-US

The Renogy Rover Li 20A is a versatile MPPT solar charge controller tailored for lithium batteries in 12V/24V systems, featuring a 20A charge current, 100V maximum PV input, and up to 260W power handling at 12V, with 98% efficiency and programmable parameters via an integrated LCD screen for easy monitoring and customization. Key features include auto voltage recognition, full system protections (overcharge, overload, short circuit, reverse polarity), RS485 communication for optional Bluetooth monitoring, and compatibility with a variety of battery chemistries, along with a 4-stage charging algorithm (bulk, boost, float, equalization). What sets it apart is its user-friendly interface with built-in diagnostics and error codes, making it accessible for DIY enthusiasts in RV or home setups, combined with a higher power capacity for medium-scale systems at an affordable price point, though with a shorter 3-year warranty compared to premium competitors.

Figure 16. *Rover Li 20 Amp MPPT Solar Charge Controller [82].*

3.1.9.5 Solar Panel Charge Controller Part Selection

GV-5-LI-14.2V: The GV-5-LI-14.2V was selected as the solar panel charge controller for *SOLARIS* due to its exceptional suitability for this compact robotic application, where reliable solar harvesting is key. Weighing just 0.176 pounds, its low weight minimizes the overall mass of the robot, enhancing mobility and energy efficiency without compromising performance. The controller's compact dimensions (4.3" x 2.2" x 1.14") and simple screw-terminal design make it incredibly easy to mount directly onto the robot's chassis, reducing installation complexity and integration time. Furthermore, its ultra-low self-consumption of only 0.19 mA ensures minimal parasitic drain on the battery during standby, preserving precious energy in a mobile system like *SOLARIS*. Finally, boasting the highest peak efficiency in its class at 99.85%, it maximizes power extraction from the solar panel even in partial shade or low-light conditions common in variable outdoor environments, providing up to 50% more energy harvest compared to PWM alternatives and backed by a robust 10-year warranty for long-term reliability.

3.1.10 Light Sensors

Table 19. *Light Sensing Technology Comparison*

Sensor Type	*Ease of Implementation*	*Can Detect Sun vs Shade?*	*Multi-Sensor Scalability (4-8 sensors)*	*Detecting Small Directional Differences*	*Cost*
Photoresistors (LDR)	**Very Easy**	**Yes**	**No issues**	**Moderate (works well with shading geometry)**	**Very Low**
Phototransistor	**Medium**	**Yes**	**No issues**	**Good (more sensitive than LDRs, but still benefit from shading geometry)**	**Low**
Photodiode	**Medium / Hard**	**Yes**	**No issues**	**Very good (with an amplifier. Raw signal is small)**	**Low - Medium**
Digital Light Sensor (LUX, I2C)	**Medium**	**Yes**	**Potential address issues (using multiple of the same sensor)**	**Excellent. Gives exact LUX values. In terms of accuracy it is the best.**	**Medium**

Sources: *[27]-[30]*

3.1.10.1 Photoresistor

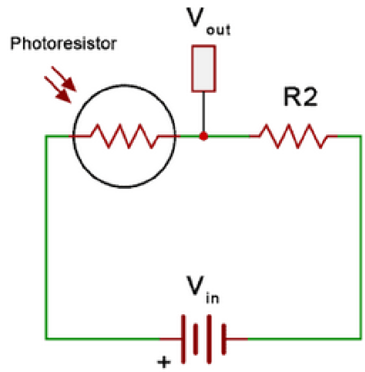


Figure 17. *LDR Light Sensing Circuit [31]*

Photoresistors, also called light dependent resistors (LDRs), are one of the simplest and most common ways to measure light in embedded systems. An LDR changes its resistance based on the amount of light hitting its surface. The most straightforward way to interface an LDR with a microcontroller is by using a voltage divider. The LDR is placed in series with a fixed resistor, and the midpoint voltage is connected to an ADC pin on the MCU. This makes the overall hardware and software implementation very simple, since the system only needs to sample an analog voltage and compare it to the other sensor readings.

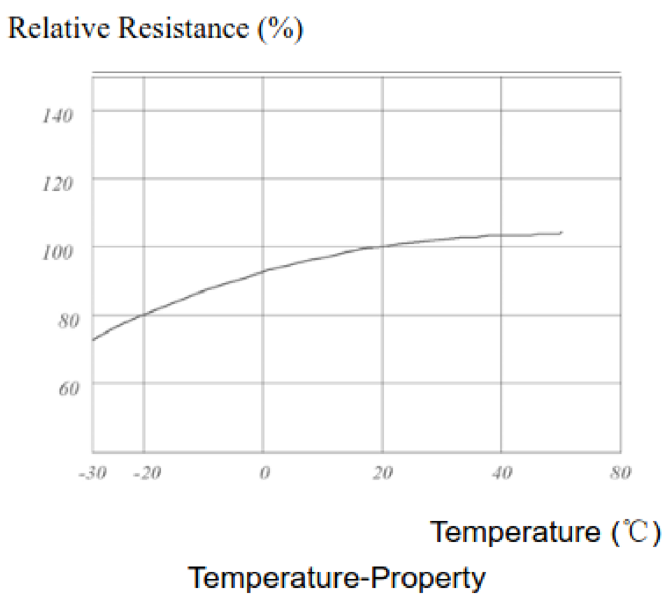


Figure 18. *LDR Resistance vs Temp [32]*

One major drawback of photoresistors is that they are not designed for accurate or repeatable light measurement. Their response to changing light is relatively slow compared to semiconductor-based sensors, and their readings can fluctuate due to temperature changes. This makes them less ideal for applications where exact or consistent intensity measurements are required.

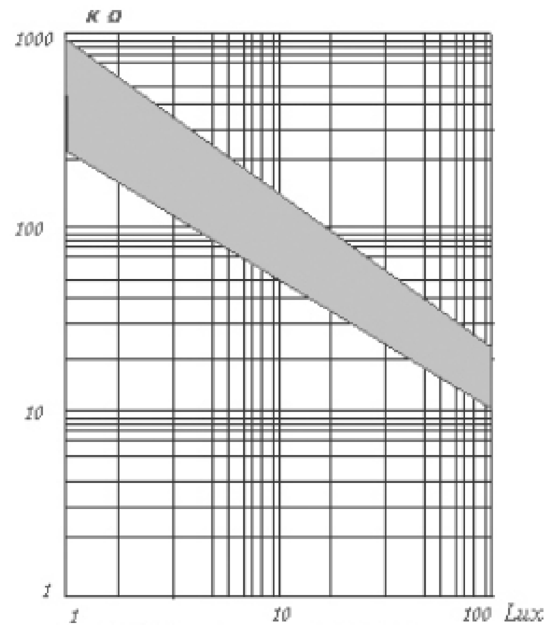


Figure 19. *LDR Non linear relationship between LUX and Resistance [32]*

Another limitation of photoresistors is that their response to light is not linear. This means that a change in resistance does not correspond to a proportional change in illumination, especially across different lighting conditions. As a result, comparing measurements between multiple LDRs can be less consistent than with semiconductor-based sensors. While this does not prevent LDRs from being used for directional tracking, it does make them less ideal for applications where consistent scaling and predictable sensor matching are desired.

However, for *SOLARIS*, the goal is not to measure absolute illuminance in lux, but instead to detect which direction contains the strongest sunlight so the robot can orient itself accordingly. In this case, LDRs are still considered "good enough," especially when used in a geometry where the chassis or a small divider structure creates shaded and unshaded regions. The sensor array can reliably detect sunlight versus shade and provide enough directional information for tracking, and the slower response time is not a major concern since the system does not need to react instantly to light changes.

Overall, photoresistors offer an extremely low-cost and low-complexity solution that can provide directional tracking information with minimal hardware overhead. Even though they are not the most accurate sensing option, their simplicity makes them a strong baseline sensor choice for the *SOLARIS* light tracking system, and they are widely used in low-cost solar tracking designs for this reason.

3.1.10.2 Phototransistor

Phototransistors can be thought of as a photodiode and a bipolar junction transistor (BJT) combined into a single device. Like a normal BJT, they have an NPN or PNP structure, but instead of electrically driving the base terminal, the base region is exposed to incoming light. When light hits the device it generates a small photo-induced current, similar to a photodiode. The key difference is that this small current is then amplified by the transistor's internal gain, which produces a much larger output current than a standalone photodiode would typically provide. Because of this built-in gain, phototransistors often do not require an external amplifier stage to generate a useful signal, making them a practical middle ground between photodiodes and photoresistors.

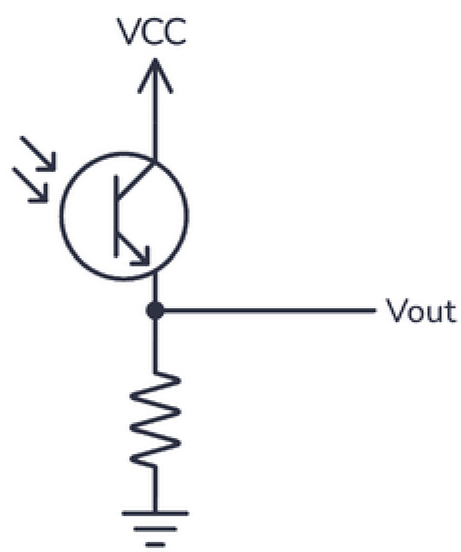


Figure 20. *Phototransistor Light Sensing Circuit [33]*

A simple phototransistor sensing circuit can be implemented using a single resistor as a load, as shown in the figure. In this configuration, the phototransistor is connected to the supply voltage and the resistor is connected to ground, with the output voltage measured at the junction between the two. As the light level increases, the phototransistor conducts more current. This increases the voltage drop across the resistor, causing the measured output voltage to rise. The output voltage can be read directly by the microcontroller's ADC, making the software side straightforward and similar to reading an LDR voltage divider.

This circuit is also safe to interface with a 3.3 V microcontroller as long as the circuit is powered from the same 3.3 V supply. Since the output voltage cannot exceed the supply voltage, the ADC input will remain within the valid 0–3.3 V range. The main design choice is selecting the resistor value so the signal does not saturate at 3.3 V too easily. A larger resistor increases sensitivity but can cause the output to peg near the supply voltage in bright sunlight, while a smaller resistor provides more headroom at the cost of reduced sensitivity. In the context of *SOLARIS*, phototransistors provide a strong, easy-to-read signal with minimal additional circuitry, while still enabling reliable detection of light differences between different directions around the robot.

3.1.10.3 Photodiode

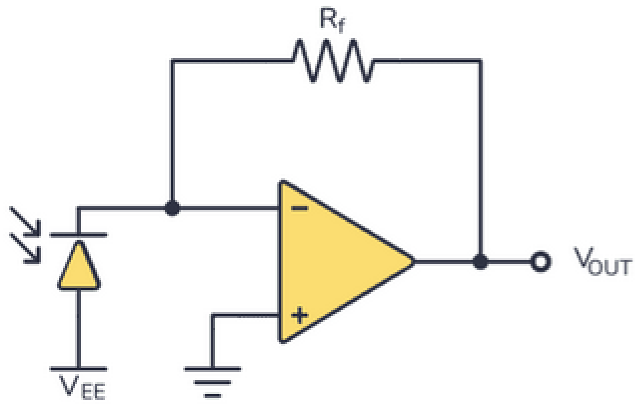


Figure 21. *Photodiode Light Sensing Circuit [34]*

Photodiodes are semiconductor light sensors that convert light into a small current. In most practical sensing applications, photodiodes are operated in reverse bias, meaning a reverse voltage is applied across the diode. Reverse bias increases the size of the depletion region, which reduces the junction capacitance. This improves response time significantly and also helps produce a more predictable, linear relationship between light intensity and output current.

One drawback of photodiodes is that the raw output signal is typically very small. While the device itself produces a photocurrent proportional to the light level, this current is often too low to directly interface with a microcontroller ADC in a stable way. As a result, photodiodes are commonly paired with an amplifier stage (such as a transimpedance amplifier) to convert the photocurrent into a readable voltage signal. This increases overall circuit complexity compared to photoresistors or phototransistors, but it also allows for improved signal-to-noise ratio and more consistent measurements.

Photodiodes also provide the fastest response time of the sensor types considered, often operating in the nanosecond range depending on the device and biasing method. This makes them ideal for high-speed optical communication and precision measurement applications. However, for *SOLARIS,* the system does not require extremely fast light sensing since solar tracking is a slow process and does not need rapid reaction to changes in illumination.

An additional benefit of photodiodes is that they can be selected based on semiconductor material and spectral response, meaning specific devices can be chosen for ultraviolet (UV), visible, or infrared (IR) detection. While this capability can be useful in specialized sensing applications, *SOLARIS* primarily needs to detect general sunlight intensity for directional tracking, so broad visible-light response is sufficient for this project.

Overall, photodiodes offer excellent speed and linearity, and can provide very high-quality measurements when paired with the proper amplifier circuitry. However, the additional analog design effort required to obtain a stable, readable output makes them less practical for the *SOLARIS* tracking system compared to simpler sensor options.

3.1.10.4 Digital Light Sensor (LUX, I2C)

Digital light sensors are integrated devices designed to measure ambient light intensity and report it as a digital value, often in units of lux. Unlike photoresistors, photodiodes, or phototransistors, these sensors typically include internal circuitry such as amplification, filtering, and an analog-to-digital converter (ADC), allowing the microcontroller to receive a stable numeric measurement rather than interpreting a raw analog voltage. This makes digital light sensors one of the most robust and repeatable options for measuring light intensity in embedded systems.

A major advantage of digital light sensors is that they provide an actual lux-based output rather than a relative signal. This can be useful for system logging, long-term analysis, and more consistent comparisons between sensors. Because the measurement and conversion are handled internally, digital sensors are also less affected by ADC noise, resistor tolerance, and unit-to-unit variation compared to analog light sensing approaches.

However, digital light sensors typically communicate over I2C, meaning they share the same two-wire bus (SDA and SCL) with other I2C peripherals. While this simplifies wiring for a single sensor, it can create scalability issues when multiple identical sensors are required. Many digital light sensors use a fixed I2C address or only provide a limited number of selectable addresses. As a result, using multiple identical devices on the same I2C bus can become problematic without additional hardware such as an I2C multiplexer. This makes digital light sensors less convenient for multi-sensor directional tracking configurations where four or more sensors are required around the robot.

Overall, digital lux sensors provide the most complete and accurate light measurement capability of the sensor types considered, and they are especially useful for systems that require precise measurement and data logging. For *SOLARIS*, digital light sensors remain a strong option for single-point illuminance measurement, but their reliance on I2C addressing can complicate the design if many sensors are needed for directional tracking.

3.1.10.5 Light Sensing Technology Selection

Phototransistors: Phototransistors were selected as the light sensing technology for *SOLARIS* because they provide high sensitivity and a strong analog output signal while still requiring minimal circuitry. Unlike photoresistors, phototransistors have built-in current gain, meaning the light-induced signal is amplified internally. This allows them to produce a larger and more stable output signal without requiring an external amplifier stage.

Phototransistors were chosen over photodiodes primarily due to implementation complexity. While photodiodes can provide fast and highly linear sensing, their output current is typically very small and often requires a dedicated amplifier circuit (such as a transimpedance amplifier) to generate a usable voltage signal. For *SOLARIS*, solar tracking is a slow process and does not require nanosecond-scale response times, so the added analog complexity of photodiodes is not justified for this application.

Digital I2C light sensors were also considered because they provide direct lux readings and can offer very repeatable measurements. However, they introduce additional software and communication complexity due to the use of the I2C bus, and multi-sensor configurations can become problematic due to device addressing limitations. Since *SOLARIS* requires multiple sensors around the robot for directional tracking, relying on multiple identical I2C light sensors would likely require extra hardware such as an I2C multiplexer.

In addition to these design tradeoffs, a published study [29] comparing LDR-based solar tracking to phototransistor-based tracking found that phototransistors produced approximately a 40% increase in voltage gain under the same sensing setup and software approach. This supports the decision to use phototransistors for *SOLARIS*, since stronger sensor output improves the ability to detect directional light differences and reduces sensitivity to noise.

Overall, phototransistors provide the best balance of sensitivity, simplicity, scalability, and tracking performance for *SOLARIS* directional sunlight detection.

3.1.10.6 Phototransistor Structural Type Comparison

3.1.10.6.1 BJT-Based Phototransistors

[59] Most discrete phototransistors are implemented using a bipolar junction transistor (BJT) structure. In this configuration, incident light generates a small photo-induced base current within the device. This base current is then amplified by the transistor's internal current gain (β), producing a larger collector current. Because of this built-in gain, BJT phototransistors can produce a usable analog signal with only a simple load resistor, making them easy to interface with a microcontroller ADC.

The operational description provided earlier in this section assumes this standard BJT-based structure, as it represents the dominant architecture used in general-purpose light sensing devices.

BJT phototransistors are widely available in silicon form, are inexpensive, and are well suited for low-frequency or DC light sensing applications.

3.1.10.6.2 FET-Based Phototransistors

Photo-FET devices work differently than BJT phototransistors. Instead of light creating a small base current that gets amplified, the incoming light affects the electric field at the gate region, which changes how much current can flow through the channel. In other words, the device behaves more like a voltage-controlled element rather than a current-amplified one.

Photo-FETs are typically used in applications requiring high input impedance, low noise, or higher frequency optical detection, such as optical communication systems and precision instrumentation. They are less common in simple discrete, through-hole packages intended for basic analog light sensing.

For low-frequency sunlight detection applications, the additional complexity and performance characteristics of photo-FET devices provide no meaningful advantage.

3.1.10.6.3 Structural Type Selection

BJT: For *SOLARIS*, the light signal we are measuring is essentially DC. Sunlight does not change rapidly, and even when it does (for example, passing clouds), the change happens very slowly relative to electronic time scales. Because of this, the circuit does not need high bandwidth or special AC performance. What matters more is having a stable, predictable analog output that can be read reliably by the 3.3 V ADC.

Due to their internal gain, ease of biasing with a single resistor, and suitability for low-frequency light detection, silicon BJT-based phototransistors are selected for this design. All further analysis and component comparison assume a standard BJT phototransistor structure.

3.1.10.7 Phototransistor Current Gain Configuration

3.1.10.7.1 Standard Phototransistor

[60] A standard phototransistor consists of a single bipolar junction transistor with a light-sensitive base region. When light strikes the base–collector junction, a photocurrent is generated and then amplified by the transistor's internal current gain, producing a collector current that increases with illumination. This configuration results in moderate gain and relatively faster response compared to multi-stage devices. Because only one transistor stage is used, standard phototransistors are a common choice for general light sensing applications where a balance of sensitivity and dynamic response is desired

3.1.10.7.2 Photodarlington

A photodarlington is a phototransistor configuration in which the emitter of a light-sensitive transistor stage drives the base of a second transistor. This provides two stages of current gain in series, resulting in overall gain that is the product of the gains of each transistor. The two-stage design yields significantly higher sensitivity, making photo darlingtons useful in low-light applications. However, the much higher gain means that under high illumination the device will generate a much larger collector current, making it more prone to producing output voltages near the supply limit in bright light, essentially clipping the output too early.

3.1.10.7.3 Gain Configuration Selection

Standard: For the *SOLARIS* project, the light sensors will be exposed to bright outdoor sunlight and used to compare relative illumination across directions. In this context, extreme internal gain is not required and can be counterproductive. With a fixed 3.3 V supply and load resistor, a phototransistor with excessively high gain will produce large collector current under bright sunlight, which can push the output voltage toward the supply and reduce the usable range, meaning small changes in brightness become harder to distinguish. A standard phototransistor, with its lower gain and faster response, is less likely to saturate the output in strong light and provides more usable headroom for ADC readings. For these reasons, a standard (non-Darlington) phototransistor is selected for *SOLARIS*.

3.1.10.8 Phototransistor Hardware Comparison

Table 20. *Phototransisotr Comparison*

	Collector	***Viewing***	***Wavelength***	***Dark***	***Max***	***Cost per***

	Light Current	*Angle*	*Range*	*Current*	*Operating Voltage*	*unit*
VTT9812FH	**0.1 mA**	**56°**	**450-700 nm**	**100 nA**	**30 V**	**$0.57**
VTT9814FH	**0.1 mA**	**50°**	**450-700 nm**	**50 nA**	**30 V**	**$0.63**
TEPT5600	**20 mA**	**20°**	**440-800 nm**	**50 nA**	**6 V**	**$0.58**
TEPT4400	**20 mA**	**30°**	**440-800 nm**	**50 nA**	**6 V**	**$0.40**

Sources: *[61]-[64]*

To select an appropriate light sensing device for *SOLARIS*, several through-hole phototransistors were evaluated. Through-hole packaging was intentionally prioritized over surface-mount options because the sensors will be mounted externally on the robot chassis rather than directly soldered to the main PCB. Using through-hole components allows flexible placement and simplifies connection through jumper wiring while maintaining mechanical robustness in outdoor conditions.

Additionally, devices with spectral responses centered within the visible light range were preferred. Since SOLARIS performs relative brightness comparisons rather than absolute irradiance measurements, selecting a phototransistor responsive to visible wavelengths improves consistency with perceived light intensity and reduces potential distortions caused by excessive infrared sensitivity.

These constraints limited the amount of phototransistors we could choose from. The options in the comparison table were the best we found.

3.1.10.8.1 VTT9812FH

The VTT9812FH offers a maximum operating voltage of 30 V and a moderate angular response of approximately 56 degrees. Its dark current is low, and its through-hole package makes it mechanically compatible with the *SOLARIS* mounting requirements. However, its wider viewing angle increases susceptibility to reflected light from surrounding surfaces, which can reduce contrast between opposing sensors. Additionally, its light current sensitivity is lower compared to the TEPT series under similar illumination conditions.

3.1.10.8.2 VTT9814FH

The VTT9814FH shares similar characteristics with the VTT9812FH, including a 30 V operating margin and through-hole packaging. It maintains a relatively wide angular response near 50 degrees. Although the higher breakdown voltage provides additional margin, this benefit is not meaningful in a 3.3 V system. Like the VTT9812FH, its broader field of view increases exposure to indirect light, which may reduce tracking contrast. As a result, it does not provide a strong advantage for the intended application.

3.1.10.8.3 TEPT5600

The TEPT5600 features a significantly narrower angular response of approximately 20°, which improves directional sensitivity and enhances contrast between illuminated and shaded regions of the robot chassis. Its spectral response range of 440 to 800 nm is centered within the visible spectrum, aligning well with natural sunlight conditions. Although its maximum collector-emitter voltage is limited to 6V, this remains fully compatible with the 3.3 V operating environment of *SOLARIS*.

3.1.10.8.4 TEPT4400

The TEPT4400 was also evaluated because it closely matches the desired characteristics for *SOLARIS*. It provides a visible-centered spectral response similar to the TEPT5600 and maintains low dark current, making it well suited for relative sunlight comparison. Its viewing angle is wider than the TEPT5600 (around 30 degrees rather than 20 degrees), which can be beneficial for tolerance to mounting misalignment and small changes in orientation while still preserving strong directional contrast. In other words, it offers a good middle ground between being directional enough to improve “brighter side vs darker side” discrimination while not being so narrow that the sensor becomes finicky. However, despite being a strong candidate, the TEPT4400 was not selected due to current availability limitations.

3.1.10.9 Phototransistor Selection

TEPT5600: The TEPT5600 was ultimately selected for *SOLARIS* based on its mechanical compatibility, spectral response, and directional characteristics. Its narrower viewing angle improves contrast between illuminated and shaded sides of the robot, which is important since the control system relies on comparing relative light intensity rather than measuring absolute values. Although the device is fairly sensitive and could approach saturation under direct sunlight, this can be managed through proper resistor selection in the signal conditioning stage. Its visible-spectrum response aligns well with natural sunlight, and the through-hole package makes it practical for external mounting and jumper-wire integration on the chassis.

3.1.11 Energy Monitoring Technologies

Table 21. *Energy Monitoring Technology Comparison*

Energy Monitoring Technology	*Accuracy*	*Power Loss*	*Complexity*	*Key Features*

Coulomb Counting	**High (short-term)**	**Low**	**Moderate**	**Drift accumulates**
Hybrid (Coulomb Counting + Open Circuit Voltage)	**Very High**	**Moderate**	**High**	**Most accurate long-term**

Sources: *[130]-[132]*

3.1.11.1 Coulomb Counting

Coulomb counting provides high short-term accuracy by integrating bidirectional current over time to track the net charge entering or leaving the battery:

$$SOC = SOC_{initial} \pm \frac{\int I \, dt}{Capacity}$$

This method requires precise current sensing and periodic recalibration to correct drift caused by measurement error, battery self-discharge, and temperature variation. Implementation complexity is moderate, as it requires accurate current sensing hardware and processing to accumulate charge over time. Power loss is minimal but present due to the shunt resistor's I^2R dissipation, which can be made negligible by selecting a very low resistance value. The primary limitation of coulomb counting is accumulated drift during long-term operation if no correction mechanism is used.

3.1.11.2 Hybrid (Coulomb Counting + Open Circuit Voltage)

The hybrid approach combines coulomb counting for real-time charge tracking with periodic open-circuit voltage (OCV) calibration to correct accumulated drift. During periods of low load or rest, the battery voltage can be compared against a pre-characterized $LiFePO_4$ OCV–SOC curve to recalibrate the estimated state of charge. Although this approach increases implementation complexity due to additional sensing and processing requirements, it provides significantly improved long-term SOC accuracy. Hybrid methods are particularly effective in systems where the battery may experience simultaneous charging and discharging conditions, such as solar applications.

3.1.11.3 Energy Monitoring Technology Selection

Hybrid: The hybrid energy monitoring approach is selected for *SOLARIS* due to its ability to provide reliable long-term SOC estimation in environments with fluctuating solar input and variable load demand. In this system, coulomb counting will continuously track charge flow to provide real-time energy monitoring, while the microcontroller periodically performs OCV-based recalibration when stable voltage conditions are detected, such as after charging cycles or during low-load periods. By referencing a pre-characterized $LiFePO_4$ OCV curve, the system can correct any accumulated drift in the coulomb counter and maintain accurate SOC estimation over extended operation. Although this approach introduces additional system complexity, the improved accuracy and reliability of the energy monitoring data provided to the user outweigh the added implementation overhead.

3.1.12 Energy Monitoring Parts

Table 22. *Energy Monitoring Part Comparison*

Energy Monitoring Part	*Current Accuracy*	*Bus Voltage Range*	*Accessible Measurements*	*Cost of One IC*
LTC2944	**±1%**	**3.6-60 V**	**Battery Voltage, Current, Die Temperature, Accumulated Charge**	**$12.82**
INA226	**±0.02-0.1%**	**0-36 V**	**Shunt Voltage, Battery Voltage, Current, Power**	**$2.99**
INA228	**±0.01-0.05%**	**-0.3-85 V**	**Shunt Voltage, Battery Voltage, Die Temperature, Current, Power, Accumulated Energy, Accumulated Charge**	**$5.27**

Sources: *[84]-[86]*

3.1.12.1 LTC2944

The LTC2944 monitors the battery’s energy by measuring the differential voltage across a high-side sense resistor located between the battery’s positive terminal and the common junction of the solar charger and loads. Its bidirectional analog integrator is specifically designed to accommodate current polarity; it "counts up" as the solar panels provide a charging current and "counts down" as the system draws power for the loads. This "true" coulomb counting occurs in real-time with 1% accuracy, allowing the chip to maintain a continuous record of the accumulated battery charge regardless of whether the system is net-charging or net-discharging.

3.1.12.2 INA226

The INA226 features an industry-standard 16-bit resolution and its ability to handle bus voltages up to 36V. In a solar charging scenario, the INA226 monitors the current flow bidirectionally. When the solar panels are generating excess power, the sensor detects a positive current flowing into the battery, and when the loads exceed solar production, it detects a negative current. Because it features a dedicated alert pin, it can be programmed to interrupt the microcontroller if the battery voltage hits a critical low or if the solar charging current exceeds the battery's 20A continuous charging limit, providing an essential hardware-level safety layer.

3.1.12.3 INA228

The INA228 represents a significant technological leap, utilizing a 20-bit delta-sigma ADC to provide extreme precision for long-term energy tracking. In solar applications where light intensity, and therefore charging current can fluctuate minutely, the higher resolution of the INA228 ensures that even the smallest energy gains from a solar panel during low-light conditions are captured without falling into the "noise" of the sensor. Furthermore, the INA228 includes an on-chip energy and charge accumulator. Unlike the INA226, which requires the microcontroller to constantly poll and integrate current data to calculate Amp-hours, the INA228 can perform these calculations internally. This reduces the processing overhead required of the *SOLARIS* MCU.

3.1.12.4 Energy Monitoring Part Selection

INA228: The INA228 was selected as the primary energy monitor over the INA226 and LTC2944 due to its combination of high-resolution data acquisition and self-contained computational features. While the INA226 is a capable industry standard, its 16-bit ADC is surpassed by the INA228's 20-bit delta-sigma ADC, which provides the superior precision required to detect minute fluctuations in solar charging current that might otherwise be lost to sensor noise. Furthermore, the INA228 offers a significant architectural advantage by integrating dedicated hardware for energy and charge accumulation. Unlike the INA226, which requires the microcontroller to constantly poll current values to manually calculate Amp-hours, the INA228 performs these complex integrations internally on-chip. This not only reduces the processing overhead and power consumption of the microcontroller but also eliminates the risk of "integration drift" caused by timing variances in software-based polling. By offloading these calculations to the sensor, the INA228 provides a more stable and accurate State of Charge (SoC) tracking system, making it the most robust choice for managing a 12.8V $LiFePO_4$ battery system in a variable solar-charging environment.

3.1.13 Inertial Measurement Unit (IMU)

Table 23. *Types of Measurement Devices*

Specifications	*Accelerometer*	*Gyroscope*	*Magnetometer*
What it Measures	Linear acceleration & gravity	Angular velocity (rotation)	Magnetic field strength

Primary Unit	$m/s^2 = gravity$	deg/s (degrees per second)	μT (microteslas)
What it Detects	Which way is down	How fast it's spinning	Which way is north
Strength	Great at sensing tilt & impact	Very precise for fast motion	Provides an absolute heading
Weakness	Noisy due to vibrations	Drifts over time	Easily distorted by metal/magnets

Sources: [71-72]

3.1.13.1 Accelerometer

An accelerometer is a key piece of hardware used to measure the inertial movement of devices like cell phones and even vehicles like cars. While a measurement like speed tells you how fast something is going, acceleration tells you how quickly that speed is changing. In particular, an accelerometer detects three things. It measures the inclination of mostly stationary objects by detecting the acceleration of gravity [69]. It also detects the acceleration of an object when it is vibrating [69]. Lastly, it detects the acceleration of an object when it experiences an impact [69]. To accomplish this, the accelerometer does not calculate speed over time but instead measures force using an elementary equation.

$$\cdot F = m \times a$$

$$\cdot a = \frac{F}{m}$$

Though this equation covers the most basic kinds of accelerometers, there exist different "flavors" of sensors used for measuring acceleration. One such type is piezoelectric, which relies on piezoelectric material like a quartz crystal to sense changes in acceleration [70]. When acceleration is applied, the material responds with a compression or strain relative to its mass. Another technology, known as piezoresistive, operates in a similar fashion but measures changes in resistance rather than electrical charge. Manufactured as a semiconductor, each axis includes multiple piezoresistors that decrease in resistance when force is applied [70]. The last accelerometer technology worth mentioning for *SOLARIS* is capacitive. Similar to piezoresistive, it is generally a semiconductor sensing device, but the key difference is a tether system that collectively acts like a spring, allowing the displacement of the mass to represent the measured acceleration [70]. Other accelerometer types are more specialized in their use cases and therefore are not relevant to *SOLARIS* as a potential technology selection.

3.1.13.2 Gyroscope

A gyroscope (gyro) is a microelectromechanical system (MEMS) sensor designed to measure angular velocity [73]. In a traditional sense, gyroscopes are generally heavy spinning wheels used to maintain orientation. The MEMS application used in electronics instead relies on tiny vibrating structures that shift when the device is rotated. While an accelerometer measures linear force, a gyroscope detects angular velocity, which can be defined as the rate at which an object rotates around an axis [73]. Within MEMS gyroscopes there are two main types: the vibrating structure and the tuning fork. Vibrating structure gyroscopes utilize the Coriolis force, which causes a secondary oscillation that can be measured to determine angular velocity [74]. The tuning fork configuration operates similarly but uses a pair of vibrating masses. When the Coriolis force acts on the tines, it can be measured in the same way to determine angular velocity [74].

$$\cdot\ \omega = \frac{V_{out} - V_{bias}}{S}$$

A gyroscope can detect three major movements: pitch, roll, and yaw. Using the equation shown above, the gyroscope receives a bias voltage (Vbias) from its power source and, given its own sensitivity (S), calculates angular velocity against its output voltage (Vout). This equation is generalized since MEMS gyroscopes can come in different configurations. Some of the main advantages of MEMS gyroscopes are their small size, low cost, low power consumption, and reliability [74]. These qualities make them useful across a wide range of application areas including consumer electronics, automotive, defense, and healthcare [74].

3.1.13.3 Magnetometer

Just like an accelerometer and a gyroscope, a magnetometer is another way for a digital device to interface with an otherwise analog world. A magnetometer measures the magnetic fields present in a surrounding environment [72]. The measurement of magnetic fields is not only useful for *SOLARIS* but is also crucial for research, navigation, and exploration. A magnetometer uses a sensor capable of measuring either magnetic flux density (B) or magnetic field strength (H) [72]. Like the sensors discussed previously, these devices also come in different configurations but operate on the same mathematical principles shown below.

$$V_H = \frac{IB}{nte}$$

The equation above references the most common type of magnetometer, the Hall Effect. When a current-carrying conductor is placed in a magnetic field, a voltage is generated perpendicular to both the direction of the field and the current flow [72]. In this equation,

(I) represents current, (B) represents magnetic flux density, (n) represents the charge carrier density of the material, (t) represents the thickness of the plate, and (e) represents the elementary charge of an electron.

$$|M| = \sqrt{M_x^2 + M_y^2 + M_z^2}$$

$$\psi = arctan^2(M_y, M_x)$$

The two equations above define the mathematical principles behind another popular magnetometer technology known as vector based. The first equation combines three directional readings to calculate total magnetic intensity using a three-dimensional Pythagorean theorem. The second equation converts these raw magnetic readings into a compass heading using trigonometry. Together, these equations mean that vector magnetometers measure magnetic flux density (B) in a specific direction within three-dimensional space [72]. This makes the magnetometer particularly valuable for its ability to provide a true heading for spatial awareness.

3.1.13.4 Degree of Axis

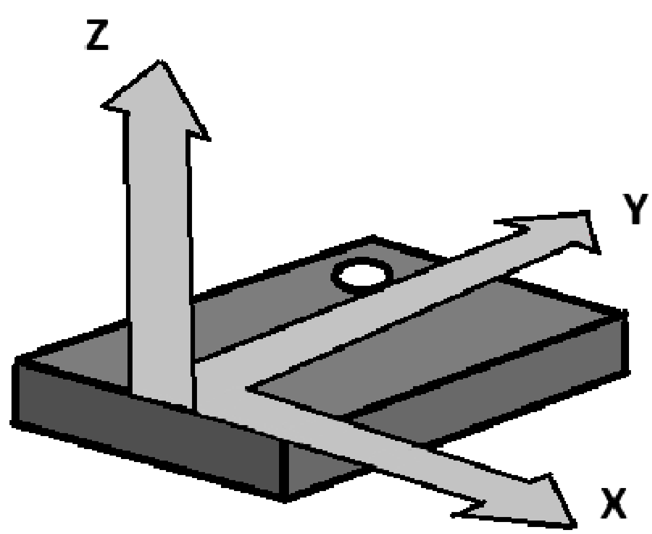


Figure 22. *Degrees of Freedom Illustrated*

In the grand scheme, the three major components that make up an inertial measurement unit (IMU) are the accelerometer, gyroscope, and magnetometer. On their own, each of these sensors measures three degrees of movement. An IMU is categorized by its Degrees of Freedom (DoF), which represents how many ways it is able to sense motion [75]. The DoF of a device expresses how many ways something can move in three-dimensional space [75]. The accelerometer is responsible for three degrees of translational movement, the gyroscope measures three degrees of rotational movement, and the magnetometer measures three degrees of direction. When referring to the DoF of a particular IMU, we are referring to how many of these sensors it contains. To obtain usable orientation information from an IMU, a process called sensor fusion is used to combine the raw readings from each sensor, which are otherwise output in different units and scales, into an accurate representation of an object's orientation in three-dimensional space [75].

Table 24. *Degrees of Freedom (DoF) Comparison*

DoF	***Sensors Included***	***What it Measures***	***Best For***
3-DoF	Accelerometer	Linear acceleration, static tilt	Screen rotation, step counting, impact sensing
6-DoF	Accelerometer + Gyroscope	Translation and rotation	VR controllers, gamepads, balancing robots
9-DoF	Accelerometer + Gyroscope + Magnetometer	Translation, rotation, and magnetic heading	Drones, autonomous vehicles, 3D mapping.

Sources: *[75]*

3.1.13.4.1 3-DoF

A 3-DoF IMU can refer to any singular sensor on its own but generally refers to an accelerometer. This means it can only sense translational movement along the X, Y, and Z axes [75]. However, it cannot detect yaw since rotating the sensor around a vertical axis does not change the force of gravity acting on it.

3.1.13.4.2 6-DoF

A 6-DoF IMU generally refers to a 3-axis accelerometer paired with a 3-axis gyroscope. With this combination, the device is capable of sensing both linear movement and angular velocity [75]. It is called 6-DoF since it combines the three axes of the accelerometer with the three axes of the gyroscope. The key advantage of this setup is the ability to calculate orientation while in motion, where the gyroscope tracks real-time rotation and the accelerometer provides a reference point.

3.1.13.4.3 9-DoF

A 9-DoF IMU adds a 3-axis magnetometer to the previous 6-DoF setup, allowing the device to sense translational movement, rotation, and direction simultaneously. What makes this configuration particularly powerful is that sensor fusion across all three sensors enables a form of self-correction within the system [75]. By sensing the Earth's magnetic field, the device has a fixed reference point for the horizontal plane, allowing for accurate heading calculation [75].

3.1.13.5 IMU Comparison

Table 25. *IMU Unit Comparison*

Specifications	*BNO-085*	*MPU-6050*	*ICM-20948*
Degrees of Freedom	9-DoF	6-DoF	9-DoF
Onboard Processing	Full Sensor Fusion (ARM Cortex-M0)	Basic Digital Motion Processor (DMP)	Advanced DMP (to offload calculations)
Power Profile	Low Power Mode	Sleep Register	Low Power Mode
Stability	High (Absolute Orientation)	Medium (No Magnetometer)	Medium-High (Relative Compass)
Use Case	Autonomous navigation and high-accuracy robotics	Budget-friendly tilt sensing and simple motion detection	Low-power mobile robots and high-speed telemetry

Sources: *[76-78]*

3.1.13.5.1 BNO-085

The BNO085 is a specialized 9-axis IMU that combines the features of a classical IMU while also functioning as a System in Package (SiP). Unlike typical IMUs that output raw data for the microcontroller to process, the BNO085 features an onboard 32-bit ARM Cortex-M0+ microcontroller that handles all sensor fusion internally [76]. Another key highlight is its ability to self-calibrate, continuously estimating and compensating for magnetometer distortions and gyroscope zero-rate offset. It also supports three communication interfaces: I2C, SPI, and UART. I2C and SPI can be used to directly configure the device, enabling pre-implemented sleep modes for low power applications and allowing individual sensors like the accelerometer, gyroscope, or magnetometer to be toggled on or off as needed [76]. This means any combination of sensors can be used interchangeably or together for maximum sensor fusion accuracy. For an application where power efficiency and data accuracy are critical, the BNO085 offloads sensor processing from the ESP32-S3, freeing it to focus on controlling the rest of the system.

3.1.13.5.2 MPU-6050

The MPU-6050 is a 6-axis sensor hub that outputs raw motion tracking data and is typically more affordable than other selections, making it a common choice for projects where the microcontroller handles all sensor fusion processing. It is marketed as one of the world's first devices to combine a 3-axis gyroscope and a 3-axis accelerometer with an onboard Digital Motion Processor (DMP) [77]. It uses six dedicated 16-bit ADCs to sample all axes simultaneously, ensuring no time skew between the two peripherals. One key feature is its auxiliary I2C master port, which allows it to act as a controller for external sensors [77]. This means it can be paired with a third-party magnetometer to automatically read compass data and fuse it with its own output for full heading calculation. The main downsides compared to other options are its lack of native magnetometer support and onboard processing, though its low price point and strong community support due to its legacy status remain notable advantages.

3.1.13.5.3 ICM-20948

The ICM-20948 is an advanced 9-axis motion tracking device designed as a successor to legacy hardware [78]. It is specifically built for low power consumption and high performance sensor fusion, making it a strong candidate for *SOLARIS*. The device is capable of consuming as little as 2.5mW in full 9-axis mode, making it one of the lowest power 9-axis IMUs available [78]. It also supports a native sleep mode where it does not draw power upon startup until it is needed. By combining a 3-axis gyroscope, 3-axis accelerometer, and 3-axis compass, it can provide a full absolute heading to magnetic north. Similar to the other options discussed, its Digital Motion Processor (DMP) supports high-frequency data transfers of up to 200Hz, offloading motion processing from the ESP32-S3 [78]. The DMP reliably fuses data from all nine axes into stable output, which can reduce both code complexity and the likelihood of unreliable readings. Like the MPU-6050, the ICM-20948 features an auxiliary I2C master port and can act as a controller for other sensors [78]. Onboard firmware also supports runtime calibration, helping maintain accuracy as environmental conditions and temperature change throughout the day.

3.1.13.6 IMU Selection

For *SOLARIS*, the BNO085 was selected as the IMU of choice due to its reliability, power efficiency, and software simplicity. While the MPU-6050 and ICM-20948 are excellent options for raw data output, the BNO085 functions as a sensor hub that simplifies autonomous navigation. By offloading sensor fusion math onto its onboard ARM Cortex-M0+, the ESP32-S3 is free to handle other critical system tasks while the BNO085 runs mathematically intensive fusion algorithms in the background [76]. Its dynamic calibration algorithms continuously monitor the magnetic environment and filter out distortions in real time, which is particularly important for keeping the vehicle on course as it traverses uneven terrain. The chip is designed with an always-on feature but excels in low-power states where proper configuration is key. With its rich feature set and zero processing load on the main controller, the BNO085 is a natural fit for a solar tracking robot. This selection reflects *SOLARIS's* priority for data integrity and system stability, ensuring the vehicle can accurately navigate toward the sun or unshaded regions.

3.1.14 Ultrasonic Sensors

3.1.14.1 How Ultrasonic Sensors Work

[55] Ultrasonic sensors measure distance by transmitting sound waves and detecting their reflections. A typical ultrasonic sensor module includes four primary connections: VCC and GND for power, a Trigger input, and an Echo output. When a short pulse is applied to the Trigger pin, the sensor emits a burst of sound waves from its transmitter. These waves travel outward in a cone-shaped pattern, commonly referred to as the beam cone, which defines the sensor's field of view. Any object located within this cone can reflect the sound waves back toward the sensor.

Once the reflected wave returns, the receiver detects it and the sensor drives the Echo pin high for a duration equal to the time it took for the sound to travel to the object and back. A microcontroller measures this time interval and calculates the distance by using the speed of sound and dividing the total travel time by two, since the signal makes a round trip. The width of the beam cone is important because it determines the coverage area and influences how accurately objects can be localized in front of the sensor.

For *SOLARIS*, ultrasonic will be used for object avoidance. By continuously sending trigger pulses and monitoring the returned echoes, *SOLARIS* can determine how close it is to obstacles. If an object is detected within a predefined threshold distance, the control system can respond by slowing down, stopping, or changing direction to prevent collisions.

3.1.14.2 Calculations for Beam Cone Coverage

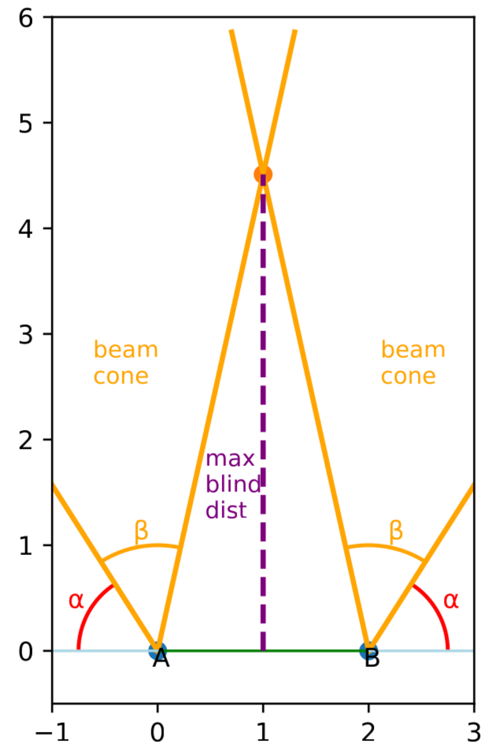


Figure 23. *Two Ultrasonic Sensor Beam Cone Coverage*

For *SOLARIS*, we plan to use four ultrasonic sensors mounted at the corners of the robot, two facing forward and two facing backward. When selecting an ultrasonic sensor, the beam cone coverage is therefore an important consideration. As shown in Figure 3.1.14.1, there are three primary regions that *SOLARIS* will be unable to detect: the far left side, the far right side, and a triangular region directly in front of the robot. While the beam cone plays a significant role in defining these blind areas, sensor placement also influences coverage. The sensors can be mounted with a slight angular offset relative to the chassis. Although not explicitly labeled in the figure, Figure 23 was created using a 10-degree offset, with the two sensors angled 10 degrees away from each other. The benefit of introducing this offset is that it reduces the undetectable angle on the far left and right sides by 10 degrees. The smaller this angle becomes, the greater the turning flexibility *SOLARIS* can be given. However, the tradeoff is that the maximum blind distance in the central region increases.

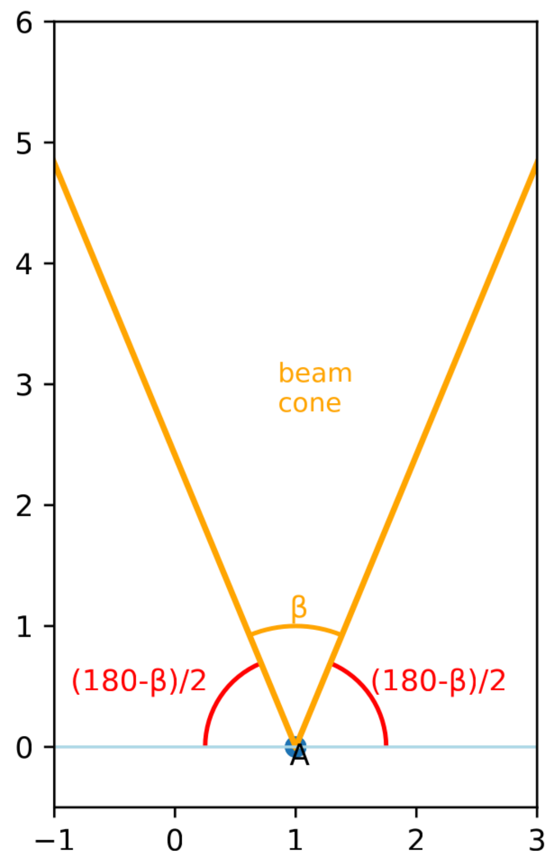


Figure 24. *One Ultrasonic Sensor Flush With X-Axis*

As already established, the two values we are looking for are the maximum blind distance in the middle region of *SOLARIS's* blind spot and the angle at which *SOLARIS* will be unable to see on its left and right sides, which we will call α. Figure 24 can be used to determine this angle. Using angle-sum rules, we know the three angles shown in the figure must add up to 180. Since the beam cone is centered, the angles on both sides are equal. In the case where the offset is 0:

$$\alpha = \frac{(180-\beta)}{2}$$

If we add an offset angle to the left sensor in this case, the angle on the left side increases by that amount, while the angle on the right side decreases by the same amount. Therefore, when we account for an offset, the angle at which *SOLARIS* will be unable to see becomes:

$$\alpha = \frac{(180-\beta)}{2} - offset$$

Now we can create an equation to determine the maximum blind distance in the middle of the robot. Looking back at Figure 23, the central blind-spot region can be split into two right triangles. The angle formed on the inside of the beam cones (not at the intersection) will be called δ. This angle can be solved in the same way as α, but in this case it increases with the offset. Therefore, the equation becomes:

$$\delta \;=\; \frac{(180-\beta)}{2} \;+\; offset$$

Before continuing, we should also define the distance between the sensors, which we will call d. We expect this distance to be 2 feet, but for the purpose of deriving the equation, we will keep it as d in case the spacing changes later. We now have everything needed to form an equation for the maximum blind-spot distance. Since the interior region forms a right triangle and we know the relevant angle while the distance between the sensors is known, we can use the following equation:

$$tan(\delta) \;=\; \frac{MaxBlindDistance}{(d/2)}$$

Thus max blind distance is:

$$MaxBlindDistance \;=\; \frac{2}{d} * tan(\delta)$$

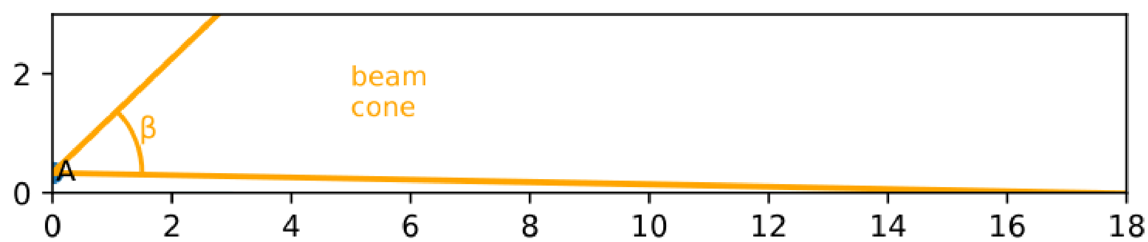


Figure 26. *Ultrasonic Sensor Beam vs Ground*

The last consideration is that the beam cone can reach the ground and create a false reading, making *SOLARIS* think there is an obstacle ahead when it is actually detecting the ground. This can be mitigated by adding a vertical (y-axis) tilt so the beam cone does not intersect the ground until a chosen distance. A reasonable target is around 5 to 6 meters (16 to 20 feet), since many ultrasonic sensors reach their effective limit around that range anyway. An example of this is shown in Figure 26, where the sensor is tilted upward so the lower edge of the cone does not hit the ground until approximately the desired distance.

One more consideration is how high off the ground the sensor should be mounted. For the figure, we chose a height of about 4 inches (or 1/3 of a foot). This height was selected because the assumption is that objects below that level are likely small enough for *SOLARIS* to drive over without issue, such as a rock, a slight incline, or a branch. This choice will likely be adjusted during testing as we determine what works best in real conditions. For now, however, assuming a sensor height of 4 inches above the ground is a reasonable starting point.

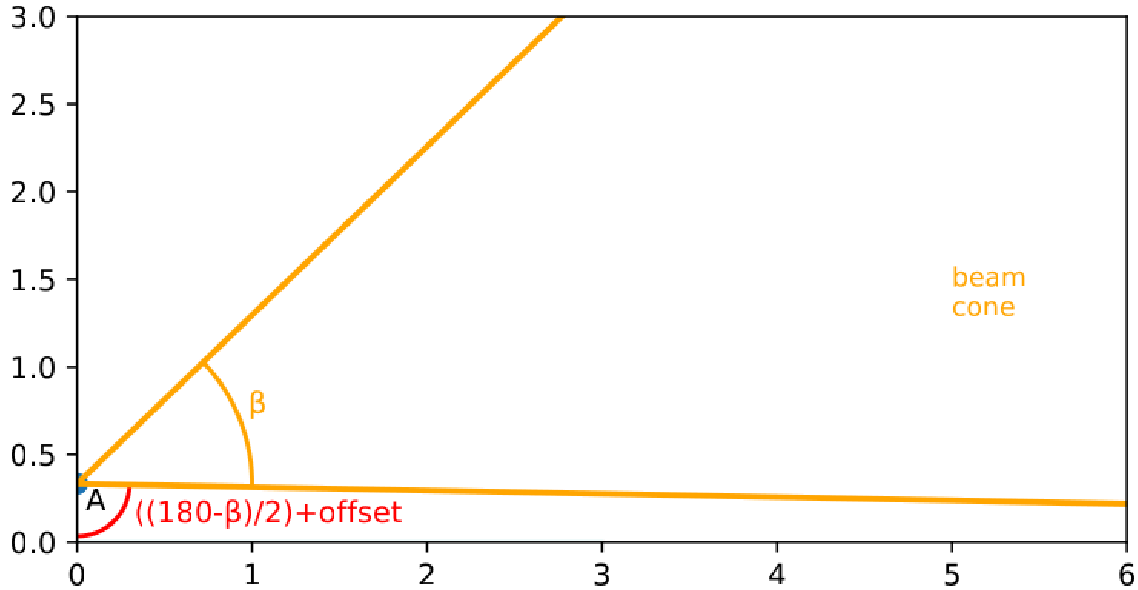


Figure 27. *Ultrasonic Sensor Angle Pointing Towards the Ground*

We now have everything needed to determine what offset we should use, given that our sensor height is 1/3 of a foot and a selected beam angle. Although it is difficult to see in the figures, the portion of the beam closest to the ground forms a right triangle with the ground surface. Similar to how α was derived using Figure 27, we can determine the angle the beam makes within this right triangle. This angle is shown as already derived in Figure 26, and we will call it θ. θ is:

$$\theta = \frac{(180-\beta)}{2} + offset$$

We can again use the properties of right triangles to develop an equation that solves for the desired offset. Let the target distance at which the beam first touches the ground be 5 meters, or 16.4 feet.

$$tan(\theta) = \frac{16.4}{(1/3)}$$

$$tan(\frac{(180-\beta)}{2} + offset) = \frac{16.4}{(1/3)}$$

$$\frac{(180-\beta)}{2} + offset = arctan(\frac{16.4}{(1/3)})$$

$$offset = arctan(\frac{16.4}{(1/3)}) - \frac{(180-\beta)}{2}$$

Now that we have finished deriving the calculations we need, we can compare our constraints and determine how the sensors should be positioned, including whether offsets are necessary, for ultrasonic sensors with different beam angles. This comparison will help us narrow down which sensor is the best fit for *SOLARIS*.

Table 26. *Beam Angle and Offset Calculation Comparison*
Assuming sensors are 2 feet apart, ⅓ foot above ground, and aiming for distance of beam touching ground to be 5m or 16.4ft

	x_offset = 0°	***x_offset = 5°***	***x_offset = 10°***	***x_offset = 15°***	***x_offset = 20°***
β = 15°	α = 82.5° MaxBlindDist = 7.6' y_offset = 6.3°	α = 77.5° MaxBlindDist = 22.9' y_offset = 6.3°	α = 72.5° MaxBlindDist = und. y_offset = 6.3°	α = 67.5° MaxBlindDist = und. y_offset = 6.3°	α = 62.5° MaxBlindDist = und. y_offset = 6.3°
β = 20°	α = 80° MaxBlindDist = 5.7' y_offset = 8.8°	α = 75° MaxBlindDist = 11.4' y_offset = 8.8°	α = 70° MaxBlindDist = und. y_offset = 8.8°	α = 65° MaxBlindDist = und. y_offset = 8.8°	α = 60° MaxBlindDist = und. y_offset = 8.8°
β = 25°	α = 77.5° MaxBlindDist = 4.5' y_offset = 11.3°	α = 72.5° MaxBlindDist = 7.6' y_offset = 11.3°	α = 67.5° MaxBlindDist = 22.9' y_offset = 11.3°	α = 62.5° MaxBlindDist = und. y_offset = 11.3°	α = 57.5° MaxBlindDist = und. y_offset = 11.3°
β = 30°	α = 75° MaxBlindDist = 3.7' y_offset = 13.8°	α = 70° MaxBlindDist = 5.7' y_offset = 13.8°	α = 65° MaxBlindDist = 11.4' y_offset = 13.8°	α = 60° MaxBlindDist = und. y_offset = 13.8°	α = 55° MaxBlindDist = und. y_offset = 13.8°
β = 35°	α = 72.5° MaxBlindDist = 3.2' y_offset = 16.3°	α = 67.5° MaxBlindDist = 4.5' y_offset = 16.3°	α = 62.5° MaxBlindDist = 7.6' y_offset = 16.3°	α = 57.5° MaxBlindDist = 22.9' y_offset = 16.3°	α = 52.5° MaxBlindDist = und. y_offset = 16.3°
β = 40°	α = 70°	α = 65°	α = 60°	α = 55°	α = 50°

	MaxBlindDist = 2.7 y_offset = 18.8°	MaxBlindDist = 3.7’ y_offset = 18.8°	MaxBlindDist = 5.7’ y_offset = 18.8°	MaxBlindDist = 11.4’ y_offset = 18.8°	MaxBlindDist = und. y_offset = 18.8°
β = 45°	α = 67.5° MaxBlindDist = 2.4’ y_offset = 21.3°	α = 62.5° MaxBlindDist = 3.1’ y_offset = 21.3°	α = 57.5° MaxBlindDist = 4.5’ y_offset = 21.3°	α = 52.5° MaxBlindDist = 7.6’ y_offset = 21.3°	α = 47.5° MaxBlindDist = 22.9’ y_offset = 21.3°
β = 60°	α = 60° MaxBlindDist = 1.7’ y_offset = 28.8°	α = 55° MaxBlindDist = 2.1’ y_offset = 28.8°	α = 50° MaxBlindDist = 2.7’ y_offset = 28.8°	α = 35° MaxBlindDist = 3.7’ y_offset = 28.8°	α = 40° MaxBlindDist = 5.67’ y_offset = 28.8°
β = 75°	α = 52.5° MaxBlindDist = 1.3’ y_offset = 36.3°	α = 47.5° MaxBlindDist = 1.5’ y_offset = 36.3°	α = 42.5° MaxBlindDist = 1.9’ y_offset = 36.3°	α = 37.5° MaxBlindDist = 2.4’ y_offset = 36.3°	α = 32.5° MaxBlindDist = 3.17’ y_offset = 36.3°

A beam angle of 75° may initially appear to be the best option since it produces the smallest α and minimizes the maximum blind distance. However, a wider beam is not always the most practical choice. As beam angle increases, range and precision typically decrease, and the likelihood of false readings rises due to reflections from unintended objects within the sensing cone.

On the other hand, a very narrow beam provides improved range, higher precision, and fewer false detections. The tradeoff is reduced environmental awareness, as the sensor covers a smaller portion of the robot’s surroundings. If the beam angle is too small, the robot may lack sufficient spatial information for reliable obstacle detection.

For *SOLARIS*, a balanced approach is preferred. The design goal is to keep the maximum blind distance near five feet while ensuring that α remains less than or equal to 60°. Based on this analysis, the most effective compromise lies in the mid-range beam angles. The selected configuration is an ultrasonic sensor with a beam angle between 40° and 45°, paired with an x_offset of 10°. This combination provides a strong balance between coverage, precision, and detection reliability, making it well-suited for the system’s design requirements.

3.1.14.3 Ultrasonic Sensor Comparisons

Table 27. *Ultrasonic Comparison Table*

	Operating Voltage	*Typical Current Draw*	*Beam Angle*	*Range*	*Minimum Time Between Readings*	*Communication Interface*	*Cost per unit*
MB1020-000	**2.5-5.5 V**	**2mA**	**42°**	**.15-6.5 m**	**50ms**	**UART + ADC**	**$19.95**
MB1232-000	**3-5.5 V**	**Not Specified**	**44°**	**.2-7.7 m**	**25ms**	**I2C**	**$37.44**
MB1013-000	**2.5-5.5 V**	**2.5-3.1 mA**	**44°**	**.3-5m**	**100ms**	**UART + ADC**	**$31.19**

Sources: *[56]-[58]*

3.1.14.3.1 Note on Limited Selection

For *SOLARIS*, the ultrasonic sensor requirements of a 3–5 V operating voltage and a 40–45° beam angle significantly narrowed the available options. Many sensors that met these specifications were priced at over $100 per unit, which is not practical for a four-sensor configuration. After evaluating the available market options, the only reasonably priced sensors that satisfied both the electrical and beam angle constraints were manufactured by MaxBotix®.

MaxBotix offers multiple sensor models, many of which belong to the same product families with similar electrical characteristics and performance specifications. While the company sells additional variants, several were either out of stock or did not align with our exact design requirements. Therefore, the sensors included in the comparison table were selected to represent their respective families while also being currently available for purchase. This approach ensures that the comparison remains both technically relevant and practically feasible for implementation in *SOLARIS*.

3.1.14.3.2 MB1020-000

One ultrasonic sensor under consideration for *SOLARIS* is the MB1020-000. This model operates within the required 3–5 V range and provides a beam angle that aligns with our 40–45° design constraint, making it electrically and mechanically compatible with the system architecture.

The sensor offers sufficient range for *SOLARIS*' expected operating speed and obstacle detection requirements. Its current draw is lower than many other commercially available sensors that meet similar beam and voltage specifications, which supports the platform's low-power, solar-driven design goals. Cost is also competitive, which is important given that multiple units will be required.

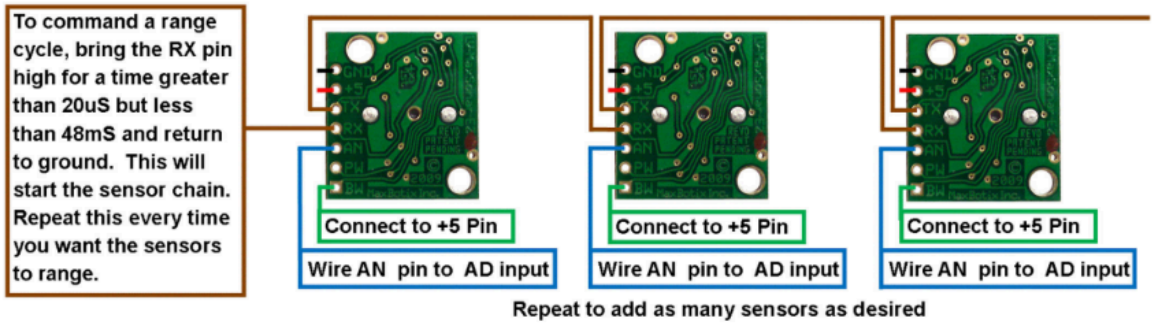


Figure 28. *MB1020-000 Sensor Chaining Diagram [56]*

An additional advantage is the documented sensor chaining capability. The ability to configure sensors to trigger sequentially helps mitigate cross-talk without requiring complex timing logic in firmware. This simplifies integration in a multi-sensor configuration.

Potential drawbacks include a lower maximum range and a slower update rate compared to some other ultrasonic sensors.

3.1.14.3.3 MB1232-000

Another ultrasonic sensor under consideration for *SOLARIS* is the MB1232-000. This model operates within the required 3–5 V range and is electrically compatible with the system architecture.

Among the sensors being compared, the MB1232-000 provides the highest maximum range and the fastest time between readings. This could allow for more frequent environmental updates and improved responsiveness in autonomous navigation.

The sensor uses an I2C interface. One consideration to make is that using multiple identical I2C sensors introduces potential addressing limitations. If the devices share a fixed address or limited configurable address range, additional hardware solutions such as I2C multiplexers may be required. This adds design complexity compared to sensors that can be independently triggered and read without bus conflicts.

Cost is another significant factor. The MB1232-000 is the most expensive option among the sensors being evaluated. Because *SOLARIS* requires multiple units, this increases total system cost substantially.

The primary tradeoff with this sensor is improved range and update speed versus higher cost and potential integration complexity related to I2C multi-device communication. Final suitability will depend on whether the performance benefits justify the added expense and system complexity.

3.1.14.3.3 MB1013-000

The MB1013-000 is also being evaluated as a potential ultrasonic sensor for *SOLARIS*. It meets the required 3–5 V operating range, making it compatible with the existing power design.

In comparison to the other sensors under review, this model has the shortest maximum range and the longest time between readings. This could reduce system responsiveness in situations where faster environmental updates are needed. Its current draw is also higher than the MB1020-000, which is a relevant consideration for a solar-powered platform where minimizing power consumption is important.

The pin configuration differs from the other candidates, which may require adjustments in PCB layout or firmware handling. However, like the MB1020-000, it supports sensor chaining, allowing multiple units to be triggered sequentially to reduce cross-talk without complex timing logic.

From a cost standpoint, the MB1013-000 is less expensive than the MB1232-000 but still significantly more expensive than the MB1020-000. Since multiple sensors are required for *SOLARIS*, this cost difference scales at the system level.

Overall, this model presents a tradeoff between cost, power consumption, range, and update speed when compared to the other options being considered.

3.1.14.4 Ultrasonic Sensor Selection

MB1020-000: The MB1020-000 was selected as the ultrasonic sensor for *SOLARIS*. Its support for sensor chaining, lowest current draw among the evaluated options, and significantly lower cost made it the most practical and well-balanced choice for a multi-sensor implementation.

While its maximum range is lower than that of the MB1232-000, it comfortably exceeds our design requirement of a minimum 5 m maximum detection distance. The reduced range does not limit system performance within the intended operating environment.

The MB1020-000 also has a slower minimum time between readings compared to the MB1232-000. For high-speed robotic systems, this could be a critical limitation due to increased latency. However, *SOLARIS* is designed as a slow-moving platform, meaning the additional latency does not meaningfully impact navigation performance.

When balancing performance, power consumption, cost, and system complexity, the MB1020-000 provides the strongest overall fit for *SOLARIS*.

3.1.15 Chassis Technology Comparison

Table 28. *Chassis Technology Comparison*

Chassis Technology	*Strength-to-Weight*	*Pros*	*Cons*	*Durability*	*Cost*
3D Printed Filament	**Moderate**	**Fast iteration, Extremely low cost, Complex geometries easily achievable, Lightweight**	**Weak in structural applications, Heat sensitivity**	**Poor(UV+ heat degrade)**	**Low**
Aluminum	**High**	**Strong and predictable behavior, Excellent for outdoor robotics, Ductile**	**Heavier than composites, Requires tools**	**Excellent**	**Moderate**
Carbon Fiber	**Very high**	**Best strength-to-weight ratio, Extremely stiff, Professional-grade performance**	**Expensive, Brittle failure, Hard to machine safely, Difficult to repair**	**Excellent**	**High**

Sources: *[109]-[111]*

The mechanical design of *SOLARIS* requires a structured approach to selecting an appropriate chassis fabrication method to house all system components. A key parameter in this trade study is strength-to-weight ratio, as higher values enable a structurally robust design while minimizing overall mass. Reducing system weight is critical for improving energy efficiency and extending operational runtime.

Another important consideration is outdoor durability. Since *SOLARIS* is intended for outdoor operation, materials must withstand environmental exposure, particularly UV radiation, which can degrade certain polymers over time and lead to premature failure. Therefore, material selection must account for long-term reliability in sun-exposed conditions.

Finally, cost is a significant factor, especially during the prototyping phase. Multiple design iterations are expected, and material and manufacturing costs can accumulate quickly. Selecting cost-effective materials and fabrication methods allows for rapid iteration without excessive resource expenditure, enabling a more efficient development cycle.

3.1.15.1 3D Printed Filament

3D printed filament is well-suited for rapid prototyping of a robot chassis due to its low cost, fast turnaround, and ability to produce complex geometries without specialized tooling. It enables tight integration of features such as mounting bosses, wire routing channels, and sensor housings directly into the structure. However, its structural performance is limited by relatively low strength and stiffness, as well as anisotropic properties caused by layer-by-layer fabrication. Additionally, common filaments are susceptible to UV degradation and thermal softening, which can compromise reliability in outdoor environments. As a result, filament-based construction is best used for non-load-bearing components.

3.1.15.2 Aluminum

Aluminum is a highly practical and widely used material for small robot chassis due to its strong balance of strength, weight, manufacturability, and durability. For a robot chassis, aluminum sheet or extrusion can provide a rigid and robust frame capable of supporting motors, batteries, and electronics without significant deflection. It is well-suited for outdoor use, offering good corrosion resistance and stable mechanical properties across a wide temperature range. Fabrication requires basic machining processes such as cutting, drilling, and fastening, which are accessible but more involved than 3D printing. Overall, aluminum offers a reliable and repeatable solution for a functional, field-ready chassis.

3.1.15.3 Carbon Fiber

Carbon fiber provides an exceptional strength-to-weight ratio and very high stiffness, making it attractive for applications where minimizing mass is critical. In a small robot chassis, carbon fiber plates or tubes can produce a lightweight yet rigid structure that enhances dynamic performance and efficiency. However, the material introduces challenges in fabrication, including difficulty in machining, the need for specialized tools, and safety concerns related to carbon fiber dust. Additionally, carbon fiber exhibits a brittle failure mode and is more difficult to repair compared to metals. Its higher cost and manufacturing complexity typically limit its use to performance-critical designs rather than general-purpose prototyping.

3.1.15.3 Chassis Technology Selection

3D Printed Filament and Aluminum: Aluminum was selected as the primary material for load-bearing and structural components due to its strong balance of mechanical performance and reliability. For a robot of this scale, the chassis must support the weight of batteries, electronics, and drive systems while maintaining rigidity under dynamic loads such as movement over uneven terrain. Aluminum provides high strength and stiffness with predictable, ductile behavior, meaning it will deform rather than fail catastrophically under stress. It also offers excellent durability for outdoor environments, with good resistance to corrosion and temperature variation. These characteristics make aluminum a dependable choice for the core structural frame where integrity and long-term performance are critical

In addition to aluminum, 3D printed filament is incorporated to enable rapid iteration and customization of non-load-bearing components. Additive manufacturing allows for quick design changes and fabrication of complex geometries such as mounts, enclosures, and cable management features without the need for specialized tooling. This is especially valuable during development, where multiple design revisions are expected. Filament-based parts are also lightweight, which helps minimize the overall system mass without compromising the primary structural integrity provided by the aluminum frame. By combining aluminum for strength and 3D printing for flexibility, the design achieves both robustness and adaptability.

3.1.16 Chassis Part Comparison

Table 29. *Chassis Part Comparison*

Chassis Part	*Density (g/cm^3)*	*Yield/Tensile Strength (MPa)*	*UV Resistance*	*Cost*
ASA Filament	**1.07**	**~45**	**Excellent**	**Low**
PETG Filament	**1.27**	**~50**	**Moderate**	**Low**
ABS Filament	**1.04**	**~40**	**Poor**	**Moderate**
PC Filament	**1.20**	**~65**	**Moderate**	**High**
5052-H32 Aluminum	**2.68**	**~193**	**Excellent**	**Low-Moderate**
6061-T6 Aluminum	**2.70**	**~255**	**Excellent**	**Moderate**
7075-T6 Aluminum	**2.81**	**~450**	**Excellent**	**High**

Sources: *[109],[112]-[114]*

The next step is selecting the specific 3D printing filament and aluminum alloy for use in *SOLARIS*. The 3D printed material will primarily serve in non-load-bearing roles such as component mounting, enclosures, and wire routing, where geometric flexibility and rapid fabrication are advantageous. In contrast, aluminum will be used for the primary chassis framework and base plates, where structural integrity and stiffness are critical to supporting system loads and maintaining alignment.

Several key material properties are considered in this selection. Density provides a measure of mass per unit volume and directly impacts the overall weight of the robot, which influences energy efficiency and mobility. Tensile strength refers to the maximum tensile (pulling) stress a material can withstand before failure, typically measured in units of pressure such as MPa. It characterizes how well a material resists being stretched or pulled apart under load, which is especially important for structural components subjected to forces during operation. Lastly, UV resistance is a critical factor for outdoor deployment, as prolonged exposure to sunlight can degrade certain materials (especially polymers) leading to embrittlement, discoloration, and reduced mechanical performance over time.

3.1.16.1 ASA Filament

ASA (Acrylonitrile Styrene Acrylate) is specifically engineered for high-performance outdoor applications due to its exceptional UV stability and weather resistance. Unlike its relative ABS, ASA does not yellow or become brittle when exposed to prolonged sunlight, making it the ideal 3D printing filament for solar-integrated projects like *SOLARIS*. It offers high mechanical strength and a high glass transition temperature of approximately 95–100°C, ensuring that external sensor mounts and solar panel clips maintain their structural integrity even in intense thermal environments.

3.1.16.2 PETG Filament

PETG (Polyethylene Terephthalate Glycol) serves as a versatile "all-rounder" filament that combines the ease of printing found in PLA with the functional durability of ABS. It is characterized by its excellent layer adhesion and low shrinkage, which allows for the production of dimensionally accurate parts without the warping issues common in other technical polymers. While it is moderately resistant to chemicals and impacts, its relatively lower heat deflection temperature (approx. 70–80°C) means it is best suited for internal structural components or electronics housings that are shielded from direct solar heating.

3.1.16.3 ABS Filament

ABS (Acrylonitrile Butadiene Styrene) is a legacy engineering plastic valued for its toughness and high heat resistance. It is highly impact-resistant and has a "ductile" failure mode, meaning it tends to bend or deform before snapping, which provides a safety margin for robotic chassis components. However, ABS is notoriously difficult to print without a heated enclosure due to its high thermal expansion coefficient, and its poor UV resistance makes it unsuitable for long-term outdoor use unless it is painted or treated with a protective coating.

3.1.16.4 PC Filament

Polycarbonate (PC) is the premier choice for 3D printed parts requiring extreme impact resistance and high-temperature tolerance. Often used in industrial applications where components must withstand significant mechanical stress, PC filament can handle operating temperatures up to 110–140°C. In a hybrid robot chassis, it is best utilized for high-torque motor housings or protective battery enclosures, as its superior tensile strength (60–72 MPa) allows it to survive collisions that would shatter standard filaments.

3.1.16.5 5052-H32 Aluminum

5052-H32 Aluminum is an alloy primarily defined by its excellent formability and high resistance to corrosion, particularly in marine or outdoor environments. As a non-heat-treatable alloy, the "H32" temper indicates that it has been strain-hardened and stabilized for a balance of strength and ductility. Its most significant feature for robotics is its ability to be bent at 90-degree angles with a small radius without cracking, making it the standard material for fabricating "U-channel" frames, base plates, and folded sheet-metal enclosures.

3.1.16.6 6061-T6 Aluminum

6061-T6 Aluminum is the most widely used structural aluminum alloy due to its excellent strength-to-weight ratio and superb machinability. The "T6" temper is achieved through solution heat-treating and artificial aging, resulting in a yield strength that is significantly higher than the 5000-series alloys. Because it can be easily drilled, tapped, and CNC-machined with high precision, it is the industry standard for 2020 T-slot extrusions and the primary load-bearing "skeleton" of most mobile robotic platforms.

3.1.16.7 7075-T6 Aluminum

7075-T6 Aluminum is a high-strength "aerospace-grade" alloy that utilizes zinc as its primary alloying element. It offers a yield strength that rivals many steels while maintaining the low density of aluminum, providing the highest specific strength of the materials considered. While it is more expensive and has lower corrosion resistance than the 6000-series, its extreme hardness makes it indispensable for critical, high-wear components such as drive axles, custom wheel hubs, and high-reduction gearbox plates where mechanical failure is not an option.

3.1.16.5 Chassis Part Selection

PETG Filament and 6061-T6 Aluminum: PETG filament was selected as the preferred 3D printing material due to its balanced mechanical properties and improved environmental resistance compared to other common filaments. It offers greater toughness and layer adhesion than ASA, reducing the likelihood of brittle failure in functional parts such as mounts and enclosures. Additionally, PETG has better thermal stability, making it less prone to softening under elevated temperatures that may occur during outdoor operation. Its enhanced UV and moisture resistance relative to many other consumer filaments also supports longer service life in sun-exposed environments. These characteristics make PETG a reliable choice for non-load-bearing components that still require durability and repeated handling during prototyping and operation.

6061-T6 aluminum was selected for the primary structural components due to its strong combination of strength, weight efficiency, and manufacturability. This alloy provides high tensile strength and good stiffness, allowing the chassis to maintain structural integrity under mechanical loads while minimizing unnecessary mass. It is also widely available and easy to machine, making it practical for fabrication processes such as cutting, drilling, and fastening. In addition, 6061-T6 exhibits excellent corrosion resistance, which is important for outdoor use where exposure to moisture and environmental conditions is expected. Overall, it offers a well-proven, cost-effective solution for building a rigid and durable chassis framework.

3.2 Programming Languages

3.2.1 High-Level Languages

3.2.1.1 C/C++

C and C++ were considered for *SOLARIS* because they are both widely used in embedded systems and provide the level of hardware access needed for a robotics project. In particular, these languages allow the software to interact closely with the microcontroller, including configuring peripherals, handling real-time I/O, and managing memory in a predictable way. This matters for *SOLARIS* because the system needs reliable low-level control for sensors, motor drivers, and communication while still being maintainable over a long development cycle.

C is the primary language considered for this project due to its simplicity, efficiency, and strong compatibility with embedded toolchains. Most microcontroller vendors provide their best support, examples, and documentation in C, and many low-level driver libraries are designed with C as the default. C is also highly predictable in terms of performance and memory usage, which is important for a system that may need to run continuously and operate under power constraints.

C++ was also considered because some embedded libraries and frameworks provide better support or cleaner interfaces when using C++. In addition, C++ can improve code organization through features like stronger type checking and better modular structure. However, since *SOLARIS* does not require complex object-oriented design, and because simplicity is a major advantage in embedded development, C remains the preferred option for most of the system. Overall, C/C++ provides an ideal balance of control, performance, and compatibility, with C serving as the primary language and C++ used only when it offers clear practical benefits.

3.2.1.2 Python

Python is a strong language to consider for *SOLARIS* for a few key reasons. First, it is simply an easy language to work with. Python is high-level, dynamically typed, and supports object-oriented programming, which makes development fast and helps keep code organized as the project grows. This makes it especially useful for higher-level software tasks where readability and development speed matter more than raw execution speed.

Second, Python has extremely strong library support across a wide range of tools. Even though we are not currently planning on using computer vision or advanced AI for this project, it is still worth considering as a future option if our initial approach is not robust enough. Python has one of the best ecosystems for intelligent systems, including libraries for computer vision, machine learning, and sensor data processing. It is also very useful for analyzing and visualizing collected data, which could be helpful if we want to evaluate performance metrics like energy gained, energy used, distance traveled, and overall system efficiency.

Lastly, Python is attractive because of its usefulness outside of the embedded side of the project. It is widely used for backend development and automation, and it can support tasks like building APIs, handling stored data, and integrating with a phone application or website. For *SOLARIS*, the performance tradeoffs of Python are negligible for these higher-level tasks, since real-time motor control and low-level peripheral access would still be handled directly on the microcontroller in a lower-level language such as C. Familiarity also plays a major role, since the entire team is comfortable with Python, and some members already have experience using it in backend web development frameworks.

3.2.1.3 Java

Java is another language that could be considered for *SOLARIS*, mostly due to how structured and portable it is. Java is a strongly object-oriented and class-based language, which can be useful for keeping a large project organized and scalable. For team-based development, Java's design naturally encourages cleaner code separation, reuse, and easier collaboration since different components can be developed as separate classes and modules.

However, Java is not a strong fit for the low-level embedded side of this project. Since *SOLARIS* requires direct interaction with microcontroller peripherals, real-time motor control, and hardware-level I/O, languages like C tend to be far more common and better supported. Java's runtime environment and memory overhead also make it less practical for most microcontroller targets.

Where Java becomes more relevant is on the software side outside of the microcontroller. Java is commonly used for backend development and is also the primary language historically used for Android app development. If *SOLARIS* ever required a native Android application or a backend service that manages stored data, system metrics, or communication between devices, Java could support those parts of the project well. Overall, Java is a strong general-purpose option, but for *SOLARIS* it is more applicable to application-level software than embedded firmware.

3.2.1.4 Javascript/Typescript

JavaScript and TypeScript are another strong option for *SOLARIS*, and they are grouped together since they are used in the same development ecosystem and typically share the same frameworks and tooling. The main difference is that TypeScript is essentially JavaScript with static typing added, which can help reduce bugs and improve maintainability in larger projects. In practice, both languages are commonly used for the same types of applications, especially when working with modern frontend frameworks.

For *SOLARIS*, the main reason to consider JavaScript and TypeScript is for the phone application. These languages are widely used with React Native, which provides a cross-platform development approach that supports both iOS and Android. This is a major advantage since it allows one shared codebase instead of needing to develop separate native applications for each platform. For a senior design project with limited time and resources, that cross-platform support can significantly reduce development overhead. React Native also has strong library support for Bluetooth Low Energy communication, which is important since BLE is a realistic option for connecting the phone application to the robot.

Another benefit is familiarity. While the team may not have direct experience with React Native specifically, React Native is based on the same general React development model used in web development. Since React and React-based frameworks are already familiar from web projects, it makes the learning curve more manageable compared to starting from a completely new mobile development environment. Overall, JavaScript and TypeScript are a practical option for building a mobile interface for *SOLARIS* due to strong ecosystem support, cross-platform compatibility, and existing team familiarity.

3.2.1.5 Swift

Swift is another language considered for the *SOLARIS* phone application. Swift is Apple's primary language for iOS development and is commonly used for building native iPhone applications. One major advantage of Swift is that it provides strong performance, direct access to iOS features, and reliable support for Bluetooth Low Energy communication.

However, the main drawback is that Swift is limited to Apple devices. Since *SOLARIS* is intended to support both iOS and Android users, using Swift would require a separate Android application to be developed in a different language, which would increase development time and complexity. For this reason, Swift is less practical as the primary language for the mobile application, even though it is a strong option for iOS-only development.

3.2.1.6 Dart

Dart is another language considered for the *SOLARIS* phone application, mainly because it is used with the Flutter framework. Flutter is a cross-platform mobile development tool that supports both iOS and Android from a single codebase, similar to React Native. This makes Dart a practical option for a senior design project since it reduces the need to maintain separate mobile applications.

Dart and Flutter also have strong support for building modern user interfaces and can support Bluetooth Low Energy communication through existing libraries. However, one downside is that Dart is less commonly used than JavaScript and TypeScript, and it is less familiar to the team. Because of this, Dart is still a valid option, but it would likely have a steeper learning curve compared to React Native development.

3.2.1.7 Code Selection

C/C++: C/C++ will be used for the embedded firmware running directly on the microcontroller. This portion of the system requires direct hardware access, predictable timing, and efficient power usage. C is expected to be the primary language for the embedded side due to its strong vendor support, large ecosystem of driver libraries, and predictable performance. C++ may be used in specific cases where libraries provide better support or cleaner integration, but the firmware will remain mostly C to keep the embedded code lightweight and reliable.

Python: Python will be used for backend development through the FastAPI framework (as further discussed in Section 3.3.3). This allows the system to expose a clean API for storing and retrieving telemetry, robot status, and logged data. Python is a strong fit for this role because it supports rapid development, has excellent library support, and integrates easily with database tools such as PostgreSQL libraries. This choice also aligns well with the team's existing familiarity with Python.

JavaScript/TypeScript: JavaScript and TypeScript will be used for the mobile application frontend through React Native (as further discussed in Section 3.3.2). This provides a practical cross-platform solution for supporting both iOS and Android users from a single codebase. TypeScript also improves maintainability through static typing, which is useful as the application grows. React Native has strong ecosystem support for BLE communication, making it a good match for *SOLARIS's* wireless protocol selection.

3.2.2 Algorithms

3.2.2.1 Reactive Obstacle Avoidance

A reactive obstacle avoidance algorithm is a navigational strategy for mobile robots that relies on local sensor data to detect and avoid collisions in real time. Rather than depending on a global system with a complete map of the environment, a reactive system makes quick decisions based on immediate sensor input, allowing it to function effectively in dynamic environments. For sensory input, reactive systems most commonly use range sensors such as ultrasonic sensors and depth sensors [89]. Ultrasonic sensors measure distance by emitting sound signals, while depth sensors use infrared lasers to generate a 3D representation of the environment. To implement this type of algorithm, two low-level AI systems are traditionally used: a Fuzzy Logic Inference System (FIS) and an Artificial Neural Network (ANN). A FIS uses defined "IF-THEN" rules to handle uncertainty and ambiguity, making soft and gradual adjustments based on sensor features [89]. An ANN is more straightforward in practice, as it can be trained via supervised learning to associate sensor inputs with specific motor actions [89]. Reactive algorithms like these allow robots to learn from experience and adapt to new situations without requiring large, rigid programs for every possible scenario. Since they do not require prior knowledge of the environment, these models are highly flexible and capable of semi-autonomous operation.

3.2.2.2 Kalman Filters

Table 30. *Kalman Filter Reference Terms*

Variable	*Description*	*Use Case*	*Input/Output*
x	State variable	n x 1 column vector	Output
P	State covariance matrix	n x n matrix	Output
z	measurement	m x 1 column vector	Input
A	State transition matrix	n x n matrix	System Model
H	State-to-measurement matrix	m x n matrix	System Model
R	Measurement covariance matrix	m x m matrix	Input
Q	Process noise covariance matrix	n x n matrix	System Model
K	Kalman Gain	n x m	Internal

Source: [88]

In applications that make use of inertial measurement units, a Kalman Filter is a common approach for combining multiple degrees of freedom through sensor fusion. The Kalman Filter works by estimating the true state of a system from two noisy sources of information. More specifically, it operates on the principle of Bayesian inference, where a probability distribution is used to represent uncertainty in both predictions and measurements [87]. The filter maintains two key pieces of information: the state estimate and the uncertainty of that estimate. Through a recursive process, the filter begins with a prediction step where the current state is estimated based on the previous state estimate [87]. An update step then follows, incorporating new sensor measurements to refine the prediction and produce an optimal estimate that balances the prediction against the reliability of the incoming data.

3.2.2.3 Solar Position

The Solar Positioning Algorithm (SPA) is a step-by-step procedure for tracking solar radiation defined by the National Renewable Energy Laboratory (NREL) [91]. The primary goal of the SPA is to calculate the solar zenith and azimuth angles based on a given location and time [91]. The algorithm operates in five steps: time scale conversion, earth position, solar coordinates, observer-specific data, and the final output of angles.

The first step calculates variables based on Universal Time (UT) and Terrestrial Time (TT) to convert raw time data into a usable format [91]. Earth position is then calculated by taking the heliocentric longitude, latitude, and radius vector into account [91]. Solar coordinates are calculated in the same manner, followed by observer-specific data where sun ascension, declination, and local hour angle are determined based on the observer's elevation and location [91]. Finally, a topocentric zenith and azimuth angle are output, which can be used in a control system to define motor behavior [91]. This algorithm has been shown to produce results within an uncertainty range of only ±0.0003°, making it one of the most precise solar positioning methods available [91].

3.3 Frameworks

3.3.1 What are Frameworks?

Before selecting frameworks for *SOLARIS*, it is important to understand the difference between the main software layers in the system and why multiple frameworks may be needed. *SOLARIS* includes both an embedded system running directly on the robot and an application layer that allows the user to monitor and control the robot. Because these parts of the system have very different requirements, they often rely on different development tools and frameworks.

The application layer is split into frontend and backend development. The frontend refers to the part of the system the user directly interacts with, such as the mobile app screens, buttons, and overall user interface. For *SOLARIS*, the frontend is also responsible for direct communication with the robot, such as connecting over Bluetooth Low Energy and sending or receiving live data. The backend refers to software hosted separately from the phone, typically on a server. The backend is responsible for storing and organizing collected information, processing requests from the app, and providing a consistent way to access stored data later. In the context of *SOLARIS*, the backend could store robot logs, historical energy metrics, usage data, and other telemetry that the phone application collects. This is important because the phone application is not intended to permanently store large amounts of data, and backend storage also makes it easier to access data from other devices such as a website, which is a stretch goal of the project.

In addition to the phone application, *SOLARIS* also requires embedded development on the robot itself. This includes the firmware running on the microcontroller, which is responsible for tasks such as reading sensors, controlling motors, managing power, handling wireless communication, and logging data. Embedded development frameworks are important because they provide the build system, drivers, libraries, and debugging tools needed to develop reliable firmware efficiently.

A framework is a set of tools and built-in structure that helps developers build software faster and more consistently. Frameworks reduce the need to write everything from scratch by providing common features such as project organization, communication libraries, hardware drivers, and platform-specific support. For *SOLARIS*, choosing the right frameworks is important because it affects development time, cross-platform compatibility, available libraries, and how easily the system can be developed and maintained.

3.3.2 Frontend Phone App Frameworks

Table 31. *Frontend Framework Comparison*

	React Native	***SwiftUI***	***Flutter***
Android & iOS Support	**Yes**	**No: iOS Only**	**Yes**
Language	**Javascript/Typescript**	**Swift**	**Dart**
BLE Support	**Yes (libraries available)**	**Yes (via CoreBluetooth)**	**Yes (via plugins)**
Team Familiarity	**Medium (react experience)**	**Low**	**Low**
Library Ecosystem	**Very Strong**	**Strong but only for iOS**	**Strong**

Sources: *[20]-[22]*

3.3.2.1 React Native

React Native is a mobile application framework developed by Meta that allows developers to build cross-platform applications for both iOS and Android using a single codebase. React Native is built around the React development model, meaning applications are designed using reusable components and structured user interfaces. Since React Native is based on JavaScript and TypeScript, it fits well with modern web development practices and has a large ecosystem of tools and libraries.

For *SOLARIS*, React Native is a strong candidate because it supports both major phone platforms while reducing development time compared to writing separate native applications. This is especially important in a senior design environment where development time is limited. Another major advantage is familiarity. While the team may not have direct experience with React Native specifically, the team does have experience with React-based web development. Since React Native follows the same general component structure and workflow, this reduces the learning curve and makes it easier to build a functional application in a short timeframe.

React Native also has reliable support for Bluetooth Low Energy communication through existing libraries. Since BLE is a practical option for communicating between the robot and the phone application, React Native provides a straightforward way to implement device scanning, connecting, and sending and receiving data. Overall, React Native is a practical framework for *SOLARIS* due to cross-platform support, strong library availability, and existing team familiarity.

3.3.2.2 SwiftUI

SwiftUI is Apple's modern user interface framework for building iOS applications using the Swift programming language. SwiftUI is designed for creating clean, responsive interfaces using a declarative development style, which makes it easier to build and maintain UI compared to older iOS approaches. Since SwiftUI is a native Apple framework, it integrates directly with iOS features and provides strong performance and stability.

For *SOLARIS*, SwiftUI is a valid framework to consider because it would allow the team to build a high quality native iPhone application. SwiftUI also has strong support for Bluetooth Low Energy communication through Apple's CoreBluetooth framework, meaning the app could reliably connect to the robot and display live data. In addition, since it is a native framework, SwiftUI would generally provide the best user experience for iOS users and would have fewer compatibility concerns compared to cross-platform frameworks.

However, the main drawback of SwiftUI is that it only supports Apple devices. Using SwiftUI would require building a separate Android application using a different language and framework if we still wanted dual compatibility, which would significantly increase development time and complexity. For a senior design project where time and resources are limited, this makes SwiftUI less practical as the primary framework, even though it is a strong option for iOS-only development.

3.3.2.3 Flutter

Flutter is a cross-platform mobile application framework developed by Google that allows developers to build applications for both iOS and Android using a single codebase. Flutter applications are primarily written in Dart, and the framework provides a large set of built-in UI components for designing modern interfaces. Flutter is known for allowing fast development through features such as hot reload, which makes it easier to quickly test and refine the application during development.

For *SOLARIS*, Flutter is a strong option because it supports both major phone platforms while reducing the need to build separate native applications. This is a major advantage for a senior design project since it lowers development time and simplifies long-term maintenance. Flutter also has support for Bluetooth Low Energy communication through existing libraries, meaning it can be used to connect to the robot and display real-time data.

One drawback of Flutter is that it introduces an additional learning curve compared to other options. Since the team is more familiar with JavaScript and React-based development, React Native may be easier to adopt than learning Dart and the Flutter ecosystem from scratch. Overall, Flutter is a strong cross-platform framework with good performance and BLE support, but its practicality depends on how quickly the team can learn and integrate it into the project timeline.

3.3.2.4 Frontend Phone App Framework Selection

React Native: React Native was selected as the mobile application framework for *SOLARIS*. React Native provides cross-platform support for both iOS and Android using a single codebase, which is a major advantage for a senior design project where development time and resources are limited. This allows the team to build and maintain one application rather than developing separate native apps for each platform.

Another major reason for selecting React Native is familiarity. React Native uses JavaScript and TypeScript, which the team is already comfortable with, and it follows the same general component-based development model as React for web development. Since the team already has experience with React-based projects, the learning curve for React Native is significantly lower compared to SwiftUI and Flutter, which would require learning either the Swift ecosystem or Dart.

React Native also provides strong library support for Bluetooth Low Energy communication, which is important since BLE is the selected wireless protocol for *SOLARIS*. This makes it straightforward to implement key features such as scanning for the robot, connecting to it, and sending and receiving control and telemetry data. Overall, React Native provides the best balance of cross-platform support, ease of development, and practical ecosystem support for the*SOLARIS* mobile application.

3.3.3 Backend Phone App Frameworks

Table 32. *Backend Framework Comparison*

	FastAPI	*Django*	*Spring Boot*	*Express.js*
Best DB framework pairing	**Postgres**	**Postgres**	**MySQL**	**MySQL**

Language	Python	Python	Java	Javascript / Typescript
Performance	High	Medium	Low to Medium	Medium
Team Familiarity	High	Medium	Low to Medium	Low to Medium
Library Ecosystem	Strong	Very Strong	Low to Medium	Medium

Sources: *[23]-[26]*

3.3.3.1 FastAPI

FastAPI is a modern Python backend framework designed for building APIs quickly and efficiently. It is commonly used for creating REST APIs and is known for having strong performance compared to many other Python web frameworks. One of the major benefits of FastAPI is that it is easy to develop with while still being scalable. It also includes automatic request validation and automatically generates interactive API documentation, which is helpful for development and testing.

For *SOLARIS*, FastAPI is a strong option because it allows the backend to be built quickly while still supporting the type of functionality the project would require. This includes storing system data, handling user requests from the phone application, and providing an organized way to access stored robot metrics from other interfaces such as a website. FastAPI also works well with common database tools such as SQLAlchemy and Alembic, making it easier to design and manage a database for storing collected robot information.

Another major advantage of FastAPI is familiarity. Since the team already has experience using FastAPI, it reduces the learning curve and makes development more efficient. Overall, FastAPI provides a strong balance of performance, ease of development, and compatibility with modern backend tools, making it a practical choice for *SOLARIS*.

3.3.3.2 Django

Django is a Python backend framework that is designed to support structured and scalable web development. Django provides many built-in tools that are commonly needed for backend systems, such as routing, authentication support, and database integration. One of the biggest advantages of Django is that it includes its own ORM and migration system, which makes it easier to design, manage, and update a database throughout development. Django is also widely used and has a very large ecosystem, meaning there is strong documentation and long-term community support.

For *SOLARIS*, Django is a valid framework to consider because it could support a backend system that stores robot metrics, organizes collected data, and serves that data to other interfaces such as the phone application or a website. Django is also useful if the project expands into a more feature-heavy platform, since it provides a strong built-in structure for managing data models and backend logic.

However, Django may be less ideal if the backend is intended to remain lightweight and focused primarily on API endpoints. Since Django is designed to support full web applications, it can introduce additional overhead compared to frameworks that are specifically optimized for API development. Overall, Django is a strong option for backend development, but it is best suited when a project benefits from its larger set of built-in features.

3.3.3.3 SpringBoot

Spring Boot is a backend framework built on the Java programming language and is widely used for developing large and structured web applications. It is especially common in industry for building REST APIs and backend services due to its reliability, long-term support, and strong ecosystem. Spring Boot provides a clear project structure and includes many built-in tools that make backend development more standardized, especially for larger codebases.

For *SOLARIS*, Spring Boot is a framework worth considering because it has strong support for database integration and backend architecture. It commonly works with tools such as Spring Data JPA and Hibernate for database management, and it supports database migration tools like Flyway or Liquibase. This makes it a strong option for building a backend system that stores robot metrics, serves data to a phone application, and scales well if the project expands.

However, Spring Boot can also be more complex than other backend frameworks, especially for smaller projects. Since it is designed with enterprise-level applications in mind, it can introduce more setup and configuration compared to lighter frameworks such as FastAPI. In addition, the team's familiarity with Java backend development may be lower compared to Python-based frameworks. Overall, Spring Boot is a strong and professional backend option, but it may not be the most efficient choice for a time-limited senior design project.

3.3.3.4 Express.js

Express.js is a backend framework for Node.js that is commonly used to build REST APIs and web servers using JavaScript or TypeScript. Express is known for being lightweight and flexible, meaning it does not force a strict project structure and allows developers to design the backend in a way that fits the needs of the application. Because of how widely used it is, Express also has a very large ecosystem of libraries and community support.

For *SOLARIS*, Express.js is a valid framework to consider because it can be used to build a backend that stores robot data, serves API endpoints to the phone application, and supports integration with a database. Express is often paired with database tools such as Prisma, Sequelize, or TypeORM, which provide structured ways to store and query system information. Since JavaScript and TypeScript are already being considered for the mobile application, using Express could also allow the project to share the same language across both frontend and backend development.

However, one drawback of Express is that it is relatively minimal by default. While this can be an advantage for small projects, it also means that more features such as validation, authentication, and project structure must be built manually or added through third-party libraries. Overall, Express.js is a strong backend option due to its simplicity, flexibility, and ecosystem, but it may require more setup and organization compared to frameworks that provide more built-in structure.

3.3.3.5 Backend Phone App Framework Selection

FastAPI: FastAPI was selected as the backend framework for *SOLARIS*. The biggest reason for this choice is familiarity. The team is already comfortable with Python, and some team members have direct experience working with FastAPI specifically. This reduces the learning curve and allows development time to be spent on implementing system features rather than learning a new backend framework from scratch.

FastAPI is also a strong fit for *SOLARIS* because the backend is primarily intended to serve as an API layer for storing and accessing robot telemetry, system status, and logged metrics. FastAPI is designed specifically for building lightweight, efficient APIs, which makes it more practical for this project than heavier full-stack frameworks. In addition, FastAPI pairs well with PostgreSQL through widely used Python libraries, making database integration and long-term data management straightforward. Overall, FastAPI provides the best balance of simplicity, performance, and development efficiency for the *SOLARIS* backend.

3.3.4 Embedded Development Frameworks

With the selection of the ESP32-S3 microcontroller (as discussed in Section 3.1.2), the embedded development framework options for *SOLARIS* can be narrowed down significantly. For this project, both PlatformIO and ESP-IDF will be used. The following sections describe each framework and how they support development for the *SOLARIS* embedded system.

3.3.4.1 Platform I/O

PlatformIO will be used as the primary development environment for the *SOLARIS* embedded firmware. It provides the overall project structure, build system, and file organization needed to keep the codebase clean and manageable throughout development. PlatformIO also simplifies the process of compiling and flashing firmware onto the ESP32-S3, which is important for fast iteration during testing.

Another major advantage is that PlatformIO integrates directly into Visual Studio Code, allowing the team to develop, build, and upload firmware from a single environment. This improves workflow efficiency and makes it easier to manage libraries, serial monitoring, and debugging tools during development.

3.3.4.2 ESP IDF

ESP-IDF (Espressif IoT Development Framework) will be used alongside PlatformIO because it provides the official software development framework for the ESP32-S3. ESP-IDF includes the core drivers, hardware libraries, and system-level tools needed to fully access the capabilities of the microcontroller. It also provides a large collection of example projects and reference implementations, which is extremely useful for learning and integrating features such as Bluetooth Low Energy, peripherals, and power management.

ESP-IDF is especially important for *SOLARIS* because it allows the project to rely on well-supported vendor libraries rather than requiring low-level code to be written from scratch. This reduces development risk and makes it easier to implement embedded features reliably.

3.4 Wireless Communication Protocols

Table 33. *Wireless Communication Protocol Comparison*

	Wi-Fi	*Bluetooth Standard*	*Bluetooth Low Energy*	*LoRaWAN*	*Zigbee*
Range (m)	**20-100**	**Up to 30**	**Up to 50**	**10,000**	**1000**
Data Rate	**Up to 46 Gbps**	**1-3 Mbps**	**.5-1 Mbps**	**27 Kbps**	**500 Kbps**
Power Consumption	**Highest**	**Medium-Low**	**Low**	**Lowest**	**Low**

Frequency Bands	**2.4GHz/5GHz/6GHz**	**2.4GHz**	**2.4GHz**	**Medium**	**915MHz / 2.4GHz**
Phone Compatibility	**Yes**	**Yes**	**Yes**	**No: separate hub required**	**ZigBee direct: Uses Bluetooth to connect to Zigbee IOT devices**

Sources: *[12]-[19]*

3.4.1 Wi-Fi

Wi-Fi is a common wireless protocol that provides strong compatibility with smartphones and relatively high usable range. It operates in the 2.4 GHz and 5 GHz bands, with newer standards also supporting 6 GHz. Although Wi-Fi is capable of very high data rates, *SOLARIS* does not require large bandwidth for normal operation, since communication is primarily limited to low-rate telemetry and control commands.

Wi-Fi is still a strong candidate because it can be straightforward to integrate into a phone application and typically provides greater range than Bluetooth-based communication, especially in open outdoor environments. Wi-Fi also does not strictly require a router. The robot can be configured to create its own local Wi-Fi network, allowing a phone to connect directly without relying on external infrastructure.

The primary disadvantage of Wi-Fi is power consumption. Wi-Fi communication generally requires more energy than Bluetooth-based alternatives, which can reduce battery life and negatively impact a solar-powered system's net energy gain. This drawback can be partially mitigated by enabling Wi-Fi only when necessary and using low-power sleep modes when communication is idle, but Wi-Fi is still typically less power-efficient than BLE for continuous operation.

3.4.2 Bluetooth Classic

Bluetooth Standard (Bluetooth Classic) is a widely supported wireless protocol that allows direct communication between the robot and a smartphone without requiring external infrastructure. It typically provides higher data rates than Bluetooth Low Energy, making it well suited for continuous streaming applications such as audio or sustained real-time data transfer.

For *SOLARIS*, Bluetooth Classic was considered because it offers native phone compatibility and does not require a router or access point. However, the primary communication needs of *SOLARIS* are low-bandwidth telemetry and control commands rather than continuous high-rate streaming. As a result, Bluetooth Classic's higher throughput does not provide a meaningful advantage for this system.

Additionally, Bluetooth Classic generally consumes more power than BLE due to its connection model and design goals. Since *SOLARIS* is a solar-powered robot where energy efficiency is a key constraint, Bluetooth Classic is less attractive than BLE for long-term operation.

3.4.3 Bluetooth Low Energy (BLE)

Bluetooth Low Energy (BLE) is a wireless protocol designed specifically for low-power, battery-operated devices. BLE supports direct smartphone connectivity without requiring external infrastructure and is well suited for applications that transmit small amounts of data intermittently. This matches the needs of *SOLARIS*, where communication primarily consists of control commands, mode selection, and periodic telemetry such as battery level, light sensor readings, and system status.

BLE is a major contender for *SOLARIS* because it offers a strong balance of energy efficiency, reliability, and ease of integration into a mobile application. Compared to Wi-Fi and Bluetooth Classic, BLE typically requires significantly less power, making it more appropriate for a solar-powered robot where minimizing energy usage is critical to meeting overall system goals.

The primary drawback of BLE is range. BLE communication generally has a shorter usable range than the other protocols, especially in outdoor environments where signal strength may vary due to obstacles, orientation, or interference. This limitation may restrict the distance at which a user can reliably connect to the robot. However, because *SOLARIS* is intended to be controlled and monitored locally by a nearby user, BLE range is considered an "acceptable" tradeoff given its power advantages. However, it may be better to use a protocol like Wi-Fi for that added benefit.

3.4.4 LoRaWAN

LoRaWAN is a low-power wide-area networking (LPWAN) protocol designed for long-range communication with very low data rates. It is commonly used in IoT applications such as remote sensing, environmental monitoring, and asset tracking. LoRaWAN is attractive for solar-powered systems because it offers extremely low power consumption and can achieve communication ranges of several kilometers under favorable conditions. It also aligns well with *SOLARIS* in terms of data requirements, since the robot primarily needs to transmit small telemetry packets rather than high-bandwidth data.

LoRaWAN was considered as a potential option because it provides the best combination of range and power efficiency among the protocols evaluated. In theory, it would allow *SOLARIS* to report status information over long distances while minimizing energy usage, which is desirable for an outdoor autonomous robot.

However, LoRaWAN has a major limitation for this project: it does not provide native smartphone compatibility. A phone cannot directly connect to LoRaWAN in the same way it can connect to Wi-Fi or Bluetooth. Supporting LoRaWAN would require an intermediary device such as a gateway or hub to relay data between the robot and the phone application. This additional infrastructure increases cost, setup complexity, and development overhead, making LoRaWAN impractical for a system intended to be controlled directly through a phone.

3.4.5 Zigbee

Zigbee is a low-power wireless protocol commonly used in IoT and smart home applications. It is designed for reliable communication between many devices and is especially effective in mesh network topologies, where multiple nodes relay messages to extend coverage and improve robustness. Zigbee operates in the 2.4 GHz band globally and can also operate in sub-GHz bands in some regions. Its low power consumption and support for multi-device networking make it a strong option for distributed sensor systems.

For *SOLARIS*, Zigbee was considered because it offers low power consumption and can provide longer range than BLE in some configurations, particularly when mesh networking is used. Zigbee is also widely used in commercial IoT ecosystems, which makes it a proven technology for low-power telemetry.

However, Zigbee has a major limitation for this project: it does not provide direct, native smartphone compatibility in the same way Wi-Fi and Bluetooth do. Zigbee Direct is a newer approach that allows a phone to communicate with Zigbee devices, but it does so by using Bluetooth as the communication method between the phone and the Zigbee device. This defeats much of the purpose of choosing Zigbee for a one-to-one robot-to-phone link, since Bluetooth is still required for the phone connection.

3.4.6 Wireless Communication Protocol Selection

Bluetooth Low Energy (BLE): After comparing the available wireless communication protocols, *SOLARIS* will use Bluetooth Low Energy (BLE). None of the protocols are truly “perfect” for this project, but BLE provided the best overall balance of advantages while having the fewest drawbacks that actually matter for our system.

One of the biggest reasons BLE stands out is native smartphone compatibility. *SOLARIS* is intended to be controlled directly through a phone, and BLE supports this without requiring any external infrastructure. Protocols like LoRaWAN and Zigbee offer strong range and power advantages, but they do not provide direct phone connectivity. Supporting them would require an additional hub or gateway to relay communication between the robot and the phone application. That added hardware and development overhead would significantly increase cost, complexity, and setup requirements, which is not justified for a one-to-one robot-to-phone link. Because of this, the only realistic contenders for *SOLARIS* were Wi-Fi and Bluetooth.

BLE is also a strong choice because it matches the communication needs of the system. *SOLARIS* does not require high bandwidth, since the data being transmitted consists of low-rate telemetry and short control commands. BLE supports this easily while staying efficient. Bluetooth Classic was considered, but its higher data rate does not provide a meaningful advantage for this project. Since Bluetooth Classic generally consumes more power than BLE, the increased throughput would be wasted for our use case.

Wi-Fi's main advantage is range, and it can also support direct backend communication when connected to a router. In theory, this would allow the robot to send data directly to a database without relying on the phone as an intermediary. However, this assumes reliable outdoor router coverage, which is not realistic for most expected use cases, especially when the robot is operating autonomously. Even with Wi-Fi , disconnects would still need to be handled, meaning *SOLARIS* would still require local storage and synchronization logic. The other Wi-Fi option is for the robot to act as its own access point so a phone can connect directly, but in that scenario neither the phone nor the robot would have internet access, making direct backend communication impossible.

Ultimately, BLE provides the most practical solution for *SOLARIS*. While it sacrifices some range compared to Wi-Fi and some power efficiency compared to LoRaWAN and Zigbee, it offers the best overall combination of low power, simplicity, and direct smartphone usability. Because *SOLARIS* is expected to be controlled locally by a nearby user, BLE range is considered an acceptable limitation, making it the most appropriate wireless communication protocol for this project.

3.5 Power Supply

Up until this point we have shared our goals and requirements for this project. Mentions of hardware usage have been stated but the reasoning for their choice have not been stated in detail. In this section we look over the valuable research used in making those decisions. It is not only valuable to look into the technologies we want to use but to also be open-minded as to why we did not choose others given sufficient investigation. This presents itself as a strong opportunity to deepen our understanding of the hardware technologies available to us for best fit within our project *SOLARIS*.

3.5.1 Brief

The *SOLARIS* battery subsystem is designed to balance weight, safety, and longevity. By prioritizing high energy density, the system maximizes runtime while maintaining a low mass profile. Safety is integrated through carefully selected chemistries and protection circuits, while cost and durability are managed by analyzing cycle life and Depth of Discharge (DoD). This multi-criteria approach ensures the robot remains operable and cost-effective over its intended lifespan.

3.5.2 Battery Considerations

SOLARIS utilizes a 12 V power supply architecture. To finalize the energy storage system, two primary factors must be determined: the optimal battery chemistry and the required capacity. The chemistry will be evaluated based on metrics such as specific energy, depth of discharge (DoD), and cost-effectiveness. Simultaneously, the capacity must be optimized to achieve a balance between system robustness and the avoidance of unnecessary weight penalties.

3.5.3 Power Distribution

Table 34. *Power Consumption Table*

	Voltage	***Current Draw***	***Quantity***	***Power Consumption***
ESP32-S3	**3.0-3.6V**	**0.108A**	**1**	**0.356W**
GV-5-LI-14.2V	**14.2V**	**0.190mA**	**1**	**2.698mW**
TEPT5600	**1.5-6V**	**0.035A**	**4**	**0.462W**
INA228	**2.7-5.5V**	**0.64mA**	**1**	**2.112mW**
BNO085	**2.4-3.6V**	**0.014A**	**1**	**0.0462W**
MB1020-000	**2.8-5.5V**	**0.004A**	**1**	**0.0132W**
goBilda 5203-2402-0188	**12.8V**	**0.3A**	**2**	**7.68W**
goBilda 5204-8002-0139	**12.8V**	**0.3A**	**4**	**15.36W**
Total Movement Power Consumption				**23.04W**

Total Sensors and Controllers Power Consumption	**0.88W**
Total System Power Consumption	**23.92W**

The power consumption for all non-locomotion components is approximately 0.9 W, totaling 10.8 Wh over a 12-hour operational window. Based on a conservative estimate of one hour of daily movement, locomotion adds 23.1 Wh to the energy budget. Combined with the 15 Wh auxiliary power allocation specified in the project objectives, the total daily energy requirement is 48.9 Wh. To ensure multi-day autonomy during periods of limited solar irradiance, a safety factor of four was applied to the daily consumption. This results in a minimum required capacity of 195.6 Wh; therefore, the system will utilize a battery with a capacity of at least 200 Wh.

3.5.4 Battery Technology Comparison

Table 35. *Battery Technology Comparison*

Battery Technology	***Specific Energy (Wh/kg)***	***Depth Of Discharge***	***Pros***	***Cons***	***Cost ($/kWh)***
Lithium Ion (NMC)	**~150-250**	**80-90%**	**Very high energy density, good power capability**	**Moderate safety risk if abused, requires BMS, shorter cycle life than LiFePO4, thermal runaway possible**	**150-300 $**
Lithium Iron Phosphate (LiFePO4)	**~90-120**	**80-100%**	**Excellent safety and thermal stability, very long cycle life (2000–5000+), good for deep cycling and solar storage**	**Lower energy density (heavier than NMC), higher upfront cost than lead-acid**	**200-400 $**
Lead-acid	**~30-50**	**50%**	**Low upfront cost, simple charging, highly recyclable**	**Very heavy, short cycle life, poor deep-discharge capability, lower efficiency**	**60-150$**

Sources: *[95],[96]*

The provided table evaluates the primary characteristics of three of the most prevalent secondary, or rechargeable, battery chemistries. To determine the most effective energy storage solution for the *SOLARIS* project, multiple battery parameters were explored.

Specific Energy is a metric that defines the energy density of the chemistry, representing the amount of energy stored per unit of mass (Wh/kg). High specific energy is critical for mobile robotics, as it directly influences the weight-to-power ratio and overall payload capacity. Depth of Discharge (DoD) indicates the percentage of the total battery capacity that can be safely discharged before recharging is required. A higher allowable DoD increases the "usable" energy available in a single cycle, effectively maximizing the utility of the battery's rated capacity without compromising its cycle life. While the base cost of raw battery cells is a factor, the total system cost is variable and includes necessary overhead such as integrated Battery Management Systems (BMS), protection circuitry, and specialized charging hardware. These elements are especially essential for maintaining the safety and longevity of high-density lithium battery chemistries.

3.5.4.1 Lithium Ion (NMC)

Lithium Nickel Manganese Cobalt (NMC) batteries are renowned for their high specific energy, typically ranging from 150 to 250 Wh/kg, making them the standard choice for weight-sensitive applications such as electric vehicles and high-end consumer electronics. While they offer a compact footprint and high power output, NMC chemistries are more susceptible to thermal runaway if punctured or overcharged, necessitating sophisticated battery management systems (BMS) for safety. Their cycle life is generally moderate, often rated for 1,000 to 2,000 full discharge cycles before significant capacity degradation occurs.

3.5.4.1 Lithium Iron Phosphate

Lithium Iron Phosphate (LFP) batteries prioritize safety and longevity over absolute energy density. With a specific energy typically between 90 and 120 Wh/kg, they are heavier than NMC but offer exceptional thermal and chemical stability, significantly reducing the risk of combustion even under extreme stress. LFP is characterized by its remarkable cycle life, frequently exceeding 2,000 to 5,000 cycles at high depths of discharge. Additionally, their cells exhibit a very flat discharge curve, maintaining a consistent voltage across the majority of the discharge cycle, which simplifies power regulation for sensitive electronics.

3.5.4.1 Lead-Acid

Lead-acid batteries are a mature technology primarily utilized for their low initial cost and high surge current capabilities. However, they possess the lowest specific energy of the three chemistries (30–50 Wh/kg), resulting in a much heavier and bulkier footprint for the same energy capacity. Their cycle life is relatively short, often limited to 300–500 cycles, and they are sensitive to deep discharges, which can lead to permanent capacity loss. Furthermore, lead-acid batteries require longer charging times and have higher self-discharge rates compared to lithium-based alternatives.

3.5.4.4 Battery Technology Selection

Lithium Iron Phosphate: For outdoor solar-powered mobile robot applications, LFP is the superior chemistry because its operational characteristics align perfectly with the challenges of fluctuating environmental conditions and repeated daily cycling. The exceptional thermal stability of LFP is a critical safety requirement for robots exposed to direct sunlight and high ambient temperatures, as it remains stable in heat that could pose a risk to NMC cells. Its high cycle life ensures that the daily charge-discharge routines dictated by solar availability do not prematurely degrade the battery, allowing for years of field deployment. Furthermore, the high charge-acceptance rate and low internal resistance of Lithium Iron Phosphate enable the system to efficiently capture and store the varying energy levels provided by solar panels, while its flat discharge curve ensures that motor performance and sensor accuracy remain consistent as the battery depletes.

3.5.5 Battery Part Comparison

Table 36. *Battery Part Comparison*

Solar Panel Charge Controller Part	*Rated Capacity*	*Operating Temperature*	*Size*	*Weight*	*Cost*
Dr.Prepare DBT1220LFP	**20Ah (256Wh)**	**-10-45°C**	**7.13 × 3.03 × 6.61 in**	**5.95lbs**	**$79.99**
Bioenno Power BLF-1220AS	**20Ah (240Wh)**	**-10-60°C**	**7.17 × 3.03 × 6.73 in**	**5.73lbs**	**$204.99**
Renogy RBT1220LFP-TM-G1	**20Ah (256Wh)**	**-20-60°C**	**7.13 × 3.03 × 6.61 in**	**5.95lbs**	**$89.99**
Ultralife URB12200	**21.6Ah (276Wh)**	**-20-60°C**	**7.12 × 3.03 × 6.57 in**	**5.97lbs**	**$285.13**

Sources: *[97]-[100]*

The candidate batteries identified for this comparison each provide a minimum capacity of 240 Wh. While this exceeds the initial 200 Wh requirement established in the power budget, the selection was dictated by the discrete capacities available in the $LiFePO_4$ market. This 20% margin serves as a beneficial "buffer" for extended autonomy while remaining the closest fit for the system's design constraints. To ensure the reliability of the *SOLARIS* unit, several battery parameters were examined. The operating temperature range is a critical constraint due to the intended deployment in Florida. With summer ambient temperatures commonly reaching 38.4°C (101°F), the battery must be capable of safe operation at these levels, especially when accounting for the internal heat generated by the power electronics and motor controllers. Additionally, since *SOLARIS* is a mobile platform, minimizing the physical footprint and total mass of the energy storage system is essential. The four batteries under consideration exhibit similar size and weight profiles, but choosing a high energy density battery is optimal for the robot's mobility and operational efficiency.

3.5.5.1 Dr.Prepare DBT1220LFP

The Dr. Prepare 12V 20Ah battery prioritizes accessible operational parameters and clear charging instructions for general-purpose solar and DC-DC systems. It specifies an optimal charging current of 10 A with a 20 A maximum limit and maintains a wide operating temperature range from -20°C to 60°C. The documentation provides a detailed overview of charging methods and safety guidelines, ensuring that users can easily integrate the 20 A continuous discharge unit into variable energy environments without exceeding the limits of the internal battery management system.

3.5.5.2 Bioenno Power BLF-1220AS

The Bioenno Power BLF-1220AS is optimized for high-power applications, featuring a robust maximum continuous discharge current of 40 A and a peak pulse capability of 60A for up to five seconds. This high current ceiling makes it particularly suitable for small electric motor use cases where transient power surges occur. The documentation emphasizes the necessity of a 14.6 V charging profile to trigger internal balancing and maintain the health of the 20 Ah capacity.

3.5.5.3 Renogy RBT1220LFP-TM-G1

The Renogy 12.8 V 20 Ah battery is designed as a deep-cycle storage solution. It operates at a nominal 12.8 V and is engineered for repeated high-depth discharge cycles, which is fundamental for solar-powered platforms. The technical literature highlights the battery's role in grid-independent systems and provides a comprehensive safety framework, positioning the unit as a reliable energy core within a larger ecosystem of solar panels and charge controllers.

3.5.5.4 Ultralife URB12200

The Ultralife URB12200 is distinguished by a slightly higher nominal capacity of 21.6 Ah and a total energy rating of 276 Wh, offering a larger energy density of 102 Wh/kg. It is engineered for high reliability and is IEC 62133 compliant, featuring a rugged exterior with integrated carry handles to facilitate mobility. Technically, it supports a maximum continuous discharge current of 20.0 A and maintains uniform voltage throughout the discharge cycle, which is a critical characteristic for applications requiring consistent power delivery.

3.5.5.5 Battery Part Selection

Renogy RBT1220LFP-TM-G1: The Renogy RBT1220LFP-TM-G1 distinguishes itself as the optimal choice for the *SOLARIS* project due to its specialized focus on deep-cycle longevity and seamless integration into solar-powered ecosystems. Unlike general-purpose lithium batteries, the Renogy unit is specifically engineered to withstand the frequent, varying charge-discharge cycles typical of a robot that relies on fluctuating solar intensity. This resilience is backed by a sophisticated internal Battery Management System (BMS) that offers comprehensive protection against overcharging from high-output panels and over-discharging during peak motor draws. While other batteries might offer higher burst currents, the Renogy provides a more stable and predictable energy density of 12.8 V and 20 Ah, which perfectly aligns with the project's requirement for a robust, high-cycle-life core. Its industry-leading reputation for durability in off-grid applications makes it the most reliable battery for ensuring the robot remains operational through years of field deployment.

3.5.6 Voltage Regulator Technology Comparison

3.5.6.1 Voltage Regulator Considerations

To properly operate the peripherals on *SOLARIS*, there must be two voltage regulators to buck the 12 V battery voltage to a usable 5 V and 3.3 V rails for usage in the smaller electronics in the robot. The dedicated 5 V rail will be used by the USB-A to charge the robot manually and the 3.3 V rail will be used to support the ESP32 and the multiple onboard sensors such as the ultrasonic sensors that will be used for ranging during navigation. The 5 V regulator must be able to support moderate current draw while having high thermal and energy efficiency to preserve power efficiency.. The 3.3 V rail mainly requires good voltage stability and low ripple to facilitate reliable ESP32 and sensor operation. One of the main goals with *SOLARIS* is power efficiency. Because of this, regulator efficiency directly impacts day-to-day operational runtime and overall energy usage. Long term component reliability can be affected if there is excess dissipation in the regulator leading to excess heat generation. Finally, noise generated by the switching regulators must be considered, especially the ones near the sensors. For these concerns, both switching and linear regulator technologies were evaluated to determine what the optimal balance was for *SOLARIS* to optimize, efficiency, thermal performance, noise, integration complexity and cost.

Table 37. *Voltage Regulator Technology Comparison*

Voltage Regulator Technology	*Power Efficiency*	*Daily Energy Capture*	*Safety*	*Complexity*	*Cost*
Switching	**High efficiency, lower power dissipation than LDO for large voltage drops.**	**Maximizes battery energy, Reduced thermal loss.**	**Lower heat generation reduces thermal stress, switching action produces ripple.**	**Moderate. Requires inductor, MOSFET, a rectifier, output filter and careful PCB layout.**	**Moderate. More components than LDO means higher cost.**
Linear	**Efficiency = Vout/Vin. Larger voltage differences result in heat dissipation**	**Reduces usable battery energy due to excess voltage dissipated as heat.**	**Low ripple with lower noise and EMI, needs thermal management at higher currents.**	**Low. Requires minimal support components and simpler PCB layout.**	**Low, Fewer external components and simple architecture.**

Sources:*[106]-[108]*

3.5.6.2 Switching Regulator

Switching regulators regulate voltage by rapidly switching a MOSFET on and off, using an inductor to transfer energy through the circuit. Because the MOSFET fully switches on and off each cycle, power dissipation across the transistor is minimized, which allows for increased efficiency depending on the load and operating conditions. This makes switching regulators particularly well suited for applications with a large voltage difference between input and output, such as stepping down from 12 V to 5 V. The higher efficiency directly reduces heat generation and improves overall energy efficiency, which is critical for maximizing battery runtime in an application like *SOLARIS*. The main drawback of switching regulators is that the MOSFET switching produces high ripple and electromagnetic interference (EMI). However, with careful PCB design and layout along with properly placed filtering capacitors, these effects can be minimized to ensure stable, low-noise operation for the onboard sensors.

3.5.6.3 Linear Regulator

Linear regulators regulate voltage by operating a pass transistor in its linear region to drop the excess voltage as heat. Unlike switching regulators, linear regulators do not store and transfer their energy via an inductor. Instead, the voltage difference between the input and output is dissipated across the regulator. This makes the theoretical efficiency = Vout/Vin. This means that stepping down from 12 V to 5 V would yield approximately a 41.7% efficiency rating meaning roughly half the input would be dissipated as heat. This makes linear regulators very inefficient for larger voltage drops in battery-powered systems such as *SOLARIS*. The advantage of linear regulators is that they have much lower ripple and EMI than switching regulators which makes them very good for powering microcontrollers and analogue sensors. They also have greater simplicity and low component count which makes them a fantastic choice for applications that can handle their efficiency drawbacks.

3.5.6.4 Voltage Regulator Technology Selection

Switching Regulator: With *SOLARIS* having a 12 V battery system and the desire to have a 5 V USB-A charging rail and the 3.3 logic rail for the MCU and sensors, the selected regulator must be able to balance efficiency, thermal performance, electrical noise, cost, and ease of integration. The 5 V rail needs to be able to support a moderate current draw and has a very sizable voltage drop from the input voltage to the output voltage from the battery, thus, a switching buck regulator was selected to ensure the desired efficiency and minimize the thermal losses on the PCB. This will ensure that the daily energy captured by the solar panel and charging system will be able to be used effectively and not simply lost as heat. The 3.3 V rail is for powering the MCU and multiple sensors. While there is higher ripple on the switching buck regulators, there will be adequate components in place on the PCB to ensure proper stable operation for these more sensitive components.

3.5.7 Voltage Regulator Part Comparison

Table 38. *Voltage Regulator Part Comparison*

Voltage Regulator Part	*Conversion Efficiency (Vin = 12 V, Vout = 5 V)*	*Output Channels*	*Key Features*	*Max Current Per Channel*	*Quiescent Current*	*Cost*
LM2679	**84% (I_{Load} = 5A)**	**1**	**High current (5A); Current limit adjustable**	**5A**	**~7mA**	**$7.42**
LM2576	**77% (I_{Load} = 3A)**	**1**	**Robust; Industry standard**	**3A**	**~5mA**	**$1.42**

Voltage Regulator Part	*Conversion Efficiency (Vin = 12 V, Vout = 5 V)*	*Output Channels*	*Key Features*	*Max Current Per Channel*	*Quiescent Current*	*Cost*
LM2679	**84% (I_{Load} = 5A)**	**1**	**High current (5A); Current limit adjustable**	**5A**	**~7mA**	**$7.42**
LM2596	**73% (I_{Load} = 3A)**	**1**	**Common; Easy to implement**	**3A**	**~5mA**	**$2.34**
LT8650S	**93% (I_{Load} = 3A)**	**2**	**Dual outputs; Synchronous; Silent Switcher 2 (Low EMI); Compact**	**4A**	**6.2µA**	**$18.31**
LTC3633	**93% (I_{Load} = 3A)**	**2**	**Dual outputs; Compact**	**3A**	**2mA**	**$12.69**

Sources: *[101]-[105]*

The *SOLARIS* power distribution network must step down the nominal 12.8 V $LiFePo_4$ battery voltage to two distinct rails: a 3.3 V rail to power the microcontroller and sensor suite, and a 5 V rail dedicated to the USB-A peripheral interface. To ensure the system meets its high-efficiency design goals, the above-listed table characteristics were considered.

As the primary objective of *SOLARIS* is to maximize solar energy harvesting and minimize parasitic losses, high conversion efficiency is the most critical metric. Synchronous buck regulators boast an increased efficiency of about 15% compared to asynchronous buck regulators.

The architecture favors dual-channel regulators. Utilizing a single Integrated Circuit (IC) to provide both 3.3 V and 5 V outputs reduces the total PCB footprint, simplifies the Bill of Materials (BOM), and streamlines the power routing logic compared to utilizing two discrete single-channel regulators. Based on power budget calculations, the 3.3 V logic rail requires a continuous average current of approximately 0.3 A. All candidate regulators provide at least an order of magnitude of current headroom (typically 3 A or greater), ensuring high reliability, lower operating temperatures, and the ability to handle transient current spikes without voltage sagging. For a solar-powered autonomous system, minimizing quiescent current is essential. Quiescent current is the power consumed by the regulator in a non-switching or no-load state. Selecting a regulator with a low I_Q ensures that the system remains efficient during idle periods.

3.5.7.1 LM2679

The LM2679 is part of the "Simple Switcher" family and is characterized by its high-current capability, providing up to 5 A of output current. Its primary advantage is a resistor-programmable peak current limit, allowing you to protect sensitive downstream loads or prevent battery surges by capping current between 3 A and 7 A. While it is highly efficient (up to 92%) and easy to design with, it is a single-channel device, meaning two separate units would be required to provide both 5 V and 3.3 V rails.

3.5.7.2 LM2576

The LM2576 is an older but extremely robust "Simple Switcher" capable of driving 3 A loads with excellent line and load regulation. It operates at a relatively low switching frequency of 52 kHz, which results in very large filter components but offers high stability and ease of layout for beginner PCB designs. While it is commonly used in the industry, its larger footprint and lower efficiency make it less ideal for high-efficiency solar-powered robotics compared to the modern synchronous regulators.

3.5.7.3 LM2596

The LM2596 is a widely used, industry-standard step-down regulator that operates at a fixed switching frequency of 150 kHz. Its key aspect is simplicity; it is internally compensated and requires very few external components, making it a reliable choice for general-purpose power regulation. However, because of its lower switching frequency, it requires larger inductors, and its efficiency (typically around 73%–80%) is notably lower than the synchronous alternatives.

3.5.7.4 LT8650S

The LT8650S is a high-performance, dual-channel synchronous regulator that is ideal for noise-sensitive applications due to its integrated "Silent Switcher 2" architecture, which drastically reduces electromagnetic interference (EMI) on the PCB. Its most standout feature is an ultralow quiescent current of 6.2 µA while regulating, ensuring that the logic rail does not become a significant parasitic drain on the 20 Ah battery. By providing two 4 A channels in a single small package, it efficiently generates both the 5 V and 3.3 V rails while maintaining up to 94.6% efficiency even at high switching frequencies.

3.5.7.5 LTC3633

The LTC3633 is a dual-channel monolithic synchronous buck regulator designed for high-frequency operation up to 4 MHz, which allows for the use of very small external inductors and capacitors. It supports a 3 A output current per channel and features a unique selectable phase shift between the two channels to reduce input and output ripple. This makes it a robust choice for compact designs where power density is a priority, though its quiescent current is higher than the LT8650S at 2 mA.

3.5.7.6 Voltage Regulator Part Selection

LT8650S: The LT8650S stands out as the superior choice for the *SOLARIS* project because it offers a rare combination of high power density, elite efficiency, and advanced electromagnetic noise suppression that the other candidates lack. Architecturally, it is the only regulator in the group featuring "Silent Switcher 2" technology, which utilizes internal bypass capacitors to minimize EMI. Furthermore, its dual-channel design provides 4 A per rail, allowing a single IC to replace two discrete units like the LM2679 or LM2596, thereby reducing PCB complexity and weight. From an energy conservation standpoint, its ultra-low 6.2 µA quiescent current is nearly three orders of magnitude more efficient than the LTC3633 or the LM-series switchers. By delivering up to 94.6% efficiency at high switching frequencies, the LT8650S enables a smaller, lighter, and more electrically quiet robot without sacrificing performance.

3.5.8 Circuit Safety

Implementing a robust circuit safety strategy for *SOLARIS* is essential to protect the $LiFePo_4$ battery and the ESP32-S3 control logic from the significant current demands of the drivetrain. A 40 A inline fuse positioned as close to the battery's positive terminal as possible serves as the final line of defense against catastrophic short circuits. Given that the four goBILDA 5204 motors have a combined maximum stall current of 36.8 A, a 40 A rating provides a functional buffer that prevents "nuisance tripping" during high-torque maneuvers on Florida's uneven terrain.

Complementing the passive fuse, a manual disconnect switch provides a reliable mechanical "hard" isolation point for both emergency situations and long-term storage. This switch should be rated for a continuous current exceeding 40 A to minimize resistive voltage drops and prevent localized heating at the contact points. From a structural standpoint, mounting this switch on an accessible exterior portion of the robot chassis allows for an immediate "kill" command that physically breaks the battery's connection, bypassing all software logic. This dual-layer combined fuse and disconnect switch approach ensures that *SOLARIS* can safely manage its high-density energy reserves while protecting the sensitive navigation sensors and processing units from back-EMF or overcurrent events.

Chapter 4 - Standards and Design Constraints

4.1 Standards

4.1.1 Brief

The development of an autonomous, solar-powered robotic system requires strict adherence to established engineering standards to ensure hardware interoperability, data integrity, and operational safety. This section outlines the communication and wireless protocols selected for the SOLARIS platform. By utilizing standardized bus architectures and RF specifications, the system can reliably interface various high-precision components- such as the ESP32-S3 controller, the BNO085 IMU, and the INA228 power monitor- within a unified framework. The following subsections detail the physical and logical constraints of the I2C, UART, and SPI wired protocols, as well as the Bluetooth Low Energy (BLE) wireless standard, providing the technical justification for their implementation in the final design.

4.1.2 Communication Protocol Standards

4.1.2.1 I2C

The I2C communication protocol is a software-based standard used for communication across digital systems. This inter-IC bus is defined as a simple bidirectional 2-wire bus for efficient inter-IC control, originally developed by Philips Semiconductors [115]. At a high level, the protocol allows multiple integrated circuits (ICs) to communicate using only two shared bus lines known as the serial data line (SDA) and the serial clock line (SCL) [115]. The bus is a true multi-controller system with built-in collision detection and arbitration procedures to prevent data corruption when more than one controller attempts to initiate a transfer simultaneously. Each device on the bus is recognized by a unique software address and operates within a controller/target relationship, meaning a controller can function as either a controller-transmitter or a controller-receiver [115]. By integrating the addressing and data transfer protocol directly on-chip, I2C enables software-defined systems and allows designers to prototype rapidly by simply connecting or disconnecting compatible ICs along the shared bus lines [115].

As discussed above, I2C is primarily a bidirectional bus but can also operate as a unidirectional bus depending on the communication mode. Bidirectional modes include Standard-mode (Sm), Fast-mode (Fm), Fast-mode Plus (Fm+), and High-speed mode (Hs-mode) [115]. I2C also supports an Ultra Fast-mode (UFm) which takes advantage of a unidirectional bus for applications such as LED controllers [115]. The different communication modes are compared below by their speed, directionality, driver type, and compatibility:

Table 39. *Communication Modes for I2C*

Feature	*Sm / Fm / Fm+*	*Hs-mode*	*Ultra Fast-mode*
Max Speed	**100kHz / 400kHz / 1MHz**	**3.4 Mbit/s**	**5 Mbit/s**
Direction	**Bidirectional**	**Bidirectional**	**Unidirectional**
Driver Type	**Open-drain**	**Open-drain/Current Source**	**Push-pull resistors**
Compatibility	**Fully downward compatible**	**Downward compatible**	**Not compatible**

Source: *[115]*

The five operating modes discussed above represent the bit-rate modes defined in the I2C standard. Beyond these, I2C protocol must also comply with other mandatory features and schemes. One such requirement is support for the START condition, STOP condition, and Acknowledge [115]. Combined with 8-bit oriented transactions using most significant bit (MSB) first ordering, these features make up a large portion of the standard [115]. All transactions begin with a START condition and are terminated by a STOP condition. A HIGH to LOW transition on the SDA line while SCL is HIGH defines a START condition, while a LOW to HIGH transition on the SDA line while SCL is HIGH defines a STOP condition [115]. These conditions are always generated by the controller. Since data is transferred in byte format, every byte placed on the SDA line must be eight bits long, and the controller must issue a START condition if it wishes to continue transmitting between individual byte messages [115]. As previously described, data is always transferred with the MSB first.

The START and STOP conditions only cover the transmitter side of the transaction. To complete the communication cycle, I2C also uses an Acknowledge (ACK) and Not Acknowledge (NACK) after every byte of data [115]. Through the ACK and NACK signals, the receiver is able to communicate back to the transmitter whether data was successfully received or not. In terms of physical bus behavior, during a transaction the transmitter releases the SDA line during the ACK clock pulse, allowing the receiver to pull SDA LOW and hold it stable through the HIGH period of the clock pulse [115].

When SDA remains HIGH during this ninth clock pulse, it is defined as a NACK signal [115]. Upon receiving a NACK, the controller has the opportunity to either generate a STOP condition to end the transaction or issue a repeated START condition to begin a new transfer [115]. Being largely software-defined, I2C remains one of the most versatile communication protocols available, supporting both bidirectional communication between two systems and unidirectional communication for simpler peripherals.

4.1.2.2 UART

Universal Asynchronous Receiver-Transmitter (UART) is a hardware-based communication protocol generally used for device-to-device data transfer. UART operates as a serial protocol where data is transferred bit by bit over a single wire, typically requiring only two wires total [117]. Unlike other communication methods, UART does not use a clock signal for synchronization. Instead, its asynchronous nature relies on a pre-configured baud rate to manage timing [117]. Both devices in the setup must be configured to the same baud rate, with an allowable difference of up to 10% before timing errors occur [117]. Unlike I2C, UART is hardware-specific and must be implemented at the hardware level to function properly. Despite this, UART remains one of the most commonly used protocols in embedded systems, microcontrollers, and computers due to its low cost and minimal wiring requirements. The standard defines a physical interface consisting of two wires, a transmitter (Tx) and a receiver (Rx), and supports only one controller and one peripheral in a full-duplex communication arrangement [117]. Data is delivered in a standard structure known as a packet, which consists of different components that work together to properly transmit and receive information.

Inside a device like a microcontroller, UART hardware acts as an intermediary between parallel and serial communication lines [117]. It receives data in parallel and wraps it into a packet containing key components such as a start bit, data frame, parity bit, and stop bit [117]. These components make up what is known as the standard UART frame structure. Packets are sent serially over the Tx wire, where the receiving UART performs the reverse process by stripping the frame bits and converting the serial data back into parallel [117]. For both devices to stay synchronized, several configuration requirements must be met: baud rates must match, an internal clock must be used to sample incoming data, and the hardware must be properly configured to define the communication parameters [117]. In a standard setup, the Tx pin of the first UART is connected directly to the Rx pin of the second UART and vice versa [117]. When not actively transmitting, UART lines are held in a high voltage state known as the idle state. This idle state allows one UART to take the role of the controller while the other acts as a peripheral to the transmission [117]. As mentioned, all of this is accomplished using just two wires to achieve reliable full-duplex communication.

Synchronization and timing play a key role in correctly communicating data between two pieces of hardware. UART achieves synchronization through start bits, baud rate matching, and timing tolerance [117]. As discussed, UART wraps communication data into a data frame that is assembled and disassembled by their respective devices. The key feature of this frame is the start bit, which signals when communication is allowed to begin [117]. During idle, when the transmission line is driven high, the controller device may drive Tx low for one clock cycle, alerting the receiver to begin reading bits. For this to work, both devices must have matching baud rates so that the sampling frequency matches the data transfer frequency [117]. The allowable difference between baud rates is limited to 10% before bits become too misaligned to be read correctly. To configure this timing, four major parameters must be determined: the Peripheral Clock (PCLK), oversample rate (OSR), baud rate divider (DIV), and fractional baud rate parameters (DIVM/DIVN) [117]. PCLK is defined as the input system clock speed at which an incoming signal can be sampled [117]. The general formula for calculating baud rate is shown below:

$$Baud\ rate\ = \frac{PCLK}{((DIVM+\frac{DIVN}{2048})\times 2OSR+2DIV}$$

4.1.2.3 SPI

The Serial Peripheral Interface (SPI) is a synchronous, high-speed, full-duplex communication protocol used to transfer data between a controller device and one or more peripherals [118]. Originally developed by Motorola in the 1980s, it has since become a de facto standard in embedded systems due to its simplicity and efficiency [118]. One of the defining characteristics of SPI is its shared clock signal, which ensures precise timing during data transfer. While it lacks the rigid standard format found in protocols like I2C, this actually gives developers the flexibility to tailor the protocol to their specific application [118]. A key strength of SPI is its bus architecture, which supports several standard configurations including independent chip select, daisy chain, and 3-wire SPI [118]. Each of these configurations highlights the customizability of SPI as a communication protocol for dynamic embedded applications. Rather than being defined by a strict data format, the standard is built around primary signals and four specific modes of operation. SPI also benefits from being an exceptionally high-speed protocol, allowing for faster data transfer compared to other common serial protocols. Typical clock frequencies reflect this, as devices like the TXE8116/24 can operate with an SPI clock period of 100ns, corresponding to a frequency of 10MHz [118].

While SPI does not define a single rigid communication format, it does establish specific hardware signals, bus architecture, and data transfer modes that serve as operational standards. SPI is primarily defined by four hardware signals. Peripheral Input Controller Output (PICO) transmits data from the controller to the peripheral [118]. Peripheral Output Controller Input (POCI) transmits data from the peripheral back to the controller [118]. Serial Clock (SCLK) is a shared signal generated by the controller to synchronize data transfer [118]. Chip Select (CS) uses an active low signal from the controller to select and enable specific peripherals [118]. These four signals make up the foundation of the SPI bus architecture in a standard four-wire setup. Within this architecture there are a few notable configurations. In a daisy chain configuration, peripherals are connected in series, allowing the entire chain to be controlled by a single CS line while data shifts through each device sequentially [118]. A simplified three-wire SPI configuration combines PICO and POCI into a single bidirectional line called Data Input/Output (DIO), reducing the total wire count [118]. Because SPI allows for tailored formats, individual devices often establish their own data frame standards. One example is the TXE8116/24, which transmits the most significant bit (MSB) first over a fixed 24-bit transaction format [118].

4.1.3 Wireless Communication Standards

4.1.3.1 Bluetooth Low Energy (BLE)

[116] Bluetooth Low Energy (BLE) was introduced in Bluetooth Core Specification version 4.0 as a low-power alternative to Bluetooth Classic (BR/EDR). Bluetooth Classic provides higher bandwidth but was designed for continuous data streaming applications like audio, making it overkill for devices that only need to transmit small amounts of data intermittently. BLE was built for battery-operated devices that communicate in bursts, prioritizing energy efficiency over raw throughput.

[116] BLE divides the 2.400 to 2.4835 GHz ISM band into 40 channels, each 2 MHz wide. To improve reliability in environments with RF interference, BLE uses a spread spectrum technique called Adaptive Frequency Hopping (AFH). Three of the 40 channels (37, 38, and 39) are reserved as advertising channels and are used for device discovery and connection establishment. The remaining 37 channels are used for data transmission. BLE maintains a channel map that classifies each data channel as used or unused, and the frequency hopping algorithm avoids channels that are performing poorly due to interference, ensuring more reliable communication.

[116] GAP or the Generic Access Profile, defines how BLE devices discover each other and establish connections. GAP assigns two roles to communicating devices, the Peripheral and the Central. The Peripheral is typically an embedded device that advertises its presence by broadcasting packets on the three dedicated advertising channels. The

Central is typically a smartphone or other host device that scans for these advertising packets and initiates a connection request when it finds a device of interest.

[116] Once a connection is established, communication moves off the advertising channels and onto the 37 data channels controlled by the Link Layer. At this point the two devices have formed a connection and can begin exchanging data. A single Central device is also capable of maintaining connections to multiple Peripherals simultaneously, which is managed at the Link Layer through the channel hopping schedule assigned to each individual connection.

[116] GATT, or the Generic Attribute Profile, defines how data is organized and exchanged once a connection has been established between two devices. GATT uses a client and server model where the Peripheral acts as the server, hosting data, and the Central acts as the client, reading or writing that data. Data on the server is organized into a hierarchy of Services and Characteristics. A Service represents a functional grouping of related data and each Service contains one or more Characteristics, which are the individual data values that can be read, written, or subscribed to by the client.

[116] The Bluetooth SIG predefines UUIDs for common Services and Characteristics such as battery level or device information, allowing devices from different manufacturers to interoperate without any custom configuration. For applications with unique data requirements, custom Services and Characteristics can be defined using application specific UUIDs, giving developers the flexibility to represent any data their application needs to exchange.

BLE has become one of the most widely adopted wireless communication standards in embedded and IoT applications due to its combination of low power consumption, native smartphone connectivity, and a flexible data model. Its layered architecture, from the physical layer up through GAP and GATT, allows developers to build wireless systems without relying on external networking infrastructure, making it well suited for a wide range of device-to-device communication uses.

4.1.4 Health & Safety Standards

4.1.4.1 UL 3100

4.1.4.2 ISO 3691-4

4.1.4.3 UL 4600

4.2 Design Constraints

4.2.1 Brief

The realization of the SOLARIS platform is governed by a complex set of design and practical constraints that balance theoretical performance with real-world engineering limitations. This chapter details the technical boundaries of the project, specifically focusing on the power distribution logic, signal integrity, and the economic factors influencing component selection. By identifying these constraints early in the design cycle, the team can ensure that the final autonomous system remains energy-positive, structurally sound, and financially viable within a student-oriented budget. The following sections outline the specific challenges associated with stepping down high-voltage battery power, managing electromagnetic interference from drive motors, and optimizing the cost-to-performance ratio of the mechanical chassis and electrical subsystems.

4.2.2 Power Constraints

The *SOLARIS* architecture must interface various peripherals and controllers that operate at various voltage rails, differing from the 12.8 V nominal output of the $LiFePO_4$ battery. To maintain a compact form factor without sacrificing performance, the system utilizes a dual-channel synchronous voltage regulator. This integrated approach provides the required 3.3 V and 5 V outputs from a single IC, achieving high conversion efficiency and minimizing thermal dissipation. Maximizing efficiency is a foundational constraint; to ensure the highest possible energy yield, the design incorporates oversized conductors to mitigate I^2R (ohmic) losses and utilizes low-power sleep modes for non-critical components during idle periods.

Central to the energy harvesting strategy is the Genasun charge controller, which employs Maximum Power Point Tracking (MPPT). Unlike standard PWM controllers, MPPT dynamically adjusts the electrical operating point of the photovoltaic cells to extract the maximum available power under varying irradiance and temperature conditions. This technology is critical for ensuring that the battery gains consistently outweigh the system's baseline consumption.

Rigorous power budgeting is essential for mission success. A sensor or peripheral is only considered a net benefit if its operational value outweighs its energetic cost. To manage this, the architecture evaluates the use of MOSFET-based power switching to programmatically isolate high-draw components. While this allows for granular power control, it introduces specific engineering trade-offs: the addition of a MOSFET increases the series resistance (R_{DS}) of the power path and requires the allocation of limited General Purpose Input/Output (GPIO) pins from the microcontroller unit (MCU).

The actuation system introduces significant constraints regarding signal integrity. Motors generate substantial current spikes and inductive back-EMF during operation, which can inject electrical noise into shared power planes. To prevent interference with sensitive I^2C sensor data or RF communications, the motor power bus must be logically and physically isolated from the logic-level power pins of the sensors and MCU.

A primary constraint of the solar subsystem is the trade-off between solar alignment and actuation cost. While the photovoltaic effect is maximized when the panel is perfectly perpendicular to the sun, constant adjustments by the pan-and-tilt motors consume significant energy. The system is therefore designed to optimize net energy gain—the difference between energy harvested and energy consumed. Rather than continuous tracking, the motors are actuated only when the predicted increase in solar harvest significantly outweighs the power cost of the maneuver, ensuring the system remains energy-positive throughout the day.

4.3 Practical Constraints

4.3.1 Economic

One of the major constraints involved in designing and building *SOLARIS* has been analyzing and minimizing the overall cost of the system build that falls within normal student budget. While we had a full-sized team to sp[lit costs, the *SOLARIS* design involves having several components that, especially if accidentally overengineered, can incur a high cost. Because *SOLARIS* is designed for off-road movement, ensuring that our components meet our technical requirements, will not fail under stress and have sufficient margin became a significant factor when selecting and procuring items.

The biggest single cost item that we've worked to minimize is the overall *SOLARIS* chassis system. The chassis needs to meet several design requirements to ensure that the chassis will not give under significant torque loads thus causing the motors to potentially spike in current usage unnecessarily thus causing excessive heating and power inefficiencies. The chassis must also be made of a material light enough to not incur an excessive contribution to the overall weight of the robot which would require larger motors, motor drivers, lower energy efficiency and even higher cost. The difficulty is in finding a pre-built frame that meets our design specifications for a low price. Research indicated that this was impossible. The economic compromise was designing the frame from scratch and using aluminum sheets to ensure *SOLARIS* had the necessary rigidity and robustness without incurring heavy costs. The cost, instead, was shifted over to the time required to research and design the chassis from scratch.

The second biggest cost factor for *SOLARIS* was the DC motor footprint on the project. To ensure that the robot could reliably navigate in the desired terrain with the expected weight, the drive DC motors were a significant design factor to attempt to minimize the cost of. Finding DC motors that were well-documented that met design requirements and were not >$80 proved challenging. The robot required 4 motors, one for each wheel and needed to be able to support the estimated 20 lb weight of the overall robot as well as navigate it over the desired terrain. The desired terrain of *SOLARIS* is grass which, when evaluating practical environmental hazards, implies that there will be roots, patches of thick grass and sticks/twigs that the robot must successfully negotiate without stalling. The other two motors were the solar panel tilt and pan motors. These motors did not have such steep requirements but were selected to ensure that, in particular, the pan motor could support the slip ring arm, solar panel, and the tilt motor. To support these larger motors, higher cost drivers were required to support the current load.

The next area of concern was the overall power and power management system, particularly the voltage regulation and energy management. Since *SOLARIS* primarily operates from a battery charged from a solar panel, efficiency is an essential consideration. This consideration meant that we chose switching regulators for both the 3.3 V and 5 V regulators to minimize complexity and cost since we would only be purchasing a single dual-buck chip. The other consideration was minimizing the solar panel cost. To meet this goal we chose a material and model that has high efficiency, did not have an excessive size, and good thermal response while also not weighing an unnecessary amount which proved to be a challenge due to the demand of solar energy and the specialty material. Additionally we needed to purchase a battery that was small enough to for it in a reasonable robot size while also having the desired capacity and ease of integration.

Chapter 5 - Comparison of ChatGPT with other Similar Platforms

5.1 Brief

As generative artificial intelligence becomes an increasingly integrated tool in the engineering workflow, evaluating the efficacy and accuracy of different Large Language Models (LLMs) is essential for technical validation. This chapter presents a series of case studies comparing OpenAI's ChatGPT against other industry-leading platforms, including Google's Gemini, xAI's Grok, and Anthropic's Claude. The objective of this comparative analysis is to determine which platform provides the most actionable, well-cited, and technically sound data for a complex robotics project like SOLARIS. By subjecting these models to prompts regarding battery chemistry, motor selection, and sensor specifications, the following sections analyze the variations in response depth, formatting, and the utility of the provided design frameworks.

5.2 Case Study 1

Please refer to Appendix B, [1a], [1b], and [1c] for the Case Study 1 prompt and the corresponding outputs generated by OpenAI's ChatGPT and Google's Gemini, both operating in deep research mode.

ChatGPT's response was concise and well-structured. The introductory paragraph clearly identified the five most common battery chemistries used in renewable solar applications, along with their key strengths and typical use cases at a high level. This was followed by one paragraph per chemistry, where each was analyzed systematically in terms of performance parameters and primary drawbacks. Notably, each paragraph included at least four in-text citations which made verification of the information straightforward and efficient.

The response by ChatGPT concluded with a horizontal bar chart comparing typical battery lifespans over time. This visualization reinforced ChatGPT's implicit conclusion that lithium iron phosphate ($LiFePO_4$) is the most favorable option, as it clearly demonstrated its superior lifespan relative to the other chemistries discussed.

In contrast, Gemini's response was significantly more detailed- approximately five times longer than ChatGPT's. It adopted a broader, big-picture perspective by examining the evolution of battery selection for renewable energy applications, including trends in the market from 2010 to 2024. While the introduction similarly listed the five chemistries covered, it remained more general and did not immediately compare them. Although this level of abstraction was not explicitly requested in the prompt, it did not detract from the overall quality of the response.

Each section in Gemini's response included tables that provided additional context, either by comparing chemistries directly or by evaluating their suitability for different applications. Furthermore, Gemini incorporated sections not present in ChatGPT's response, including a comprehensive comparison of economic factors, safety considerations, environmental impact, and installation requirements. The response concluded with a "Strategic Deployment Summary," which identified $LiFePO_4$ as the industry benchmark.

Both responses included a total of approximately 20 cited sources, offering strong opportunities for further research into battery chemistries. Although Gemini's response was less concise and included tables contrary to the prompt's instructions, it was ultimately preferred due to its more comprehensive comparative analysis and its clear articulation of the advantages and disadvantages across all five chemistries.

5.3 Case Study 2

Please refer to Appendix B, [2a], [2b], and [2c] for the Case Study 2 prompt and the corresponding outputs generated by ChatGPT and Grok.

ChatGPT's response was extremely expansive and detailed in its analysis. It began by summarizing why DC motors are desirable in mobile robotics platforms. It went on to break down various motor types and their use cases citing controllability, efficiency and compatibility as a main reason. ChatGPT then went on to explain how DC motors are integrated into, and how they fit into the larger overall system.

ChatGPT then concluded its research by discussing various motor control technologies and methods commonly used in mobile robotics. It described basic controls such as PWM and H-Bridges for brushed DC motors and inverters for brushless motors. It also discussed more advanced control techniques such as using closed loop feedback control to allow the robot to maintain speed over any kind of terrain or laid that hinders its movement.

In contrast, Grok had a much shorter response but with far more pictures included in its explanation. It began by giving a succinct and directly to the point overview of the types of DC motors used in mobile robotics platforms but included a more technical and low level approach that included the use of schematics and drawings to aid understanding. The main difference here was Grok got more into the details on construction and technical operation rather than wider robot design.

Grok then talked about motor control methods and technologies. Grok spoke about H-Bridge and PWM control for Brushed motors and inverter control for brushless DC motors like ChatGPT did as well has discussed the use of PID, FOC and sensorless controls. THe main difference was Grok treated it as more of an assorted list of methods. One thing that Grok did include was a discussion of the use of AI based tuning for control systems.

Overall, ChatGPT's explanation was more generalized system level and application focused explaining how motor choices will affect a given project's performance. Grok's response was more based on component level descriptions, principles of operation and how they apply to the scope of mobile robots.

5.4 Case Study 3

Please refer to Appendix B, [3a] and [3b] for the Case Study 3 prompt and the corresponding output generated by Claude.

Claude [1b] provided a great starting point on what should be considered while looking for ultrasonic sensors for a design project. The response provided a comprehensive breakdown of the most critical selection criteria for an ultrasonic sensor including detection range, beam width, resolution, output type, (UART, I2C, PWM, etc..) update rate, environmental sensitivity, power consumption, cross-talk, physical form factor, and cost and availability. While all of these factors were acknowledged and considered during the selection process, the response helped the team establish a clear prioritization, identifying range and beam width as the two most impactful criteria for *SOLARIS'* use case, while confirming that the remaining factors, though still relevant, were secondary concerns that the chosen sensor addressed.

Detection range was the first major filter applied. *SOLARIS* is a robot intended to navigate in the real world, meaning the sensor needed to provide enough advance warning for it to reroute before colliding with said obstacle. The MB1020-000 offers a maximum range of 6.5 meters which gives ample time for *SOLARIS* to see obstacles and change course. Claude framing range as a top tier concern reinforced that this should be a hard requirement rather than a preference.

Beam width was the second primary consideration and required the most analysis. The MB1020-000 has a beam width of 42 degrees, which the team evaluated mathematically to confirm it was appropriate for *SOLARIS'* geometry and obstacle detection goals. A beam that is too narrow risks missing obstacles that fall outside the sensor's cone, while a beam that is too wide can introduce picking up on objects not in front of *SOLARIS*. The 42 degree requirement was a good middle ground to choose from. Claude's response's called out the beam width as another top tier priority, which again reinforced this as another hard requirement.

Beyond the two primary criteria, Claude's response also prompted the team to verify other things such as the update rate, which turned out to be a non-issue for the MB1020-000. *SOLARIS* is designed to drive at slow speeds, and the minimum read time of 50ms means we should reasonably be able to read from each sensor multiple times per second. Although this was a small consideration, all of Claude's responses helped make sure we didn't miss any specification during the verification process.

Overall, Claude's [1b] response served as a useful decision making framework that aided in the sensor selection process.

5.5 Case Study 4

Chapter 6 - Hardware Design

6.1 Brief

6.2 Subsystem Block Diagram

6.3 Schematic Diagram

6.4 Architecture

6.5 Structural Illustration

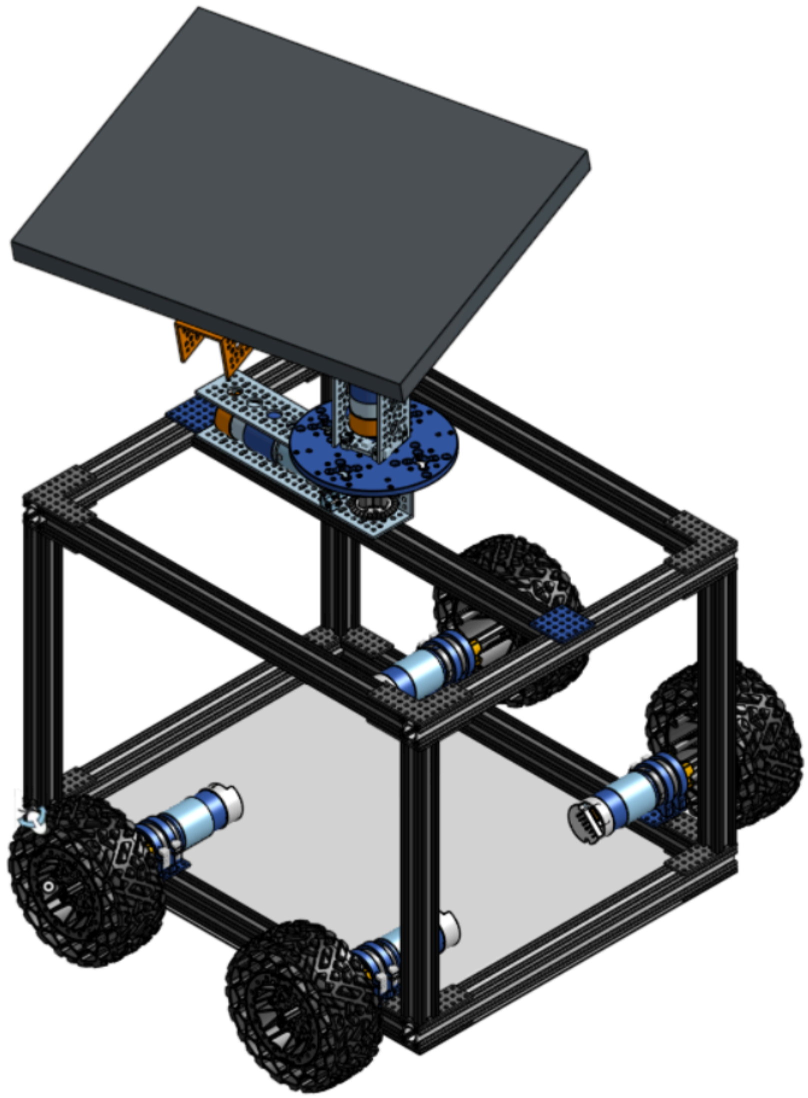


Figure 29. *SOLARIS Chassis Without Covers*

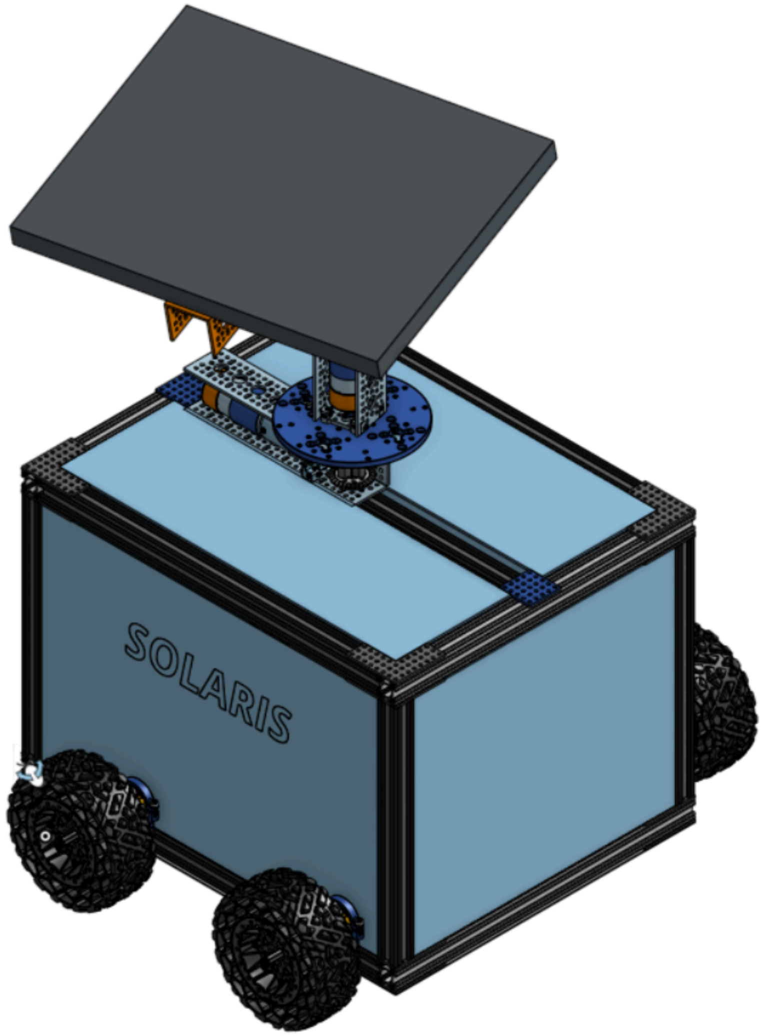


Figure 30. *SOLARIS Chassis With Covers*

Chapter 7 - Software Design

7.1 Brief

7.2 Case and State Diagrams

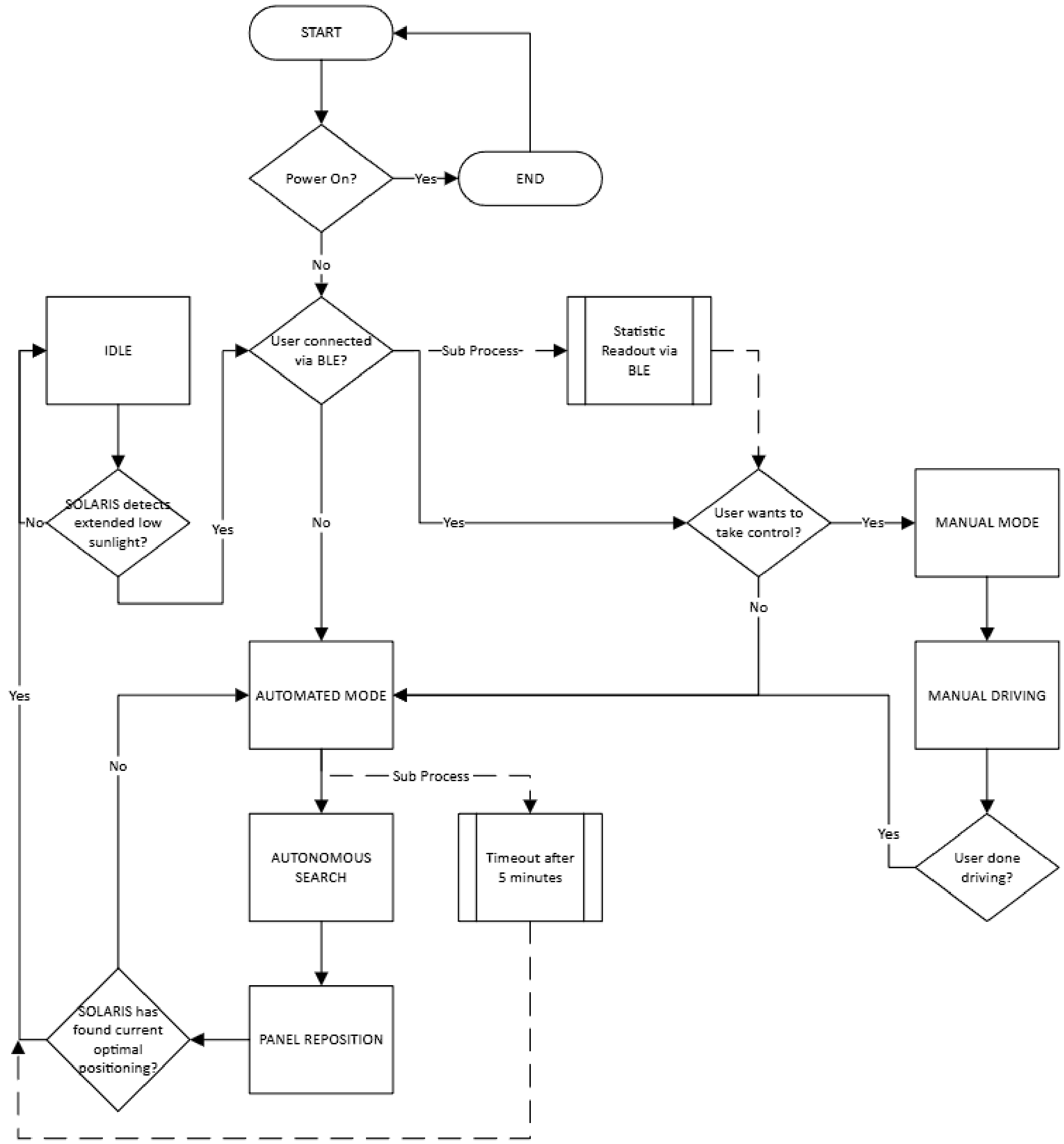


Figure 31. *Driving Software State Diagram*

7.3 Software Flowcharts

7.4 Data Transfer

7.5 User Interface

Figure 32. *SOLARIS Phone App Home Screen*

The home page presents two options: Sign In and Sign Up. Tapping either button navigates to the respective authentication screen. If the app detects an existing session, it skips the home page entirely and routes the user directly to the main dashboard.

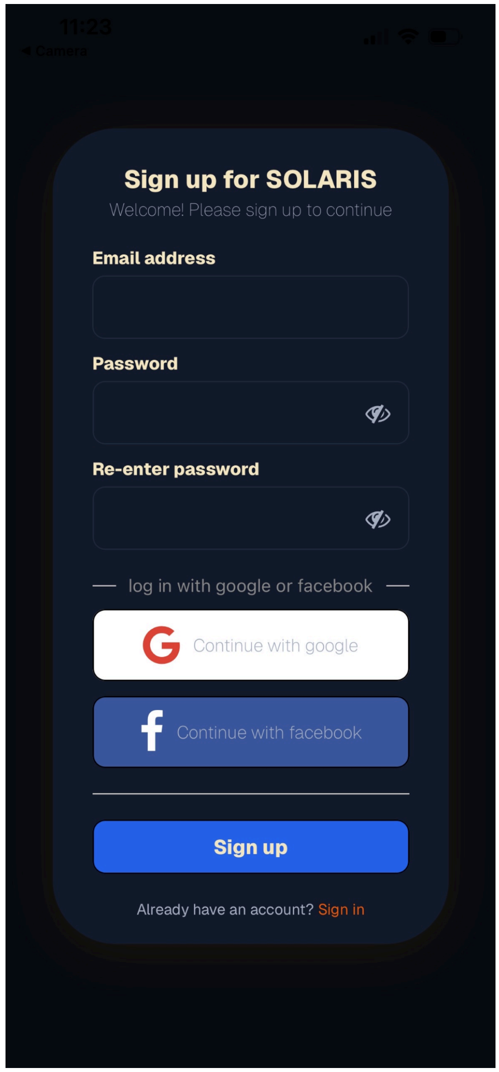
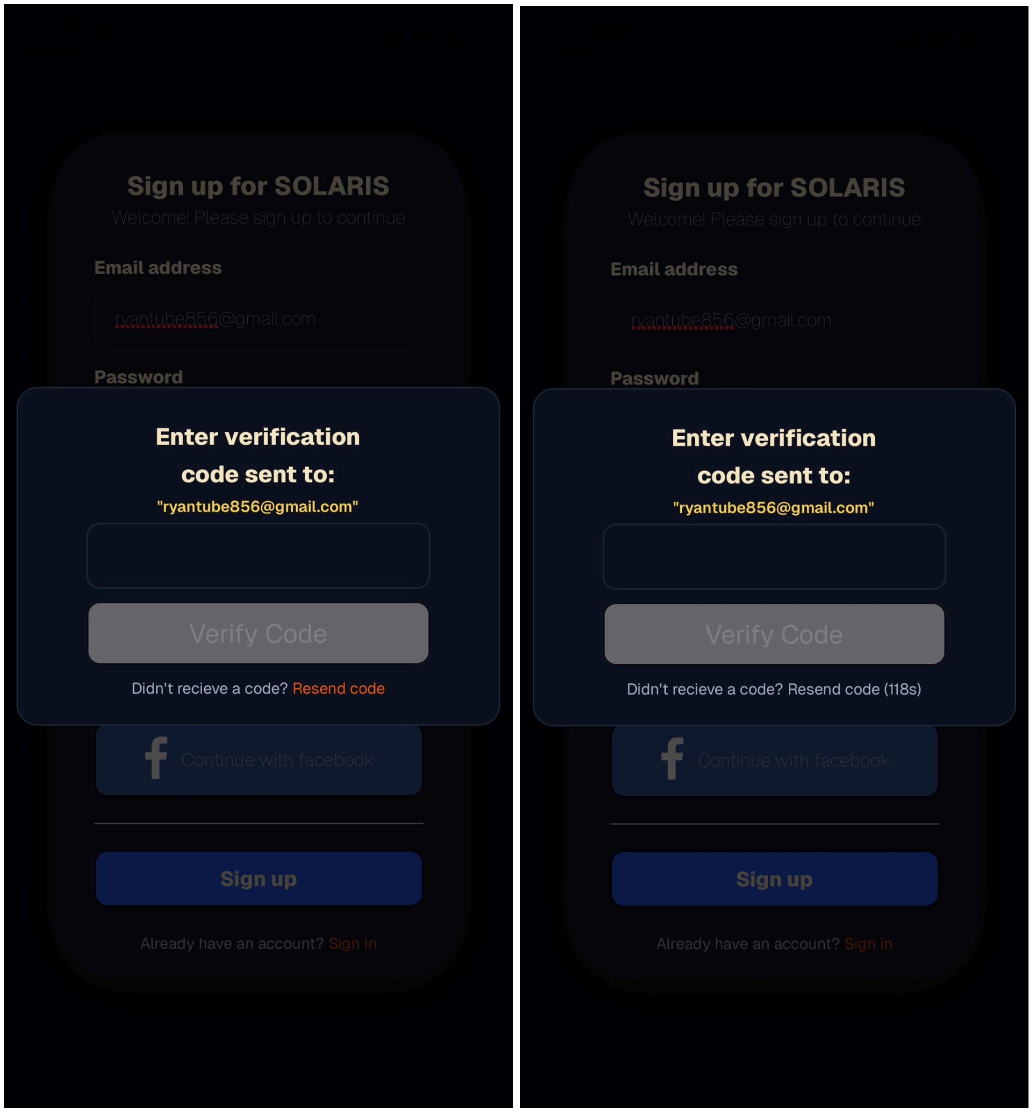


Figure 33. *SOLARIS Phone Sign Up Screen*

The sign-up screen offers two paths: OAuth via Google or Facebook, or manual registration with an email and password. For manual sign-up, the user enters their email, creates a password, and confirms it. Authentication is handled by Clerk. Dynamic inline alerts provide real-time feedback for input errors such as invalid email format or passwords that don't meet minimum length requirements.



Figures 34 and 35. *SOLARIS Phone Verification Functionality*

After submitting valid credentials, the user is prompted to enter a 6-digit verification code sent to their email. The verify button remains disabled until all 6 characters are entered. If the code wasn't received, a resend option is available, though it's rate-limited to once every 2 minutes. Once a valid code is submitted, Clerk verifies it server-side and the user is granted access as a registered account.

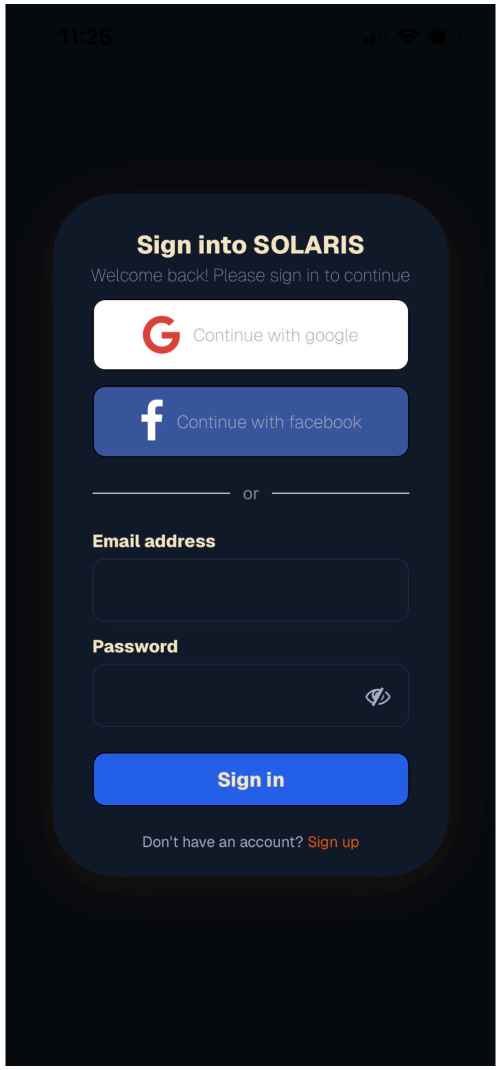


Figure 36. *SOLARIS Phone Sign In Page*

The sign-in screen mirrors the sign-up flow, offering OAuth via Google or Facebook or manual entry with a registered email and password. Dynamic alerts surface credential errors in real time, such as an incorrect password or unrecognized email. Once Clerk verifies the credentials, the user is routed into the authenticated portion of the app.

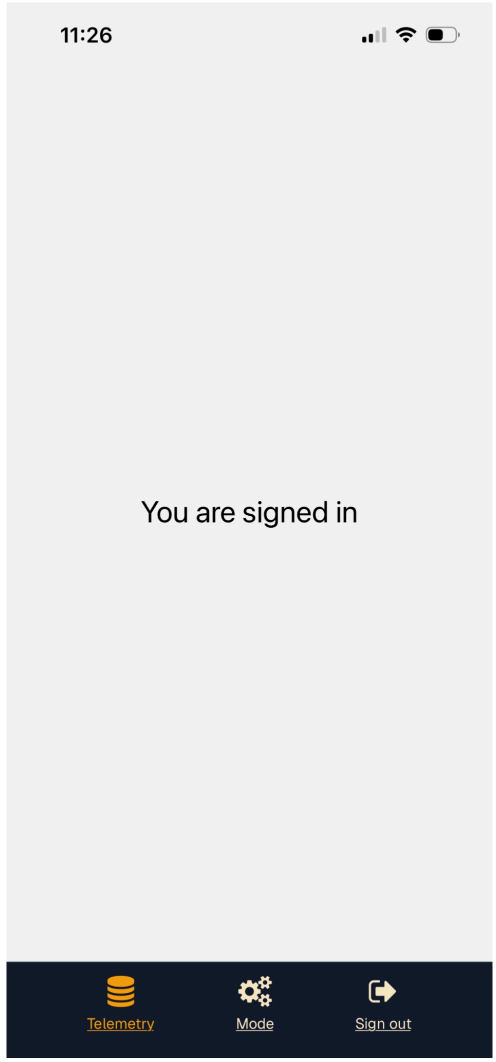


Figure 37. *SOLARIS Signed In Page*

The signed-in page is still under active development and will expand as the companion app is completed. Currently, the footer bar is functional, offering navigation between the Telemetry and Mode Selection pages, along with a sign-out button. Full documentation for this section will be added once the app is finalized.

Chapter 10 - Administrative Content

10.1 Budget

The total budget allocated for the *SOLARIS* project is approximately $1,000. This budget is intended to cover all major system components, including mechanical hardware, electronic components, sensing and actuation devices, power management hardware, and prototyping materials. Funds are also reserved for iterative development and testing, allowing for component replacement or redesign as needed throughout the project lifecycle. Establishing a defined budget constraint ensures that design decisions remain practical and encourage cost-effective engineering tradeoffs while still meeting the project's functional and performance objectives.

10.2 Bill of Materials

***Table 40.** Itemized Bill of Materials*

Part Name	Part Number	Market Availability								Total # Ordered		Cost	
		DigiKey	Min Qty.	Amazon	Min Qty.	Mouser	Min Qty.	Third Party	Min Qty.	Minimum Needed	Total Purchased	Minimum Cost	Total Cost
ESP-32-S3 Microcontroller Chip	ESP32-S3-WROOM-1-N16R8	$6.56	1	N/A		$6.71	1	N/A		1			
ESP-32-S3 (Dev Kit)	ESP32-S3-WROOM-1-N16R8(Dev Board)	N/A		$23.74	3	N/A		N/A		0	3		
DC Drive Motor	5204-8002-0100	N/A		N/A		N/A		goBILDA: $56.99	1	4			
Pan/Tilt Motor	5203-2402-0188	N/A		N/A		N/A		goBILDA: $54.99	1	2			
STMicroelectronics VNH5019ATR-E	VNH5019ATR-E	N/A		N/A		$9.22	1	N/A		6			
Newpowa Solar Panel	NPA20S-12J	N/A		$37.99	1	N/A		Newpowa: $39.9	1	1			

								9					
Genasun Multi-Power-Point-Tracker	GV-5-LI-14.2V	$99.00	1	$104.00	1	N/A		N/A		1			
TEPT Phototransistor	TEPT5600	$0.70	1	$21.08	10	$0.61	1	N/A		8			
Energy Monitor Unit (Dev Kit)	INA228EVM 5832 var.	$14.95	1	$19.98	1	$14.95	1	N/A		0			
TI Energy Monitor Unit (Automotive Grade)	INA228AQDGSRQ1	$5.27	1	N/A		$5.27	1	N/A		1			
Inertial Measurement Unit (IMU)	BNO085	$24.95	1	$43.12	1	$24.95	1	N/A		1			
Voltage Regulator 3.3V & 5V	LT8650SPJV	$21.47	1	N/A		N/A		N/A		1			
Ultrasonic Sensor	MB1020-000	$19.95	1	N/A		N/A		MaxBotix: $24.95	1				

10.3 Distribution of Work

Table 41. *Distribution of Work*

Electrical Engineering	***Responsibilities***
Rafael Puig	Input and output port configuration from the user
	Battery safety and power system design
	Lead PCB design
Electrical Engineering	***Responsibilities***
Nathan Hammond	Servo and motor technology
	Design of the power delivery system
	Co-Lead PCB design for drivetrain control
Computer Engineering	***Responsibilities***
Ryan Markowitz	Phone app design for interface with the MCU

	Software lead for IoT connection
	Co MCU firmware design
Computer Engineering	***Responsibilities***
Garrett Fortier	Lead MCU firmware design and debug
	Automation for the self-driving algorithm and modularity
	Hardware/Software integration

10.4 Milestones

Table 42. *Milestone Progress*

Milestone Number	*Milestone Description*	*Start Date*	*Anticipated End Date*	*Dependencies & Requirements*
#1	Recruitment	01/12/2026	01/16/2026	Scout for potential team members interested in working together
#2	Brainstorming	01/16/2026	01/22/2026	Talk with one another briefly over a week to discuss ideas
#3	D&C Document	01/22/2026	01/30/2026	Divide up the task and work independently to add relevant information
#4	Mentor Meeting	02/04/2026	02/04/2026	Meet with Dr. Suboh for D&C revision
#5	60-Page Milestone	02/06/2026	03/27/2026	Work will be divided up, and then relevant portions of the paper will be written

#6	Final Document	03/01/2026	04/28/2026	All group written portions will be formatted onto one document

Appendices

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Appendix B – AI References

[1a] **Case Study 1 Prompt:** "Write a concise explanation comparing the five most common battery chemistries used in solar systems, using natural paragraphs instead of structured tables."

[1b] **Case Study 1 ChatGPT Deep Research Outcome:** "***Solar Battery Chemistries: An Executive Summary***
Among common solar batteries, lithium iron phosphate ($LiFePO_4$) leads for long life and safety, while standard lithium-ion (NMC) packs more energy per weight. Traditional lead–acid types cost less upfront but age faster and need upkeep. AGM/gel (sealed lead–acid) improve on flooded types with less maintenance. NiMH batteries are rarely used in large solar systems, finding niche use in small solar lights and electronics.

$LiFePO_4$ (Lithium Iron Phosphate)
$LiFePO_4$ batteries have become the go-to choice for home solar storage because they last a long time and are very stable. A good $LiFePO_4$ battery usually gives 8–12 years of life (often 10–15) with a very high cycle count – typically 2,000–6,000 full cycles at about 80% depth-of-discharge. They retain voltage well across discharge, can be drained deeply daily (most systems use 20–80% each cycle), and require essentially no maintenance (no watering or cleaning corrosion). $LiFePO_4$ chemistry is very thermally stable and safe (far less fire risk than other lithium types). Round-trip efficiency is excellent – often above 92%, meaning most of the solar power you store comes back out. The trade-off is higher initial cost (more than lead-acid), but because they last so much longer with no upkeep, their cost per usable kilowatt-hour is low.

NMC (Nickel-Manganese-Cobalt Lithium-Ion)
Standard lithium-ion (NMC) batteries are common in electronics and some solar products. They pack more energy into a given space than $LiFePO_4$ (higher energy density) but have some tradeoffs. A typical NMC home battery can cycle a few thousand times (often up to 3,000–5,000 partial cycles), giving roughly a 10–12 year lifespan with

good care. NMC units have very high round-trip efficiency (around 90–95% or even up to ~98% under ideal conditions), so you lose little energy charging them. However, they are more expensive upfront (especially since they use cobalt) and their chemistry is less forgiving: they require strong battery management to avoid overheating or overcharging. In practice, NMC batteries deliver large bursts of power (good C-rate) and are lighter, making them attractive when space is tight (e.g. wall-mounted storage or electric vehicles). But for typical home solar use, their slightly shorter life and greater sensitivity mean $LiFePO_4$ is often preferred unless weight/size is critical.

Flooded Lead–Acid

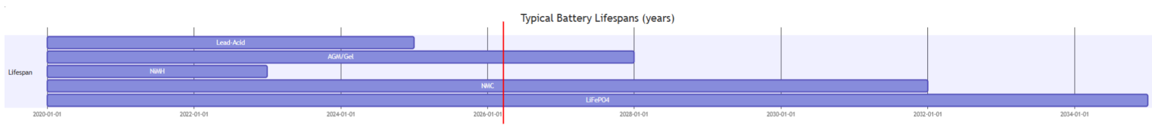
Flooded (wet cell) lead–acid batteries are the classic, lowest-cost option. You'll often see them in off-grid cabins or diesel backup because you can get a lot of capacity for cheap. They perform adequately but have important drawbacks: their round-trip efficiency is only about 75–85% (a lot of energy is lost as heat), and you should never regularly discharge them beyond ~50% without hurting their life. Typical cycle life is low (hundreds of cycles), so in a daily solar-use scenario they usually last only 3–5 years before they fall below ~80% capacity. Maintenance is high: flooded cells must be kept topped up with distilled water, cleaned of corrosion, and equalized regularly. They also emit hydrogen gas when charged, so they need ventilation and safe handling. On the plus side, flooded lead–acid has the lowest upfront cost of all chemistries, making it a budget-friendly starter choice. But owners should expect more frequent replacements, lower efficiency, and a lifespan of just a few years unless the system is very carefully managed.

AGM/VRLA (Sealed Lead–Acid – AGM and Gel)

AGM and gel batteries are sealed varieties of lead–acid. They use absorbed glass mats or gelled electrolyte so you don't have to add water. They bridge the gap between flooded lead–acid and lithium. Efficiency is still on the order of 80–85% (similar to flooded), but sealed construction means much lower routine maintenance (no watering, no acid spills). Sealed lead–acid typically achieves a bit longer life – roughly 800–1,500 cycles or about 4–8 years in a solar cycling setup. They handle deep discharges somewhat better than flooded, but manufacturers still often limit you to ~50–70% DoD to maximize life. Safety is fair: AGM/gel will not leak, but they can overheat if overcharged and lack the intrinsic safety of $LiFePO_4$. Upfront cost is higher than flooded lead–acid (usually a medium price point), yet they remove the fuss of maintenance. In practice, VRLA batteries are often used for small to mid-size off-grid kits or backup systems where you want sealed convenience, but they will still need replacement roughly every 5–8 years under normal solar usage.

NiMH (Nickel–Metal Hydride) and Other Nickel-Based

Nickel–metal hydride batteries are unusual in home solar. They were common in older consumer electronics and hybrid cars, but rarely used for whole-house storage. NiMH cells are safer than the old NiCd (no toxic cadmium) and have decent charge efficiency (charging up to ~70% state is nearly 100% efficient). However, they have lower energy density and higher self-discharge, so they aren't a great fit for large stationary use. NiMH cycle life is modest – on the order of 300–500 full cycles. In practical terms that means if used daily, they might only last ~1–3 years before needing replacement. Typical round-trip efficiency is lower than lithium (charging generates heat near end of charge, wasting energy), and they still require proper charging termination to avoid overheat. Because of this, NiMH tends to show up only in very small solar applications: for example, garden lights, sensors, or portable devices often use AA- or C-size NiMH packs powered by a solar panel. For a homeowner looking at a solar battery bank, NiMH is not a competitive choice – it's cheap to buy, but it has short life, modest efficiency, and high self-discharge, meaning it's rarely used for home energy storage.



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[1c] **Case Study 1 Gemini Deep Research Outcome:** "***The Global Transition of Energy Storage: A Comparative Analysis of Primary Battery Chemistries in Solar Systems***

The global trajectory of renewable energy integration has reached a pivotal juncture in 2026, where the efficacy of solar power is no longer defined solely by photovoltaic efficiency but by the sophistication of the associated energy storage systems. As the decentralized energy landscape matures, the selection of battery chemistry has evolved from a secondary technical consideration into a primary strategic pillar for residential, commercial, and utility-scale stakeholders. This transformation is underpinned by a staggering 93% decline in system-level battery storage costs between 2010 and 2024, falling from $2,571 per kilowatt-hour (kWh) to approximately $192/kWh$.[1] This economic shift, coupled with intensified research into material science, has solidified five primary chemistries as the dominant architectures for solar storage: Lithium Iron Phosphate ($LiFePO_4$ or LFP), Lithium Nickel Manganese Cobalt Oxide (NMC), Lead-Acid variants, Redox Flow systems, and the burgeoning Sodium-Ion (Na-ion) technology.[2]

The fundamental role of energy storage in a solar-integrated grid is the temporal shifting of power—capturing generation during peak solar irradiance for use during evening *peak demand or providing critical resilience during grid outages.*[5] *However, the physical and chemical constraints of each battery type dictate its suitability for specific operational profiles. While high energy density remains the priority for the mobile transportation sector, the stationary storage market has pivoted toward safety, cycle life, and levelized cost, leading to a significant divergence in chemical adoption.*[7] *By 2025, the global residential battery storage market was valued at $6.85 billion, with projections suggesting a rise to $8.73 billion by 2026, reflecting a compound annual growth rate (CAGR) of 17.80%.*[5]

Lithium Iron Phosphate ($LiFePO_4$): The Technical and Economic Benchmark

Lithium Iron Phosphate (LFP) has emerged as the unequivocal leader in the 2026 stationary storage market, capturing an estimated 85% share of utility-scale projects and a dominant portion of new residential installations.[1] *The ascent of LFP is not a result of superior energy density—it actually offers less capacity per kilogram than other lithium variants—but rather its exceptional safety profile and lifecycle economics.*[2]

Chemical Architecture and Thermal Stability

LFP batteries utilize an iron phosphate cathode, which provides a robust olivine crystalline structure. This structure is significantly more stable than the metal-oxide layers found in chemistries like NMC or LCO.[8] *In practical terms, the* $P - O$ *bond in the phosphate cathode is much stronger, which prevents the release of oxygen during chemical decomposition or thermal stress.*[8] *This characteristic virtually eliminates the risk of thermal runaway, a self-sustaining heating cycle that can lead to fires or explosions.*[2]

The thermal stability of LFP is a decisive factor in the 2026 regulatory environment. LFP cells remain stable at temperatures exceeding $200^\circ C$ *(*$392^\circ F$*), whereas other lithium chemistries may reach critical instability at temperatures as low as* $150^\circ C$*.*[9] *This inherent safety simplifies permitting processes and reduces the need for expensive, high-capacity fire suppression systems, lowering the balance-of-system (BOS) costs for commercial and industrial (C&I) projects.*[8]

Performance Metrics and Lifecycle Longevity

From an operational standpoint, LFP offers a superior value proposition over its lifespan. While its upfront cost can range from $600 to $900 per kWh *for high-performance residential systems, it delivers between 6,000 and 10,000 full charge-discharge cycles before dropping below 80% of its original capacity.*[7] *In a typical daily solar cycling application, this translates to a service life of 15 to 20 years, far* *outlasting the 10-year industry standard of the previous decade.*[7]

LFP batteries also allow for a significant depth of discharge (DoD), often rated between 95% and 100%.[2] *This means a homeowner with a* $10kWh$ *LFP system can utilize nearly all the stored energy every day without damaging the internal chemistry. When compared to lead-acid systems that are limited to a 50% DoD to preserve health, the usable energy of an LFP system is effectively double for the same nameplate capacity.*[13]

Metric	***Lithium Iron Phosphate (LFP)***	***Industry Context (2025-2026)***
Cycle Life	*6,000 – 10,000 cycles*	*Highest among lithium variants* [7]
Depth of Discharge	*95% – 100%*	*Maximizes usable capacity* [7]
Round-Trip Efficiency	*95% – 98%*	*Minimal energy loss* [7]
Thermal Stability	*Stable up to* $>$	*Safe for indoor installation* [9]
Energy Density	*120 – 160* Wh/kg	*Lower than NMC, higher than lead-acid* [7]
Lifespan	*15 – 20 years*	*Ideal for long-term solar ROI* [7]

Lithium Nickel Manganese Cobalt Oxide (NMC): The High-Density Specialist

Lithium Nickel Manganese Cobalt Oxide (NMC) represents the primary alternative within the lithium-ion family for solar applications. Historically favored for its high energy density, NMC has seen its share of the stationary storage market decline in favor of LFP, yet it remains a critical choice for space-constrained environments.[7]

Energy Density and Space Constraints

NMC batteries utilize a mixed-metal cathode that enables significantly higher energy density than LFP, typically ranging from 150 to 250 Wh/kg*.*[7] *This density allows manufacturers to pack a larger amount of energy into a smaller, lighter enclosure. For* *urban residential settings or commercial retrofits where physical space is at an absolute premium, NMC remains a vital technology.[7] A standard NMC-based storage unit might be 30% to 50% more compact than an LFP unit of equivalent capacity, facilitating easier wall mounting and reducing the footprint in garages or utility rooms.[10]*

Trade-offs: Lifecycle and Safety

The high energy density of NMC comes with inherent trade-offs in longevity and safety. NMC batteries generally offer a shorter cycle life, typically between 3,000 and 5,000 cycles.[7] In a daily cycling solar application, this results in a service life of 10 to 15 years, requiring replacement sooner than an LFP alternative.[7] Additionally, to maintain the health of the chemistry, installers often recommend limiting the depth of discharge to 80% or 90%, which reduces the "true" usable energy compared to the nameplate rating.[13]

Safety remains a significant point of comparison. NMC chemistries are more susceptible to thermal runaway due to the nature of the metal-oxide bonds in the cathode, which can release oxygen during decomposition and fuel a fire.[8] In the 2026 market, this has led to increased scrutiny from insurers and local fire authorities, often necessitating more complex thermal management systems and increased physical spacing between units, which can offset some of the initial space-saving advantages.[8]

Feature	***Lithium NMC***	***Strategic Implication***
Energy Density	*150 – 250* Wh/kg	*Best for space-constrained sites* [7]
Cycle Life	*3,000 – 5,000 cycles*	*Shorter ROI window than LFP* [7]
Depth of Discharge	*80% – 90%*	*Limited usable energy per cycle* [13]
Safety Rating	*Moderate*	*Requires advanced BMS and cooling* [7]
Cost per kWh	*$500 – $700*	*Historically cheaper, now competing with LFP* [7]

Lead-Acid Storage: Evolutionary Persistence in a Lithium-Dominated Era

Lead-acid batteries are the oldest form of rechargeable energy storage and, despite the rapid rise of lithium, they maintain a presence in the 2026 solar market.[12] Their *persistence is driven by low upfront costs and a deeply established global infrastructure for manufacturing and recycling.*[7]

Variations: Flooded, AGM, and Gel

The lead-acid market for solar is divided into three primary categories: Flooded (Wet Cell), Absorbed Glass Mat (AGM), and Gel.

1. ***Flooded Lead-Acid (FLA):*** *These are the most economical upfront but require significant maintenance. Users must regularly check and replenish distilled water levels and perform equalization charges to prevent sulfation—the buildup of lead sulfate crystals on the plates.*[7] *They also emit hydrogen gas during charging, necessitating well-ventilated dedicated enclosures.*[7]
2. ***AGM (Absorbed Glass Mat):*** *A sealed, maintenance-free variant where the electrolyte is held in fiberglass mats. AGM batteries offer better efficiency than flooded cells and handle high discharge rates more effectively.*[7] *They are common in budget-conscious backup systems that are not cycled daily.*[7]
3. ***Gel Batteries:*** *These use a silica additive to turn the electrolyte into a stiff gel, making them exceptionally resistant to spills and more stable in extreme temperatures.*[7] *Gel batteries are particularly suited for off-grid applications in harsh climates, although they are highly sensitive to overcharging.*[12]

Efficiency and Depth of Discharge Constraints

The primary limitation of lead-acid technology in a solar context is its inefficiency and short lifespan. Lead-acid batteries generally operate at 70% to 85% round-trip efficiency, compared to over 95% for lithium.[11] *This means a significant portion of the solar energy collected is lost as heat during the storage process.*

Furthermore, to prevent permanent chemical damage, lead-acid batteries should not be discharged below 50% of their capacity.[7] *A user requiring* $10kWh$ *of usable daily energy would need to install a* $20kWh$ *lead-acid bank. When combined with a cycle life that often falls below 1,000 cycles for flooded cells, the frequent replacement requirements often make lead-acid the most expensive option on a levelized cost basis over 10 to 15 years.*[11]

Type	***Cycle Life (50% DoD)***	***Maintenance Requirement***	***Ideal Use Case***

Flooded (FLA)	*500 – 1,000*	*High (Water/Venting)*	*Budget off-grid* [12]
AGM	*1,000 – 1,500*	*None (Sealed)*	*Backup/Standby* [12]
Gel	*1,000 – 1,500*	*None (Sealed)*	*Extreme temperatures* [12]
Efficiency	*60% – 85%*	*N/A*	*Lowest upfront cost* [13]

Redox Flow Battery Systems: Architectures for Multi-Hour Grid Resilience

Redox Flow Batteries (RFB) represent a fundamental departure from the "closed-cell" design of lithium and lead-acid. Instead of storing energy in solid electrodes, flow batteries store energy in liquid electrolyte solutions held in external tanks.[2] These liquids are pumped through a central membrane stack where the chemical reaction occurs.[2]

Decoupling Power and Energy

The unique advantage of flow batteries is the ability to decouple power capacity (determined by the size of the membrane stack) from energy capacity (determined by the volume of the electrolyte tanks).[16] For a solar farm needing to bridge 10 hours of darkness, a flow battery can be scaled simply by increasing the size of its storage tanks, providing a lower marginal cost for long-duration storage.[14]

In 2026, the two primary flow technologies are Vanadium Redox Flow Batteries (VRFB) and Iron Flow Batteries. Vanadium is the more mature technology, offering high electrochemical stability and the ability to reuse the electrolyte indefinitely.[16] Iron flow batteries are emerging as a cheaper alternative, utilizing abundant, non-toxic materials, although they are still moving toward large-scale commercial saturation.[16]

Lifespan and Safety Profile

Flow batteries are unparalleled in their longevity. They can be cycled over 20,000 times with zero degradation, often lasting 25 to 30 years—effectively matching the lifespan of the solar panels themselves.[14] They also allow for 100% depth of discharge every single day without any damage to the system.[14]

Safety is another hallmark of flow technology. Because the electrolytes are typically water-based and non-flammable, they possess zero fire risk.[14] This allows them to be classified as "non-hazardous" for indoor or near-building installation, saving significant costs on fire suppression and specialized insurance.[14] However, their low energy density

(20 to 35 Wh/kg) and the footprint required for liquid tanks make them unsuitable for residential garages, relegating them to utility-scale solar farms or heavy industrial sites.[11]

Characteristic	***Vanadium Flow (VRFB)***	***Strategic Advantage***
Cycle Life	*20,000 – 30,000+*	*"Buy once" for 25-30 years* [14]
Depth of Discharge	*100%*	*Full utilization of tanks* [14]
Safety	*Non-flammable*	*No thermal runaway risk* [17]
Degradation	*Near zero*	*Electrolyte never wears out* [19]
Physical Size	*Very Large*	*Scalable for grid-level storage* [14]
Efficiency	*70% – 80%*	*Lower than lithium due to pumps* [16]

Sodium-Ion (Na-ion) Technology: The Frontier of Resource-Agnostic Storage

As of 2026, Sodium-Ion (Na-ion) technology has emerged as a disruptive contender to the lithium-ion monopoly. Its value proposition is centered on resource abundance and supply chain stability, as sodium is approximately 500 times more abundant than lithium and can be sourced globally from seawater or salt deposits.[4]

Material Advantage and Manufacturing Synergy

Sodium-ion batteries function on a principle similar to lithium-ion, but utilize sodium ions ($Na+$ *) which are larger and heavier than lithium ions (* $Li+$ *).*[21] *This necessitates different cathode and anode materials—typically hard carbon for the anode instead of graphite.*[21] *A critical economic advantage for sodium-ion is the ability to use aluminum current collectors for both the cathode and the anode, whereas lithium cells require more expensive copper for the anode.*[4]

Remarkably, sodium-ion is a "drop-in" technology that can be manufactured on existing lithium-ion production lines with only minor modifications.[22] *This has allowed major* *manufacturers, particularly in China (which accounts for over 60% of global storage production), to rapidly scale sodium-ion as a hedge against lithium price volatility.*[6]

Performance in Extreme Conditions

Sodium-ion batteries exhibit superior performance in extreme temperatures compared to standard lithium chemistries. They can discharge normally in environments as cold as $-40^\circ C$ *and maintain 80% capacity at* $-20^\circ C$*, a range where LFP batteries often struggle or require energy-intensive heating systems.*[20] *Furthermore, they are inherently safer to transport; because they can be discharged to zero volts without damage, they can be shipped as "non-dangerous" goods without the residual fire risk associated with lithium batteries.*[4] *While their current energy density (110-160* Wh/kg*) is on the lower end of the lithium spectrum, they represent the most promising path toward low-cost, sustainable mass storage.*[20]

Attribute	***Sodium-Ion (Na-ion)***	***LFP Comparison***
Raw Material Cost	*Very Low*	*Higher (Lithium/Iron)* [20]
Low-Temp Performance	*Excellent (*$-40^\circ C$*)*	*Moderate* [20]
Shipping Safety	*Safe at 0 Volts*	*Risk at 0 Volts* [4]
Charge Speed	*Up to 90% in 15 mins*	*Slower (typically 1-2 hours)* [20]
Cycle Life	*2,000 – 6,000 (Current)*	*6,000 – 10,000* [20]
Current Status	*Early commercialization*	*Mature market leader* [22]

Comparative Economic Analysis and Levelized Cost of Storage (LCOS)

To evaluate these chemistries for professional deployment, one must look beyond the initial capital expenditure ($CAPEX$*) and calculate the Levelized Cost of Storage (*$LCOS$*). This metric represents the cost per unit of energy delivered over the entire lifetime of the system, accounting for degradation, efficiency, and replacement* *frequency.*[14]

Upfront Price vs. Long-Term Value

In 2026, LFP systems often appear more expensive upfront ($650-$850/kWh installed) compared to lead-acid ($150-$300/kWh).[13] *However, when viewed over a 25-year horizon, the picture flips. In regions like New Zealand, the* $LCOS$ *for LFP is estimated at 18–28 cents per* kWh*, while lead-acid exceeds 30–40 cents due to the need for multiple pack replacements and lower usable capacity.*[14]

Flow batteries present the most dramatic long-term savings. While they have the highest upfront cost ($900-$1,200/kWh for smaller C&I systems), their 25-year $LCOS$ *drops to 11–17 cents per* kWh *because they essentially never wear out and require minimal maintenance.*[14] *For large-scale utility projects, the* $LCOS$ *for flow systems is increasingly the benchmark for grid stability tenders.*[14]

Future Cost Trajectories (2030-2050)

Projections for sodium-ion suggest that it will eventually provide the lowest $LCOS$ *for medium-duration storage. By 2050, sodium-ion batteries are expected to deliver storage at 11–14 Euros per* MWh*, significantly cheaper than lithium-ion projections of 16–22 Euros per* MWh*.*[22] *This trend is driven by high learning rates and the avoidance of critical mineral scarcity.*[22]

System Type	***2026 Upfront ($/kWh)***	***25-Year LCOS ($/kWh)***	***Service Frequency***
Lead-Acid	*$150 – $300*	*$0.25 – $0.40*	*Monthly* [13]
Lithium LFP	*$600 – $900*	*$0.18 – $0.28*	*Minimal* [7]
Vanadium Flow	*$900 – $1,200*	*$0.11 – $0.17*	*Annual/Bi-annual* [14]

NMC	*$500 – $700*	*$0.20 – $0.30*	*Low* [8]
Sodium-Ion	*$400 – $600*	*TBD (Falling)*	*Minimal* [20]

Safety, Regulation, and the Built Environment

The integration of solar batteries into residential and commercial buildings is governed by increasingly strict safety codes, notably UL 9540 (Energy Storage Systems) and NFPA 855 (Standard for the Installation of Stationary Energy Storage Systems).[7] *These regulations dictate minimum clearances from property lines, ventilation requirements, and fire suppression protocols.*[7]

Thermal Management and Installation

For lithium-ion (particularly NMC) and lead-acid, thermal management is a critical design constraint. These chemistries require active cooling or heating to maintain an optimal operating range of $15^\circ C$ *to* $35^\circ C$.[11] *Operating outside these bounds significantly accelerates degradation and increases safety risks. LFP and sodium-ion offer more flexibility, operating effectively in wider temperature windows (* $-4^\circ F$ *to* $140^\circ F$ *for LFP), which reduces the energy consumed by the system's own climate control—often called "parasitic load"—by 5% to 10%.*[7]

Environmental and Ethical Considerations

The sustainability of battery chemistry is becoming a differentiator for corporate buyers. Lead-acid batteries are the standard for recyclability, with a mature closed-loop system.[13] *Lithium-ion recycling is a growing but complex industry, with LFP being harder to recycle profitably because it lacks high-value nickel and cobalt.*[10] *However, LFP and sodium-ion are considered more "ethical" as they avoid the environmental and human rights issues associated with cobalt mining.*[2] *Flow batteries are perhaps the most sustainable, as the electrolyte is 100% reusable and the mechanical parts are easily salvaged.*[14]

Strategic Deployment Summary

Choosing the correct solar battery chemistry in 2026 requires balancing physical constraints against lifecycle economic goals. For standard residential and C&I applications where safety and lifespan are paramount, Lithium Iron Phosphate ($LiFePO_4$ *) remains the industry benchmark. For space-critical retrofits, Lithium NMC continues to provide the necessary density, albeit with a higher safety and replacement overhead.*

Redox Flow batteries are the definitive choice for long-duration, utility-scale projects where the goal is grid-level resilience over a 30-year timeframe. Lead-acid technologies occupy a niche for low-use backup or remote, budget-constrained off-grid sites. Finally, Sodium-Ion represents the most promising frontier for resource-secure, low-cost mass deployment, particularly in climates with extreme temperatures.

The convergence of these five chemistries allows the 2026 solar market to address a diverse range of needs, from a single home in a suburb to a multi-megawatt industrial solar park. As manufacturing continues to scale and costs continue their downward trajectory, energy storage has transformed from a luxury addition into the essential heartbeat of the global energy transition."

[2a] **Case Study 2 Prompt: Do in depth research on DC motor, and motor control technologies and their applications in mobile robotics platforms. You may use tables but minimize their use.**

[2b] **Case Study 2 ChatGPT Deep Research Response:**

1) Why DC motors dominate mobile robotics

For mobile robots, the actuator has to do four things well: produce usable wheel torque at low speed, respond quickly to control commands, fit inside a battery-powered system, and interface cleanly with embedded electronics. That is why DC motor families—mainly brushed DC motors and brushless DC / PMSM motors—show up so often in mobile robotics. Brushed DC motors are easy to drive and model because torque is closely tied to armature current, which makes them attractive for simple wheeled robots and low-cost drivetrains. Brushless motors usually win when a robot needs better efficiency, longer service life, higher power density, smoother operation, or more advanced torque/speed control. In practice, that is why you see brushed DC on many small educational and budget differential-drive robots, while BLDC/PMSM solutions are common in higher-performance AMRs, AGVs, drones, and industrial mobile platforms.

A second reason DC motors are so common is that they fit the way mobile robots are controlled. A robot controller usually thinks in layers: "I want the chassis to move at this linear and angular velocity," then "I need each wheel at this speed," then "I need each motor to generate this torque/current." Differential-drive and car-like robots both map body motion into wheel motion in a very direct way, and wheel encoders can feed that back into odometry and low-level motor loops. ROS 2's mobile robot kinematics documentation is a good reflection of how this is done in modern robotics software stacks: wheel speeds are derived from body twist, and odometry can be computed from encoder readings.

2) The main DC motor types used in mobile robots

Brushed DC motors

A brushed DC motor uses mechanical commutation through brushes and a commutator. Its biggest strengths in mobile robotics are simplicity, low controller complexity, easy reversal, and low entry cost. With a brushed motor, you can get bidirectional motion with an H-bridge and control speed with PWM. That makes brushed DC especially appealing in student robots, hobby platforms, compact service robots, and lower-cost AGV drive modules. TI's brushed motor control application note shows a very typical embedded setup: a controller reads speed and current, runs PI/PID logic, and updates PWM duty cycle and direction through an H-bridge.

The tradeoff is that brushes wear, generate electrical noise, and become a maintenance and lifetime limitation. Brushed motors also tend to be less efficient and less scalable for very high-duty-cycle fleets than brushless alternatives. That does not make them bad for robots; it just means they are strongest where cost, design speed, and implementation simplicity matter more than maximum efficiency or maintenance-free life. In many mobile robots, especially small differential-drive systems, brushed DC motors with gearboxes and encoders remain a completely valid choice because they are easy to integrate into closed-loop speed control.

Brushless DC motors / PMSMs

"BLDC" in practice overlaps heavily with permanent-magnet synchronous motors (PMSMs) in robotics hardware. These motors replace brushes with electronic commutation, which usually improves efficiency, power density, reliability, and noise performance. TI and ST both emphasize that advanced control methods like FOC improve efficiency, control accuracy, torque smoothness, and dynamic response in brushless drives. That makes brushless motors especially attractive in autonomous mobile robots that run for long periods from batteries, where drivetrain losses directly reduce range and usable runtime.

Brushless systems do ask more from the electronics and software. Instead of one H-bridge, a typical 3-phase BLDC drive uses a three-phase inverter, rotor position information or position estimation, commutation logic, and often a faster control MCU. That added complexity is the price you pay for better performance. On modern robotic platforms, though, that price is increasingly worth paying, especially when the platform needs precise torque control, quiet operation, regenerative braking, trajectory tracking, or high efficiency. ODrive's robotics-oriented product line is a good real-world example of how the market has moved: it explicitly emphasizes FOC current control, torque/velocity/position modes, regenerative braking, and encoder support as core *robotics features.*

3) Why gearmotors are so common on mobile robots

Most mobile robots do not drive wheels directly from the raw motor shaft. They use a gearbox, often planetary or spur, because the wheel needs much more torque and much less speed than the motor naturally provides. This matters a lot in mobile robotics because the robot almost always operates near low or moderate wheel speeds where traction force, obstacle crossing, slope climbing, and start-up acceleration dominate. Real robotics case studies from maxon repeatedly show the same pattern: motors are paired with gearheads to reach practical output torque and packaging targets, whether on hull-cleaning robots, service robots, or articulated systems.

Planetary gearboxes are popular in mobile robots because they offer high reduction in compact volume and tend to handle torque better than many simple spur layouts of comparable size. Spur gear reductions can still make sense when cost, simplicity, and efficiency at lower reduction ratios matter. In mobile robotics, the gearbox choice is really a system trade: more reduction gives better low-speed torque and easier control authority, but it can also limit top speed, increase reflected inertia, add backlash, and reduce backdrivability. That becomes important if the robot relies on odometry accuracy, precise low-speed motion, or compliant interaction with terrain.

4) The control stack: from battery to wheel motion

A lot of confusion in robotics comes from mixing up motor, driver, controller, and feedback sensor. They are related, but not the same.

- *The motor converts electrical energy into mechanical rotation.*
- *The power driver / inverter switches current into the motor.*
- *The controller computes what the driver should do.*
- *The sensor reports speed, position, or current so the controller can close the loop.*

For a brushed DC wheel drive, the power stage is usually an H-bridge. NXP describes this class of device as a programmable H-bridge power IC designed to drive DC motors or bidirectional actuators, and TI's brushed-control note shows the normal software architecture: measure speed and current, run a speed loop and current loop, then update duty cycle and direction. That is the heart of modern low-level motion control.

For a brushless drive, the power stage is a 3-phase inverter. The controller must know when and how strongly to energize each phase. That can be done with simple six-step commutation, or with more sophisticated sinusoidal/FOC methods. The more advanced the commutation, the more the robot can get smooth torque, quiet operation, and *accurate low-speed control.*

5) Open-loop versus closed-loop motor control

Open-loop control

Open-loop control means the controller sends a command—often just PWM duty cycle—without verifying that the wheel actually achieved the intended speed or torque. This is the simplest possible approach and can work for toys, very low-cost platforms, or situations where accuracy is not important. But mobile robots rarely live in such a clean world. Terrain changes, floor friction changes, battery voltage sags, wheel wear, and left/right motor mismatch all push the robot away from the commanded motion. That is why a differential-drive robot controlled open-loop often drifts instead of moving straight.

Closed-loop control

Closed-loop control adds feedback—usually encoder speed/position and often current sensing. This is what most serious mobile robots use. A low-level controller compares commanded wheel speed to measured wheel speed and keeps correcting the PWM/inverter output. The result is more repeatable velocity, better straight-line tracking, better turning accuracy, and usable odometry. The ROS 2 kinematics documentation explicitly notes that odometry can be calculated from encoder readings, and TI's encoder note explains how incremental encoders provide pulses that indicate motion, speed, and direction.

In practice, most mobile robots benefit from nested loops:

1. *an inner current/torque loop,*
2. *a middle speed loop,*
3. *and sometimes an outer position/trajectory loop.*

That structure matters because current changes motor torque quickly, while speed and position evolve more slowly. TI's recent brushed-motor note is a clear example of this thinking: speed is controlled through feedback, the current is limited or regulated, and the duty cycle is computed from those loops.

6) PWM, H-bridges, and why they matter so much in robotics

For brushed DC motors, pulse-width modulation (PWM) is the standard method for controlling average motor voltage and therefore speed/torque behavior. The H-bridge lets the system reverse polarity across the motor so the wheel can move forward or backward. This is foundational in mobile robotics because nearly every wheeled robot needs reversible traction. NXP's H-bridge documents and application material describe these *drivers as the basic means to apply voltage across a motor in either direction, while TI's note shows how that switching is integrated with sensed speed/current loops.*

The important engineering point is that PWM is not the same thing as closed-loop control. PWM is just the actuator command method. A robot can use PWM open-loop, which is crude, or PWM inside a closed-loop controller, which is much better. In mobile robotics, the second approach is the one that actually delivers consistent path following and odometry. This matters especially for differential-drive platforms, where tiny left/right speed differences accumulate into heading error very quickly.

7) Motor feedback technologies used on mobile robots

Incremental encoders

Incremental rotary encoders are the most common feedback device on mobile robots. TI describes them as devices that translate rotational movement into electrical signals, with pulses that can indicate speed and direction. In practice, they are popular because they are affordable, easy to read with microcontrollers, and directly useful for wheel speed control and odometry. ROS-based wheeled robots commonly use encoder counts to estimate travel distance and heading change.

The limitation is that incremental encoders tell you motion relative to a starting point, not absolute shaft angle at power-up. For many wheeled robots that is perfectly fine. You command the robot to start from a known state, then integrate wheel motion forward. But if you need instant startup position knowledge or you need to recover from resets gracefully, incremental sensing alone can be limiting.

Absolute encoders

Absolute encoders report shaft angle directly, even after power cycling. maxon's ENX encoder documentation notes that some compact encoder families can provide incremental square-wave signals and also absolute angular values via interfaces such as SSI or BiSS-C. In mobile robotics, absolute sensing is especially attractive when a platform needs precise startup alignment, accurate wheel module initialization, or robust position knowledge in higher-end servo systems.

Hall sensors

Hall sensors are common in BLDC motors for commutation and coarse position feedback. They are cheaper and simpler than high-resolution encoders, but they provide much lower angular resolution. maxon's EPOS4 application notes point out that observers become especially helpful when using low-resolution feedback like Hall *sensors or relatively low-count incremental encoders, because they can produce better velocity estimation and improved low-speed regulation. That is very relevant in mobile robots that need quiet, smooth crawl speeds.*

8) Brushed motor control technologies in mobile robotics

For brushed wheel motors, the most common control architectures are:

a) Voltage / PWM control

This is the simplest: command PWM and accept whatever speed results. It is cheap and easy but sensitive to load and battery variation. It is usually acceptable only for the simplest robots.

b) Closed-loop speed control with encoder feedback

This is the standard practical upgrade. The encoder measures wheel speed, and a PI or PID controller adjusts PWM so the wheel tracks the target speed. The 2022 mobile robot study on a brushed DC motor with a low-cost magnetic quadrature encoder is a good example of how useful even modest encoder-based PID control can be for overshoot, undershoot, and velocity response in a robot application.

c) Current-limited or current-controlled brushed drive

Here the system monitors motor current to protect hardware and sometimes to regulate torque more directly. This matters in mobile robots because wheel stall, curb impacts, carpet transitions, and startup acceleration can pull much higher current than steady cruising. TI's sensored brushed-motor control note explicitly uses speed and current information together, which is a strong reflection of how real robotic drives are implemented.

d) Position control

This is more common for steering actuators, pan/tilt mechanisms, and robotic appendages than for pure traction wheels, but it can also appear in wheel modules or docking mechanisms. maxon's position-control materials show how encoder interfaces and servo controllers are used when precise commanded position matters more than just raw wheel speed.

9) Brushless motor control technologies in mobile robotics

a) Six-step / trapezoidal commutation

This is the classic BLDC control method. The controller energizes the three phases in six discrete sectors per electrical cycle. Microchip's and ST's application notes describe six-step / trapezoidal / 120° commutation as a standard sensorless BLDC method. Its biggest strengths are simplicity, lower compute burden, and easier implementation compared with FOC. That makes it attractive in cost-sensitive robots, fans, pumps, and some basic traction systems.

The downside is higher torque ripple, more acoustic noise, and weaker low-speed smoothness than FOC. For a mobile robot, that can show up as jerkier crawl behavior, poorer precision at very low wheel speed, and more difficulty holding smooth motion under changing load. On a robot that only needs coarse speed control, six-step may still be enough. On a robot that needs precise indoor navigation, smooth docking, or silent operation, it is often not the best end state.

b) Sensorless BLDC control

Sensorless control removes physical rotor-position sensors and estimates commutation timing from motor signals such as back-EMF. Microchip notes that its sensorless six-step approach works on a wide range of motors, does not require detailed motor parameters, and is relatively insensitive to manufacturing variation. That is attractive for cost and wiring reduction.

The catch is that sensorless methods are usually weakest at very low speed or standstill, because back-EMF becomes hard to measure there. For mobile robots, that is a real concern: robots spend a lot of time starting, stopping, creeping, docking, and making small corrections. So sensorless BLDC can be excellent in the right operating region, but sensored systems usually behave better when a robot must deliver confident low-speed traction and repeatable startup torque.

c) Sensored BLDC with Hall or encoder feedback

Adding Hall sensors or an encoder improves startup and low-speed control. Hall sensors are often enough for commutation and moderate speed regulation, while high-resolution encoders support much better velocity and position estimation. NXP's older BLDC application note explicitly targets quadrature-encoder-based 3-phase BLDC control, which is still conceptually aligned with how robotic servo drives are built today.

d) Field-oriented control (FOC)

FOC is the most important advanced motor-control technology to understand for modern mobile robotics. TI describes FOC as making the stator and rotor magnetic fields orthogonal to achieve maximum electromagnetic torque, while improving efficiency, *smoothness, and dynamic response; TI also notes that it can support operation above nominal speed through field weakening. ST similarly positions FOC as a way to improve control and accuracy for brushless motors.*

Why that matters on a robot:

- *smoother low-speed motion,*
- *lower torque ripple,*
- *better efficiency from a battery pack,*
- *better torque response when load changes,*
- *better suitability for torque, speed, and position loops.*

That is a big reason FOC is so common in modern robotic wheel drives, gimbals, and servo modules. ODrive's robotics products, for example, highlight FOC current control together with torque, velocity, position, trajectory control, encoder support, and regenerative braking—exactly the feature set you expect when the motor controller is being designed for serious robot motion rather than just "spin the motor."

10) Regenerative braking and energy management

In mobile robots, the drivetrain is part of the power system, not just the motion system. When the robot decelerates, goes downhill, or is externally driven, a motor can act as a generator. Some higher-end brushless controllers support regenerative braking, returning energy to the DC bus or safely dissipating it through a brake resistor if the battery cannot absorb it fast enough. ODrive explicitly advertises regenerative braking and brake-chopper capability for robotics applications.

This matters more as robots get heavier and faster. On a tiny classroom robot, regen may not be worth the complexity. On an AGV or AMR carrying useful payload, it can improve energy efficiency and thermal behavior. The design challenge is that regenerated energy has to go somewhere safe. If the battery is full or cannot accept charge fast enough, the DC bus can overvoltage unless the controller provides a dissipation path. That is one reason power electronics and battery management become inseparable in professional mobile robot design.

11) Applications by robot class

Here is the most useful way to think about actuator choice: the right motor technology depends on the robot's operating regime.

Robot/application class	*Typical motor choice*	*Why*

Small educational or hobby differential-drive robot	*Brushed DC gearmotor + incremental encoder*	*Cheapest path to closed-loop wheel control and odometry; easy H-bridge drive.*
Student design robot / light service platform	*Brushed DC gearmotor or low-voltage BLDC*	*Brushed for simplicity, BLDC if efficiency/runtime/noise matter more.*
Industrial AMR / AGV	*BLDC/PMSM with sensored FOC*	*Better efficiency, longer life, smoother motion, stronger low-speed control.*
Harsh-environment or long-duty-cycle mobile robot	*BLDC + gearbox + sealed/robust feedback*	*Better reliability and lower maintenance under sustained operation.*
Precision mobile platform or wheel module	*BLDC/PMSM + encoder + servo drive*	*Better torque, velocity, and position regulation.*

That table is simplified, but the pattern is real: brushed DC tends to win on simplicity and cost; brushless tends to win on performance, efficiency, and lifecycle.

12) Motor control and robot kinematics are tightly linked

On mobile robots, motor control cannot be separated from locomotion architecture.

For a differential-drive robot, the controller independently sets left and right wheel speeds. ROS 2's equations show that body linear velocity is the average of left/right wheel velocities, while angular velocity depends on their difference divided by track width. That means wheel-speed control quality directly determines path quality. If one wheel is poorly controlled, the robot's heading will drift.

For a car-like or Ackermann robot, the traction motor and the steering actuator play different roles. The drive motor largely determines longitudinal motion, while the steering motor determines curvature. That often means the drive motor can be speed-focused while the steering motor needs tighter position control. Again, encoder placement matters: the ROS 2 documentation explicitly distinguishes odometry cases based on *where encoders are mounted.*

For tracked robots, drivetrain control gets harder because track-ground interaction introduces more slip and uneven resistance. In those robots, strong current limiting and good wheel/track speed feedback become even more important, because open-loop voltage control alone is usually not enough to keep motion predictable. The hull-cleaning robot case study is a good reminder that motor selection on battery-operated mobile robots is inseparable from friction, sealing, payload, and environmental loading.

13) The role of observers, estimation, and sensor fusion

A common beginner assumption is that "an encoder solves everything." It does not. Encoders give you shaft motion, but mobile robots still suffer from wheel slip, backlash, compliance, and sensor noise. That is why practical systems often combine low-level motor feedback with higher-level estimation from IMUs and localization. maxon's EPOS4 application note notes that observers can improve velocity estimation and low-speed regulation when feedback resolution is limited, and the Nav2 documentation shows how odometry is commonly fused with other sources for smoother robot state estimation.

So the real control story in mobile robotics is layered:

- *motor current control for torque,*
- *wheel-speed control for traction,*
- *wheel odometry for local motion estimate,*
- *IMU/localization fusion for a better pose estimate,*
- *planner/controller above that.*

That layered architecture is one reason motor and encoder selection matter so much more than they first appear to.

14) What usually drives actuator choice in real projects

In real mobile robot design, the choice is rarely "which motor is theoretically best?" It is usually "which motor-control stack best fits the platform constraints?" The most important constraints are:

Battery efficiency and runtime.
Brushless + FOC often wins for sustained duty because better efficiency means less heat and more runtime per charge.

Control simplicity and development time.
Brushed DC + H-bridge + encoder is often the fastest path to a working robot.

Low-speed smoothness.
If the robot must crawl, dock, or move precisely indoors, sensored BLDC with good control algorithms becomes much more attractive.

Maintenance and lifecycle.
Brushless systems usually win for long-lived fleets because they eliminate brushes as a wear item.

Cost and integration burden.
The cheapest motor is not always the cheapest drivetrain once driver complexity, sensing, firmware, and debugging are included. That is why many student teams still choose brushed gearmotors even when BLDC looks better on paper.

15) Bottom-line conclusions

If you strip away the marketing language, the practical picture is pretty clear.

Brushed DC motors are still highly relevant in mobile robotics because they are easy to drive, cheap to prototype with, and pair naturally with H-bridges, PWM, and encoder-based PID speed loops. They are especially good for educational robots, prototypes, low-cost differential drives, and systems where fast implementation matters more than peak efficiency.

Brushless DC / PMSM motors are the stronger long-term solution for many serious mobile robots because they offer better efficiency, smoother torque, better durability, and better compatibility with modern control methods like FOC. They are especially attractive in AMRs, AGVs, service robots, and battery-powered platforms that need long duty cycles, precise low-speed motion, or advanced trajectory control.

The most important control technologies to know for mobile robotics are:

- *PWM + H-bridge for brushed drives,*
- *encoder-based closed-loop speed control for repeatable wheel motion,*
- *current limiting/current control for torque and protection,*
- *six-step commutation for simpler BLDC systems,*
- *FOC for higher-performance brushless traction and servo motion,*
- *odometry from encoders tied into the robot's kinematic model,*
- *and sensor fusion/observers when raw encoder data is not enough.*

For most mobile robotics platforms, the best mental model is this:
the motor is only one part of the drivetrain; the real capability comes from the motor + gearbox + driver + sensors + control algorithm acting as a single system.

If you want, I can turn this into a report-style section with headings and references, or rewrite it specifically for a senior design mobile robot paper.

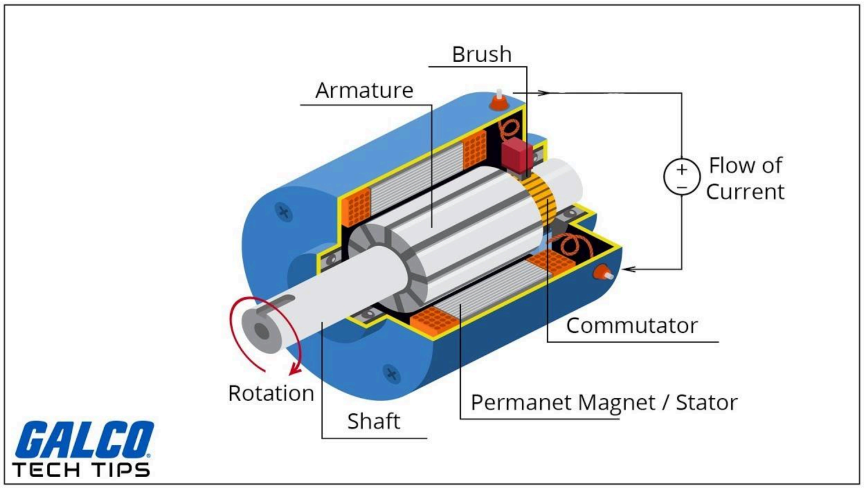
[2c] **Case Study 2 Grok Deep Research Response:**

DC motors serve as the workhorse actuators in mobile robotics platforms, prized for their high starting torque, variable speed control, compact size, and direct compatibility with battery power sources. Mobile robots—whether hobbyist differential-drive bots, warehouse autonomous mobile robots (AMRs), or research platforms like TurtleBot—rely on them for propulsion, precise maneuvering, and reliable operation in dynamic environments. This research draws from engineering fundamentals, control theory, and real-world robotic applications to explore DC motors, their control technologies, and integration in mobile systems.

Fundamentals of DC Motors: Brushed vs. Brushless

DC motors convert electrical energy into mechanical rotation through the interaction of magnetic fields (Lorentz force principle). Current in a conductor within a magnetic field produces torque, causing the rotor to spin. Two primary types dominate robotics: brushed and brushless (BLDC) DC motors. Both are permanent-magnet designs, but they differ fundamentally in commutation—the process of switching current to maintain continuous rotation.

Brushed DC motors use a mechanical commutator (segmented copper rings on the rotor) and carbon or precious-metal brushes to switch current direction in the armature windings as the rotor turns. The stator holds permanent magnets. This self-commutating design allows simple direct connection to a DC supply, making them inexpensive and easy to drive. However, brushes wear out over time (typically 1,000–5,000 hours), cause electrical arcing/sparking, generate electromagnetic interference (EMI), and produce audible noise. Efficiency is lower (around 60–80%) due to friction and brush losses, and they require more maintenance.



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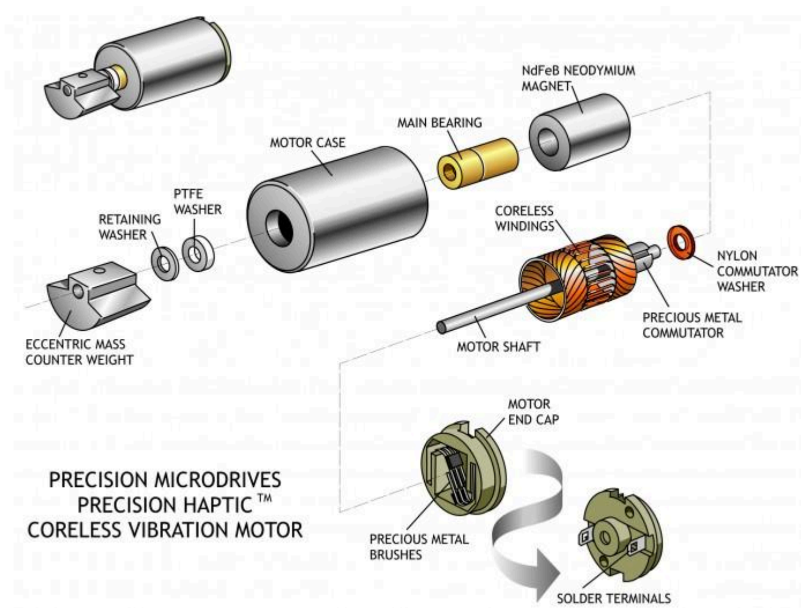


precisionmicrodrives.com

Brushless DC (BLDC) motors reverse the configuration: permanent magnets are on the rotor, and windings are on the stator. Electronic commutation (via an external controller) replaces brushes, using Hall-effect sensors, encoders, or sensorless back-EMF detection to time phase energization. This yields higher efficiency (often 85–95%), greater torque density, higher speeds (up to tens of thousands of RPM), quieter operation, longer life (20,000+ hours with minimal maintenance), and better thermal characteristics since heat-generating windings are stationary. Drawbacks include higher upfront cost and the need for sophisticated drivers. BLDC motors excel in battery-powered mobile robots *where efficiency directly extends runtime.*

lamnow.com

Brushless DC Motor Stator Winding | Lamnow

In practice, both types are often paired with gearboxes (e.g., planetary or spur) to increase torque and reduce speed for robotic traction needs. Gear motors with encoders are standard for odometry in mobile platforms.

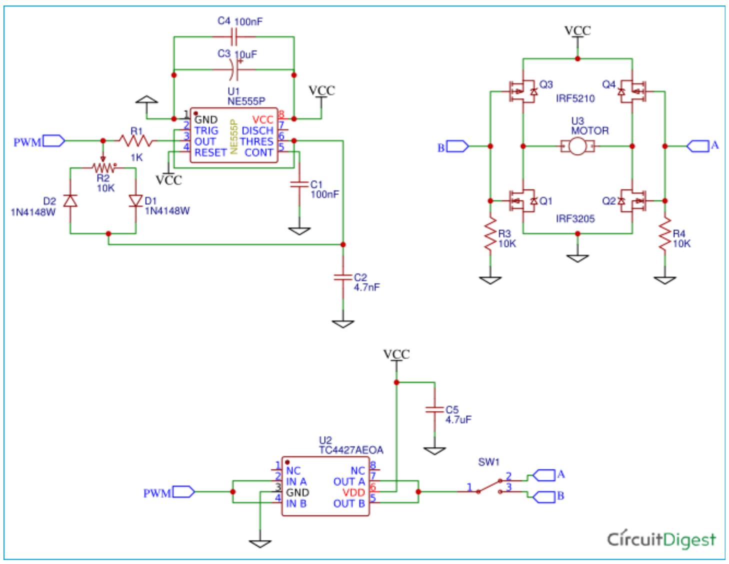
Motor Control Technologies

Raw DC motors respond to voltage for speed (via back-EMF) and current for torque, but practical control in robotics demands precision for velocity, position, direction, and load compensation. Key technologies include:

Pulse Width Modulation (PWM) is the cornerstone for speed control. A microcontroller or driver rapidly switches full voltage on/off at a fixed frequency (typically 1–20 kHz to avoid audible noise while keeping switching losses low). The duty cycle (percentage of "on" time) sets the average voltage delivered to the motor, effectively controlling speed without wasting power as heat (unlike linear regulators). In mobile robots, PWM enables smooth acceleration and energy-efficient cruising.

H-bridge circuits handle bidirectional control (forward/reverse) and braking. Four switches (transistors or MOSFETs) form an "H" shape around the motor. Activating opposite pairs (e.g., top-left and bottom-right) reverses polarity. Modern implementations use complementary PWM with dead-time insertion to prevent shoot-through (simultaneous conduction causing shorts). Integrated drivers like the L298N (for *brushed) or three-phase bridge drivers for BLDC simplify this. Braking modes (coast, slow decay, fast decay) further tune responsiveness.*

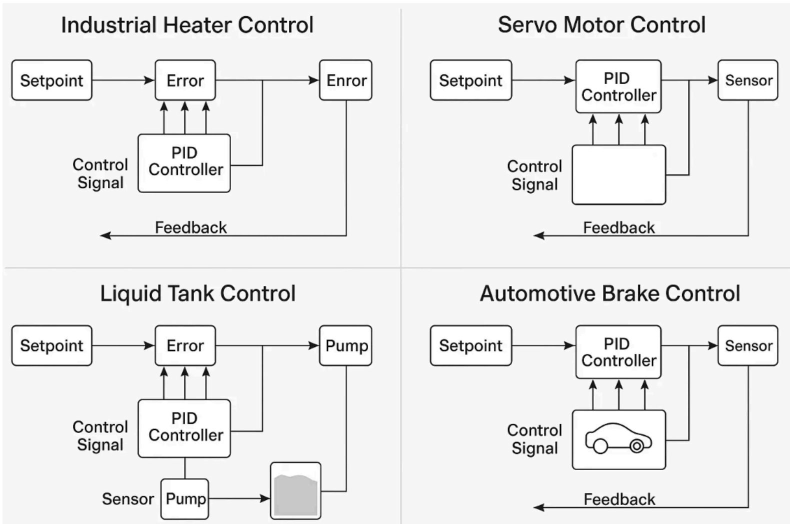
hackatronic.com



circuitdigest.com

Feedback systems close the control loop. For BLDC commutation, three Hall-effect sensors (60° or 120° spacing) detect rotor position via magnetic fields. For finer control, quadrature encoders (optical or magnetic) on the shaft provide high-resolution position/speed feedback (hundreds to thousands of counts per revolution). Encoders enable odometry—calculating robot pose from wheel rotations—in SLAM and navigation. Sensorless techniques for BLDC estimate position from back-EMF voltage during rotation, reducing wiring, cost, and failure points, though they perform poorly at very low speeds.

PID (Proportional-Integral-Derivative) control is ubiquitous for closed-loop performance. It compares measured speed/position (from encoders) against a setpoint, computing an error signal. The proportional term reacts to current error, integral eliminates steady-state offset (e.g., from friction or slope), and derivative damps overshoot. Tuned via Ziegler-Nichols or manual methods (often with ROS2 diff_drive_controller), PID ensures a robot maintains velocity despite uneven terrain, payload changes, or wheel slip. Cascaded loops (outer position, inner velocity) are *common in advanced platforms.*



keyirobot.com

Deep Dive: Understanding PID Control for Robot Motor Stabilization

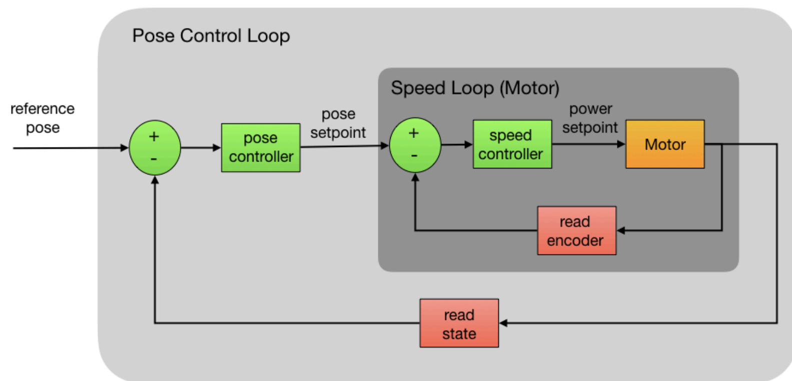
For BLDC motors, advanced algorithms shine: trapezoidal (6-step) commutation is simple but causes torque ripple; sinusoidal reduces ripple; Field-Oriented Control (FOC)—also called vector control—decouples torque and flux currents (like separately excited DC motors) using Clarke/Park transforms and PI regulators. FOC delivers smooth, efficient, high-dynamic performance ideal for precise robotic motion. Modern microcontrollers (e.g., with hardware FOC peripherals) or dedicated ICs make it accessible. Sensorless FOC variants are a growing trend for compact AMRs.

Brushless Electronic Speed Controllers (ESCs) integrate these functions for hobby robotics, while industrial drivers add CAN/EtherCAT for fleet coordination.

Applications in Mobile Robotics Platforms

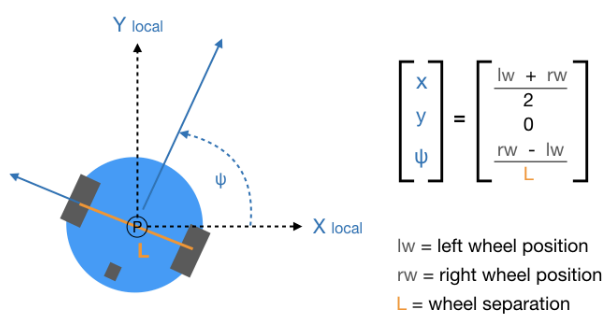
Mobile robots emphasize locomotion, so DC motors drive wheels, tracks, or legs with emphasis on torque for acceleration, efficiency for battery life, and controllability for navigation. The most common configuration is differential drive: two independently controlled motors (usually geared DC or BLDC) on opposite sides, plus casters or omniwheels for stability. Varying relative speeds enables forward/backward motion, in-place rotation (zero turning radius), and curved paths—kinematics are straightforward via wheel odometry.

Pose Control Loop



forum.dronebotworkshop.com

Wheeled Robot Internal Kinematics



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Examples across scales:

- *Hobby/educational platforms (e.g., Arduino or Raspberry Pi bots) use low-cost brushed gear motors with H-bridge shields and basic PWM/PID for line-following or obstacle avoidance.*
- *Research robots like TurtleBot employ BLDC or brushed motors with high-resolution encoders for ROS-based SLAM, mapping, and autonomous navigation.*
- *Industrial AMRs/AGVs in warehouses (e.g., for logistics) favor BLDC gearmotors with FOC or sensorless control for quiet, efficient 24/7 operation under varying loads. High starting torque handles inclines or sudden stops.*
- *Tracked robots (e.g., search-and-rescue) use dual-motor differential steering for*

rough terrain.
- *Delivery or agricultural robots benefit from BLDC's power density for longer range.*

Advantages in mobile contexts: DC motors provide instant high torque for starting heavy payloads or climbing, precise speed variation for smooth trajectories, and regenerative braking for energy recovery. Battery compatibility eliminates bulky inverters needed for AC motors. In differential setups, independent motor control simplifies software (e.g., pure pursuit or PID-based path following).

Challenges include wheel slip (mitigated by PID + IMU fusion), EMI from brushes (favoring BLDC in sensor-heavy robots), and thermal management in enclosed chassis. Gear reduction trades speed for torque but introduces backlash, addressed by high-quality encoders.

Current trends (as of 2026): The shift toward BLDC with FOC and sensorless control accelerates due to efficiency gains, falling controller costs, and integration in compact drives. Market growth reflects demand in robotics and EVs, with AI-tuned adaptive PID or neural controllers emerging for uncertain environments. Frameless or integrated gearmotors reduce weight, while functional safety features (e.g., safe torque off) appear in collaborative AMRs.

In summary, DC motors—especially BLDC variants with modern electronic controls—enable the mobility, precision, and endurance that define today's mobile robotics. From simple PWM/H-bridge hobby bots to sophisticated FOC-driven industrial fleets, these technologies balance cost, performance, and reliability, with ongoing advances in sensorless and intelligent control pushing the boundaries further. For custom platforms, motor selection hinges on torque/speed requirements, battery voltage, and control complexity, always validated through simulation and real-world testing.

[3a] Case Study 3 Prompt: When selecting an ultrasonic sensor for a design project, what are some of the most important considerations to be aware of?

[3b] Case Study 3 Claude Sonnet 4.6 Response:

Here are the key considerations when selecting an ultrasonic sensor for a design project:

Range The sensor's minimum and maximum detection distance must match your application. Most ultrasonic sensors have a blind spot (minimum range) where they can't detect objects — for close-range work, this matters a lot. The MaxBotix sensors, for example, have variants tuned for short vs. long range.

Beam Width / Beam Pattern Ultrasonic sensors emit a cone-shaped beam. A narrow beam gives more precise directional detection; a wide beam covers more area but can pick up unintended objects (walls, floors, etc.). Know what your sensor's beam angle is and whether it suits your environment.

Resolution and Accuracy These are different things — resolution is the smallest change the sensor can detect; accuracy is how close the reading is to the actual distance. For precise obstacle avoidance or measurement tasks, both matter.

Output Type Sensors typically output via one of: PWM (pulse width), analog voltage, UART serial, or I2C. Choose based on what your microcontroller supports and how many sensors you're running simultaneously.

Update Rate How fast does the sensor report readings? For a moving robot, a slow update rate means stale distance data, which can cause missed obstacles or jittery control. Higher update rates are better for dynamic environments.

Environmental Sensitivity Ultrasonic sensors are affected by temperature (speed of sound changes with temperature), humidity, and surface material (soft/angled surfaces absorb or deflect the signal). Outdoor or variable-temp environments need more care here.

Power Consumption Relevant if you're on a battery-powered platform. Some sensors have sleep/low-power modes; others are always-on.

Cross-Talk (Multi-Sensor Interference) If you're running multiple ultrasonic sensors simultaneously, their signals can interfere with each other. You'll need staggered firing (multiplexing) or sensors designed for multi-sensor setups.

Form Factor and Mounting Physical size, mounting hole pattern, and whether the sensor faces forward-only vs. has a transducer you can aim — all affect integration into your mechanical design.

Cost and Availability For a student/prototype project, the difference between a $4 HC-SR04 and a $25+ MaxBotix unit is significant. Higher-end sensors earn their price in better beam control, more reliable output, and built-in filtering.